

The Health and Economic Impact of a Global System for Pragmatic Clinical Trials

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Abstract

Clinical trials cost \$2.60B (95% CI: \$1.50B-\$4B) and take a decade per drug. Most humans die before the drugs that could save them finish paperwork. A decentralized framework for drug assessment could transform this by enabling: (1) subsidized patient participation making trials accessible and incentivized, (2) universal trial access from home via phone or computer, (3) real-world data aggregation into unified databases, (4) treatment rankings (“Consumer Reports for drugs”) with every treatment ranked by real-world effectiveness, and (5) outcome labels showing exactly what happened to real patients. Pragmatic trials cost 44.1x (95% CI: 39.4x-89.1x) less than traditional trials per patient, enabling vastly more parallel research. The framework generates health benefits through timeline acceleration: exploring the therapeutic search space would take ~443 years (95% CI: 324 years-712 years) under the status quo, but treatments get discovered ~212 years (95% CI: 135 years-355 years) earlier with expanded capacity. Combined with eliminating the 8.2 years (95% CI: 4.85 years-11.5 years) regulatory efficacy delay (via opt-in access to ubiquitous trials after Phase I safety), the average treatment reaches patients significantly sooner. This timeline shift saves 10.7B deaths (95% CI: 7.39B deaths-16.2B deaths), eliminates 1931T hours (95% CI: 1362T hours-2616T hours) of suffering, and generates \$84.8 quadrillion (95% CI: \$62.4 quadrillion-\$97.3 quadrillion) in economic value at \$0.841 (95% CI: \$0.242-\$1.75)/DALY - competitive with the most effective global health interventions (bed nets at \$89 (95% CI: \$78-\$100)/DALY) while operating at vastly greater scale.

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i Understanding These Figures: The Untested Treatment Backlog

The bottleneck: 6.65k diseases (95% CI: 5.70k diseases-8.24k diseases) currently lack effective treatments. We have 9.50k compounds (95% CI: 7.00k compounds-12.0k compounds) proven-safe (FDA-approved drugs + GRAS substances), yet only 0.342% (95% CI: 0.21%-0.514%) of drug-disease combinations have ever been tested. At the current discovery rate of 15 diseases/year (95% CI: 8 diseases/year-30 diseases/year), systematically testing all 9.50M combinations plausible pairings would take ~443 years (95% CI: 324 years-712 years). Most will never be tested.

The figures below represent **cumulative benefits over the entire acceleration period** - the total lives saved by addressing this backlog faster. This is the same methodology used to value smallpox eradication (program cost → total future lives saved) and climate infrastructure investments. These are not annual figures; they are the one-time benefit of permanently accelerating medical progress.

1 Executive Summary

The Problem: Clinical trials cost billions and require a decade per drug. The majority of patients with treatable conditions die before potentially effective treatments complete regulatory approval. Of 2.40B people (95% CI: 2.00B people-2.80B people) people with chronic disease, only 1.90M patients/year (95% CI: 1.50M patients/year-2.30M patients/year) participate in trials annually (0.06%).

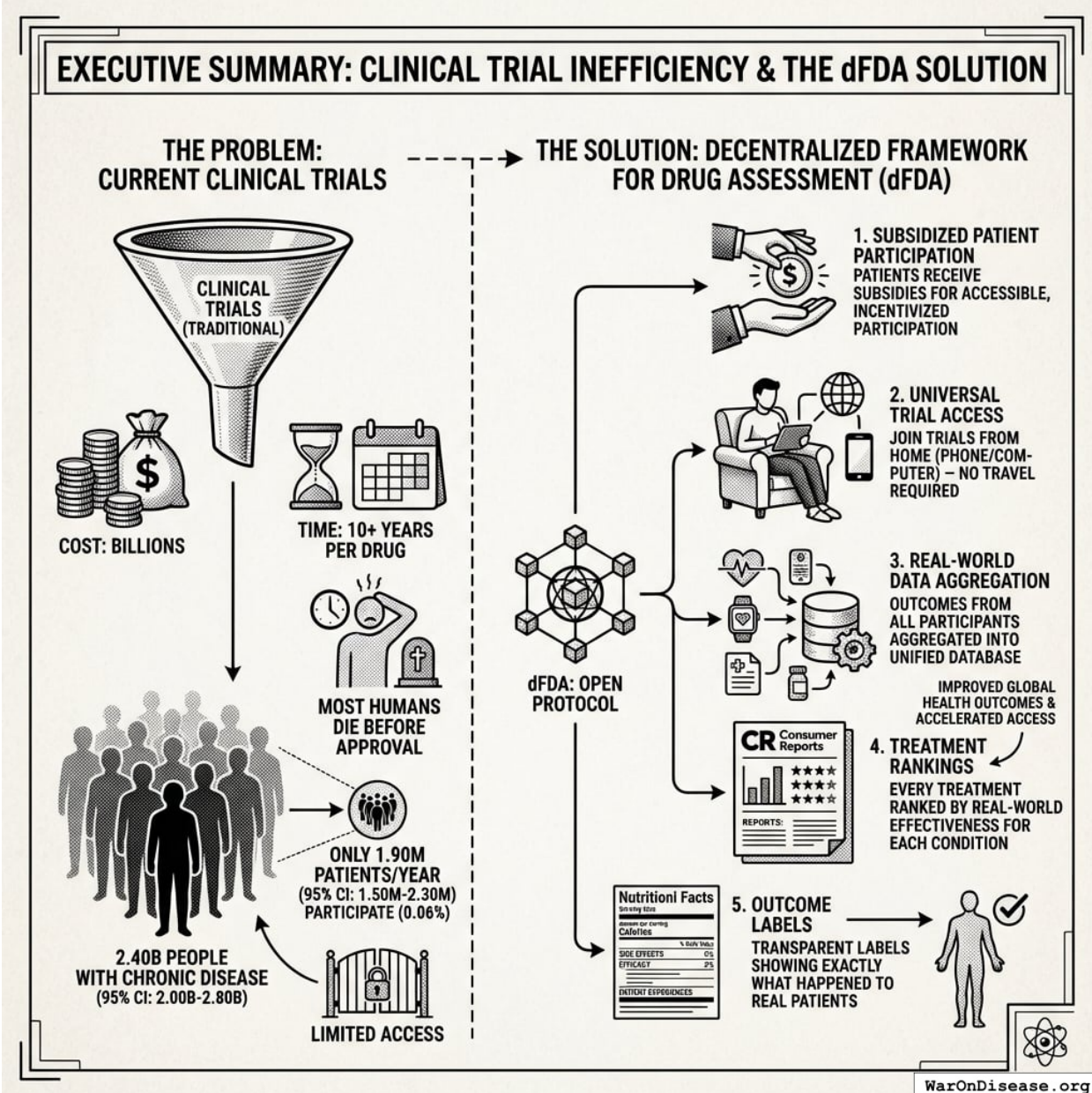


Figure 1: Architecture of the dFDA protocol showing the flow from decentralized patient participation and mobile data collection to aggregated treatment rankings and outcome labels.

The Solution: A decentralized framework for drug assessment (dFDA) - an open protocol that enables:

- 1. Subsidized Patient Participation:** Patients receive subsidies to participate in trials, making participation accessible and incentivized
- 2. Universal Trial Access:** Any patient can join trials from home via their phone or computer - no travel to research centers required
- 3. Real-World Data Aggregation:** Outcomes from all participants are aggregated into a unified database

4. **Treatment Rankings:** Standardized effectiveness rankings for every treatment-condition pair, updated continuously with real-world evidence
5. **Outcome Labels:** “Nutrition facts for drugs” showing exactly what happened to real patients who tried each treatment

i Protocol, Not Platform

The dFDA is **protocol infrastructure** - open standards that existing systems adopt, not a competing service:

- **What it IS:** Open standards for trial data exchange, outcome reporting, and treatment ranking - like HTTP for clinical trials, or FHIR for health data
- **What it is NOT:** A SaaS product that competes with Medable, Science 37, or other clinical trial platforms
- **How existing systems benefit:** DCT platforms, EHR systems, and health apps integrate the protocol to gain interoperability, access larger patient pools, and contribute to standardized treatment rankings

Who builds on it: Medable, Science 37, Epic, Cerner, academic medical centers, Apple Health, Fitbit - any system collecting health outcomes.

How it works: Companies register treatments through any protocol-compliant system. Patients search for their condition, see treatments ranked by effectiveness, and can join trials through their preferred app or platform. Patient-reported outcomes flow back into the aggregated rankings. The result is a self-improving ecosystem where every participant’s experience helps the next patient make better decisions.

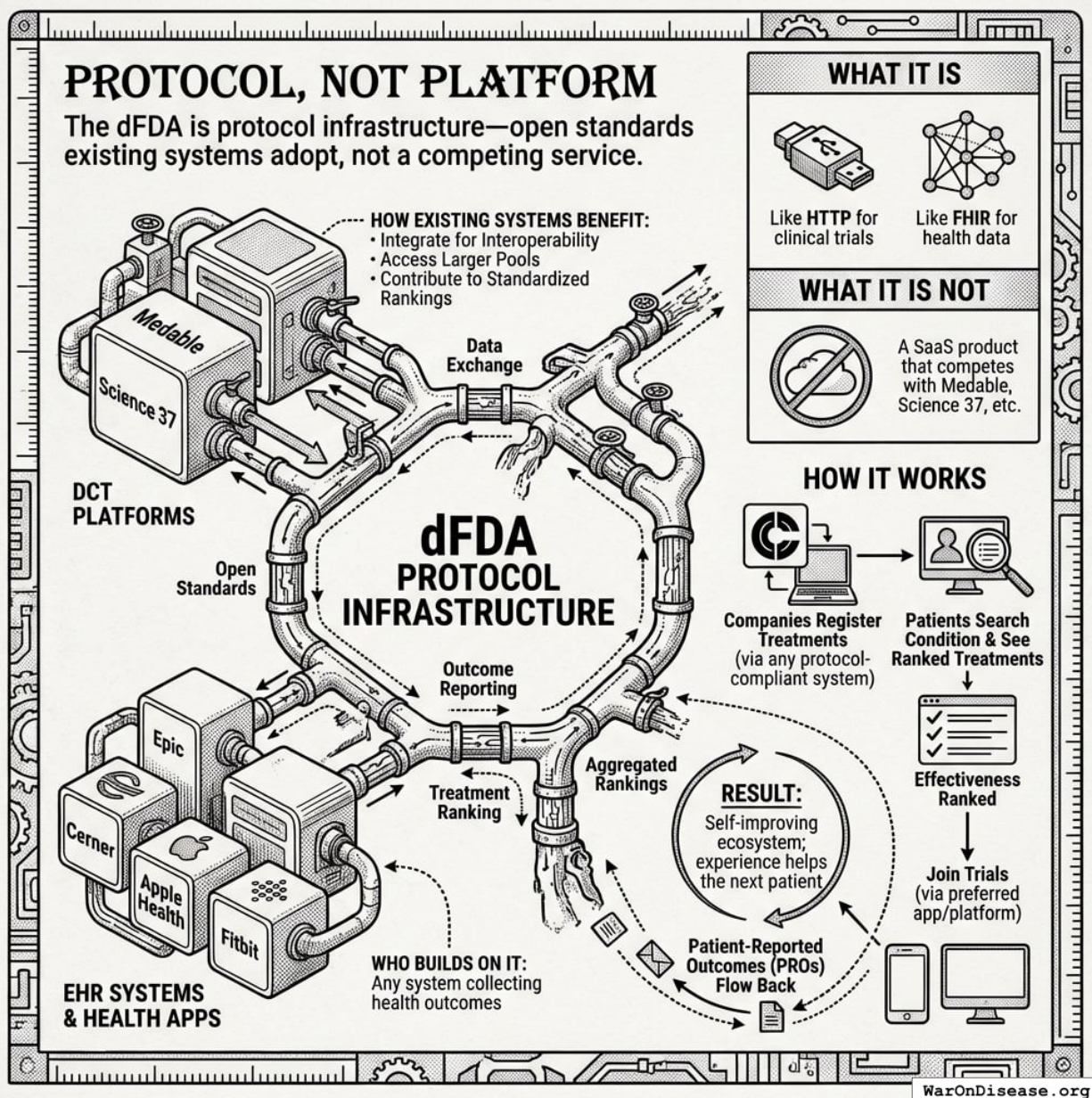


Figure 2: System architecture diagram showing the dFDA protocol as a foundational layer that enables interoperability between EHRs, clinical trial platforms, and consumer health apps through a shared data ecosystem.

1.1 What You Get

- **Cost Reduction:** Clinical trial costs decrease by 97.7% (95% CI: 97.5%-98.9%) (44.1x (95% CI: 39.4x-89.1x)). Pragmatic trials like ADAPTABLE (\$929 (95% CI: \$929-\$1.40K)/patient) and systematic reviews (median \$97 (95% CI: \$19-\$478)/patient) demonstrate feasibility. Applied to the \$60B (95% CI: \$50B-\$75B) annual trial market, this yields tens of billions in savings.
- **Expanded Therapeutic Coverage:** Lower costs enable testing of rare diseases and treat-

ments that lack commercial incentives. Treatments reach patients earlier in disease progression.

- **Mortality Reduction:** The framework generates 565B DALYs (95% CI: 361B DALYs-877B DALYs) additional life-years through the 212 years (95% CI: 135 years-355 years) timeline shift (12.3x (95% CI: 4.19x-61.3x) trial capacity plus efficacy lag elimination), enabling faster access, improved prevention data, and research for neglected diseases.

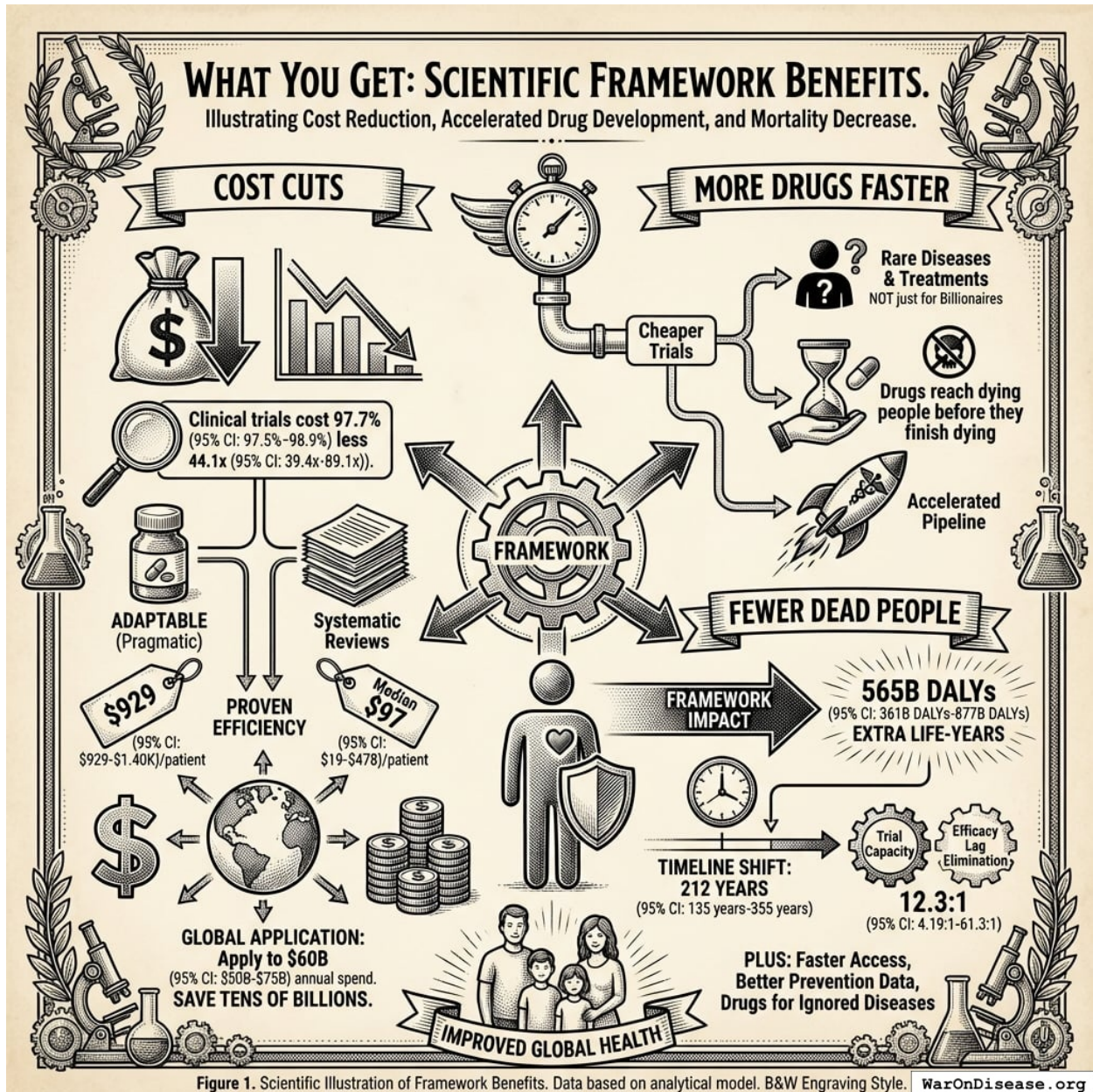


Figure 3: Before/after comparison showing 97.7% cost reduction, 12.3x trial capacity increase, and 565B DALYs averted from traditional to decentralized framework.

1.2 Key Findings

Metric	Value	Context
Cost-Effectiveness	\$0.841 (95% CI: \$0.242-\$1.75)/DALY	Competitive with bed nets (\$89 (95% CI: \$78-\$100)/DALY) at vastly greater scale
Lives Saved	10.7B deaths (95% CI: 7.39B deaths-16.2B deaths)	One-time benefit from 212 years (95% CI: 135 years-355 years) timeline shift
DALYs Averted	565B DALYs (95% CI: 361B DALYs-877B DALYs)	Captures both mortality and morbidity
Suffering Eliminated	1931T hours (95% CI: 1362T hours-2616T hours)	Human suffering averted from timeline shift
Total Economic Value	\$84.8 quadrillion (95% CI: \$62.4 quadrillion-\$97.3 quadrillion)	10.7B deaths (95% CI: 7.39B deaths-16.2B deaths) standard QALY valuation
Efficacy Lag Eliminated	8.2 years (95% CI: 4.85 years-11.5 years)	Post-Phase I access via trial participation
ROI (R&D Savings)	637:1 (95% CI: 569:1-790:1)	44.1x (95% CI: 39.4x-89.1x) cheaper trials
Annual R&D Savings	\$58.6B (95% CI: \$49.2B-\$73.1B)	From 97.7% (95% CI: 97.5%-98.9%) cost reduction
Trial Capacity Increase	12.3x (95% CI: 4.19x-61.3x)	Enabling parallel therapeutic space exploration

Key Metric Derivations

Lives Saved:

$$\begin{aligned}
& Lives_{max} \\
&= Deaths_{disease,daily} \times T_{accel,max} \times 338 \\
&= 150,000 \times 212 \times 338 \\
&= 10.7B
\end{aligned}$$

$$\text{where } T_{accel,max} = T_{accel} + T_{lag} = 204 + 8.2 = 212$$

$$\begin{aligned}
& \text{where } T_{accel} \\
&= T_{first,SQ} \times \left(1 - \frac{1}{k_{capacity}}\right) \\
&= 222 \times \left(1 - \frac{1}{12.3}\right) \\
&= 204
\end{aligned}$$

$$\text{where } T_{first,SQ} = T_{queue,SQ} \times 0.5 = 443 \times 0.5 = 222$$

$$\text{where } T_{queue,SQ} = \frac{N_{untreated}}{Treatments_{new,ann}} = \frac{6,650}{15} = 443$$

$$\text{where } N_{untreated} = N_{rare} \times 0.95 = 7,000 \times 0.95 = 6,650$$

$$\text{where } k_{capacity} = \frac{N_{fundable,ann}}{Slots_{curr}} = \frac{23.4M}{1.9M} = 12.3$$

$$\begin{aligned}
& \text{where } N_{fundable,ann} \\
&= \frac{Subsidies_{trial,ann}}{Cost_{pragmatic,pt}} \\
&= \frac{\$21.7B}{\$929} \\
&= 23.4M
\end{aligned}$$

$$\begin{aligned}
& \text{where } Subsidies_{trial,ann} \\
&= Treasury_{RD,ann} - OPEX_{dFDA} \\
&= \$21.8B - \$40M \\
&= \$21.7B
\end{aligned}$$

$$\begin{aligned}
& \text{where } OPEX_{dFDA} \\
&= Cost_{platform} + Cost_{staff} + Cost_{infra} \\
&\quad + Cost_{regulatory} + Cost_{community} \\
&= \$15M + \$10M + \$8M + \$5M + \$2M \\
&= \$40M
\end{aligned}$$

$$\begin{aligned}
& \text{where } Treasury_{RD,ann} \\
&= Funding_{treaty} - Payout_{bond,ann} - Funding_{political,ann} \\
&= \$27.2B - \$2.72B - \$2.72B \\
&= \$21.8B
\end{aligned}$$

$$\begin{aligned}
& \text{where } Funding_{treaty} \\
&= Spending_{mil} \times Reduce_{treaty} \\
&= \$2.72T \times 1\% \\
&= \$27.2B
\end{aligned}$$

Suffering Hours Eliminated:

$$\begin{aligned}
& Hours_{suffer,max} \\
&= DALYs_{max} \times Pct_{YLD} \times 8760 \\
&= 565B \times 0.39 \times 8760 \\
&= 1930T
\end{aligned}$$

$$\begin{aligned}
& \text{where } DALYs_{max} \\
&= DALYs_{global,ann} \times Pct_{avoid,DALY} \times T_{accel,max} \\
&= 2.88B \times 92.6\% \times 212 \\
&= 565B
\end{aligned}$$

$$\text{where } T_{accel,max} = T_{accel} + T_{lag} = 204 + 8.2 = 212$$

$$\begin{aligned}
& \text{where } T_{accel} \\
&= T_{first,SQ} \times \left(1 - \frac{1}{k_{capacity}}\right) \\
&= 222 \times \left(1 - \frac{1}{12.3}\right) \\
&= 204
\end{aligned}$$

$$\text{where } T_{first,SQ} = T_{queue,SQ} \times 0.5 = 443 \times 0.5 = 222$$

$$\text{where } T_{queue,SQ} = \frac{N_{untreated}}{Treatments_{new,ann}} = \frac{6,650}{15} = 443$$

$$\text{where } N_{untreated} = N_{rare} \times 0.95 = 7,000 \times 0.95 = 6,650$$

$$\text{where } k_{capacity} = \frac{N_{fundable,ann}}{Slots_{curr}} = \frac{23.4M}{1.9M} = 12.3$$

$$\begin{aligned}
& \text{where } N_{fundable,ann} \\
&= \frac{Subsidies_{trial,ann}}{Cost_{pragmatic,pt}} \\
&= \frac{\$21.7B}{\$929} \\
&= 23.4M
\end{aligned}$$

$$\begin{aligned}
& \text{where } Subsidies_{trial,ann} \\
&= Treasury_{RD,ann} - OPEX_{dFDA} \\
&= \$21.8B - \$40M \\
&= \$21.7B
\end{aligned}$$

$$\begin{aligned}
& \text{where } OPEX_{dFDA} \\
&= Cost_{platform} + Cost_{staff} + Cost_{infra} \\
&\quad + Cost_{regulatory} + Cost_{community} \\
&= \$15M + \$10M + \$8M + \$5M + \$2M \\
&= \$40M
\end{aligned}$$

$$\begin{aligned}
& \text{where } Treasury_{RD,ann} \\
&= Funding_{treaty} - Payout_{bond,ann} - Funding_{political,ann} \\
&= \$27.2B - \$2.72B - \$2.72B \\
&= \$21.8B
\end{aligned}$$

Cost per DALY:

Interpreting the Timeline Figure: The 212 years (95% CI: 135 years-355 years)-year figure represents a **discovery capacity model** of medical research. Think of the therapeutic search space as the set of all untested drug-disease combinations, with trial capacity determining how fast we can explore it.

The Discovery Capacity Model: {#the-discovery-capacity-model}

Parameter	Status Quo	With dFDA	Impact
Untreated diseases	6.65k diseases (95% CI: 5.70k diseases-8.24k diseases)	6.65k diseases (95% CI: 5.70k diseases-8.24k diseases)	Same backlog
Discovery rate (first treatments/year)	15 diseases/year (95% CI: 8 diseases/year-30 diseases/year)	185 diseases/year (95% CI: 107 diseases/year-490 diseases/year)	12.3x (95% CI: 4.19x-61.3x) faster
Time to explore search space	443 years (95% CI: 324 years-712 years)	36 years (95% CI: 11.6 years-77.2 years)	Centuries saved
Expected time to first treatment	~443 years (95% CI: 324 years-712 years)/2	~36 years (95% CI: 11.6 years-77.2 years)/2	204 years (95% CI: 123 years-350 years) earlier

Why treatments arriving sooner saves lives:

A disease that would receive its first effective treatment in year 200 under the status quo might receive it in year 16 with dFDA. During those 184 years, people die from that disease who could have been saved. The 10.7B deaths (95% CI: 7.39B deaths-16.2B deaths) figure captures the cumulative lives saved across all diseases during their acceleration periods.

The two components:

1. **Discovery acceleration** (204 years (95% CI: 123 years-350 years)): Higher discovery rate explores the therapeutic space faster, moving treatments forward
2. **Efficacy lag elimination** (8.2 years (95% CI: 4.85 years-11.5 years)): Once discovered, treatments reach patients immediately instead of waiting for Phase II/III

Total timeline shift: 212 years (95% CI: 135 years-355 years) = 204 years (95% CI: 123 years-350 years) + 8.2 years (95% CI: 4.85 years-11.5 years)

How the 12.3x (95% CI: 4.19x-61.3x) capacity increase works: With \$21.8B/year in trial funding at \$929 (95% CI: \$97-\$3K)/patient (based on ADAPTABLE trial), the system enables 23.4M patients/year (95% CI: 9.44M patients/year-96.8M patients/year) annual trial participants vs. current 1.90M patients/year (95% CI: 1.50M patients/year-2.30M patients/year), increasing trial completion rate from 15 diseases/year (95% CI: 8 diseases/year-30 diseases/year) to 185 diseases/year (95% CI: 107 diseases/year-490 diseases/year). This removes the primary bottleneck to medical progress: currently less than 0.06% of willing patients can access trials, and over 9.50k compounds (95% CI: 7.00k compounds-12.0k compounds) proven-safe (FDA-approved drugs + GRAS substances) remain untested for most conditions they could improve.

2 Vision and Capabilities

2.1 Core Model: An Open Coordination Protocol

The dFDA protocol enables existing systems (DCT platforms, EHRs, health apps) to:

For Treatment Providers (via compliant platforms):

- Register treatments through any protocol-compliant system
- Access aggregated effectiveness data across all participating platforms
- Receive automatic liability coverage through protocol governance
- Benefit from standardized outcome reporting across the ecosystem

For Patients (via participating apps and platforms):

- Search any condition, see treatments ranked by real-world effectiveness
- Join trials through their preferred platform or health app
- Receive subsidies to offset participation costs
- Report outcomes through any protocol-compliant interface
- Access “Outcome Labels” showing what happened to similar patients

The Result: A self-sustaining research ecosystem where participating platforms collect outcome data using standardized schemas, and the protocol aggregates this into continuously-updated treatment rankings available to all participants.

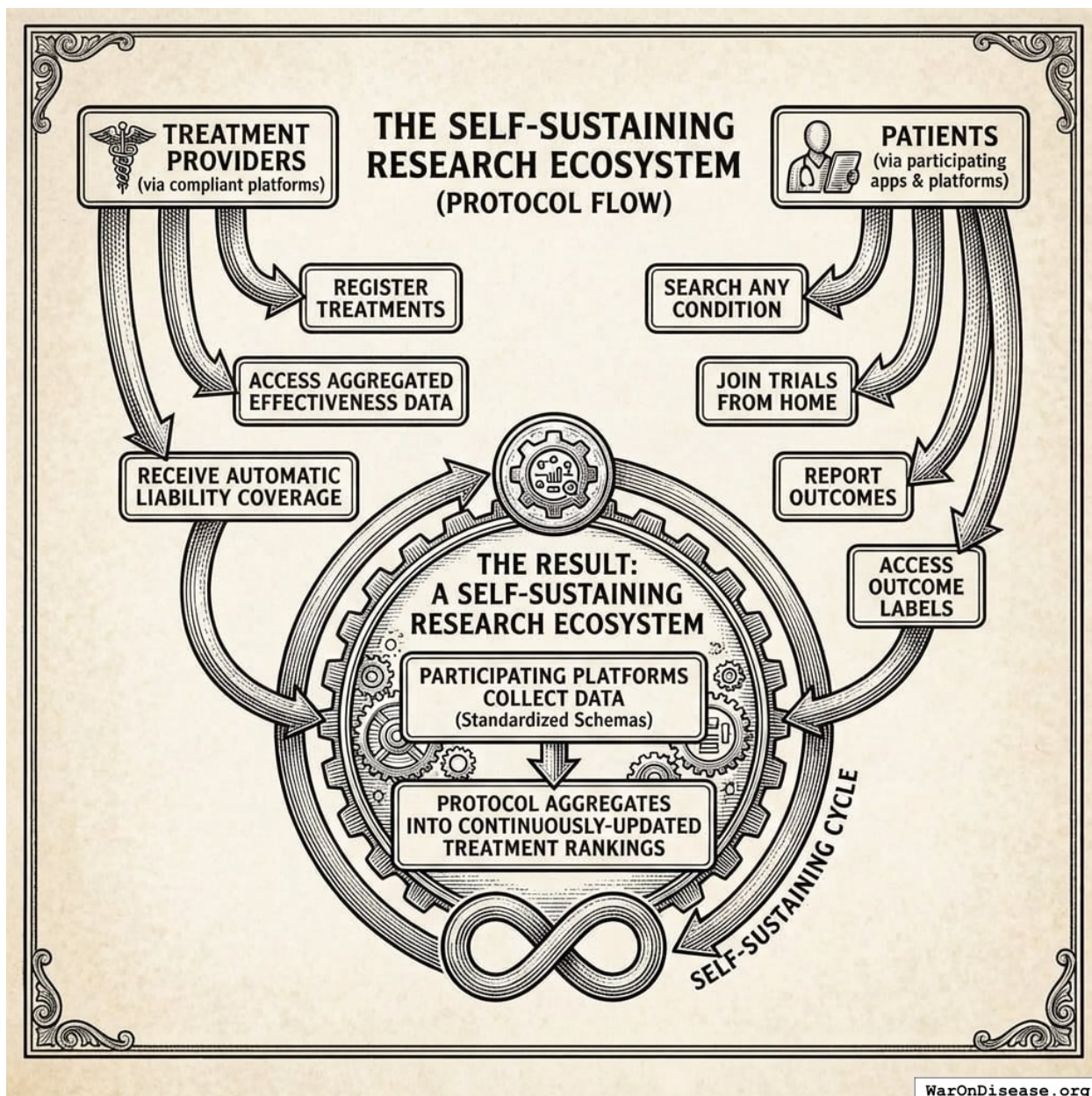


Figure 4: Operational flow of the decentralized framework, illustrating the cycle between patient participation, data aggregation via the protocol, and the generation of treatment rankings.

2.2 Key Capabilities

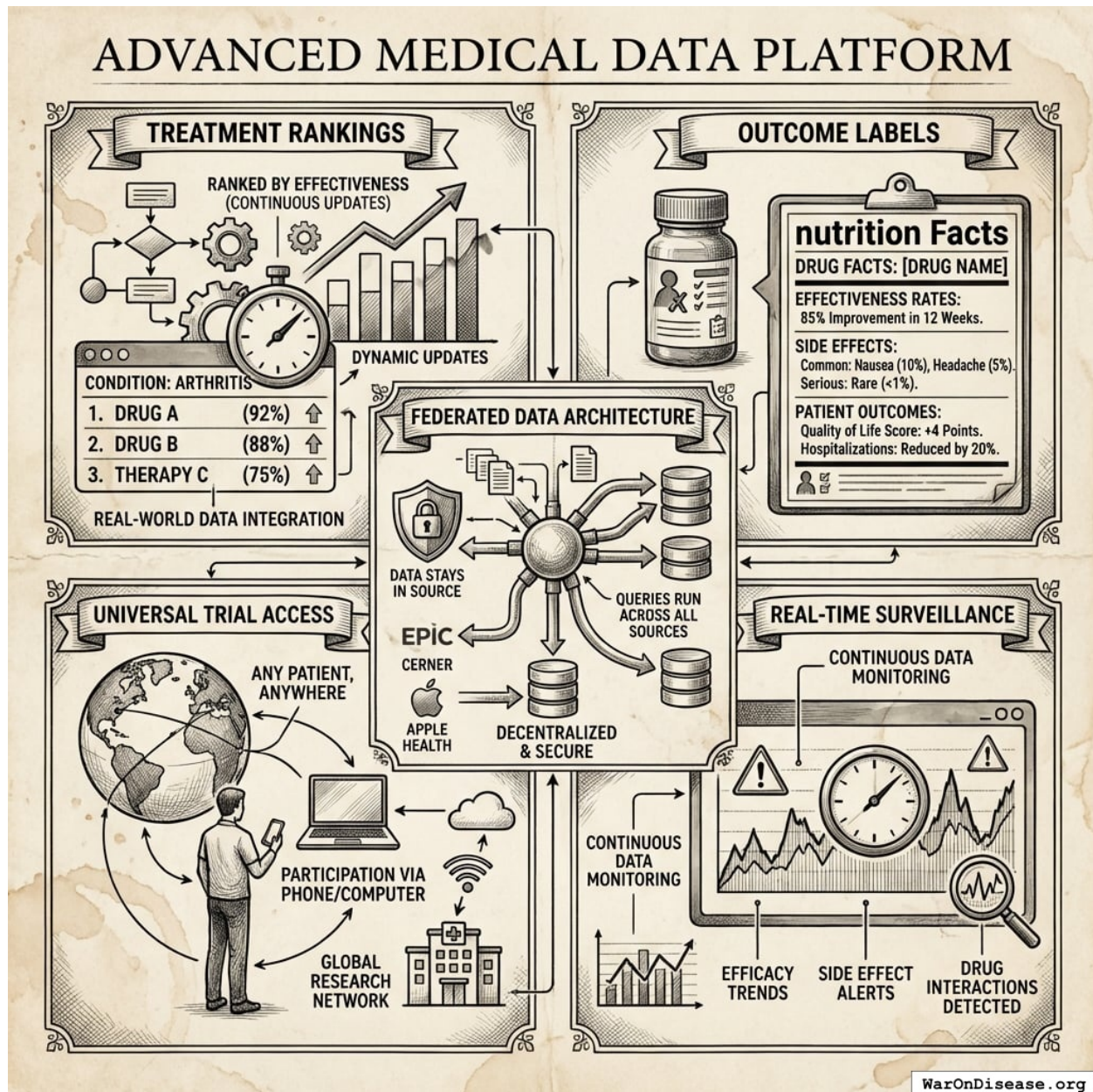


Figure 5: A federated data architecture diagram showing a central query engine interacting with distributed data sources like Epic, Cerner, and Apple Health, where data remains at the source while insights are aggregated.

- **Treatment Rankings:** Every treatment for every condition ranked by real-world effectiveness, updated continuously as new data arrives
- **Outcome Labels:** Standardized “nutrition facts for drugs” showing effectiveness rates, side effects, and outcomes from real patients
- **Universal Trial Access:** Any patient can participate from anywhere via phone/computer
- **Real-Time Surveillance:** Continuous data on efficacy, side effects, and drug interactions

- **Federated Data Architecture:** Data stays in source systems (Epic, Cerner, Apple Health) while queries run across all sources

2.3 Potential Impact on the Status Quo

- **Speed of Trials:** Reduced overhead and automated data capture compresses timelines.
- **Cost of Trials:** Using existing healthcare encounters, telemedicine, and EHR data to drastically cut per-patient costs (modeled on pragmatic trials like Oxford RECOVERY and the US-based ADAPTABLE trial).
- **Scale & Scope:** Enables testing many more drugs, off-label indications, unpatentable treatments, nutraceuticals, and personalized medicine approaches.
- **Innovation Incentives:** Lower R&D costs increase profitability and encourage more entrants/innovation in the life sciences.

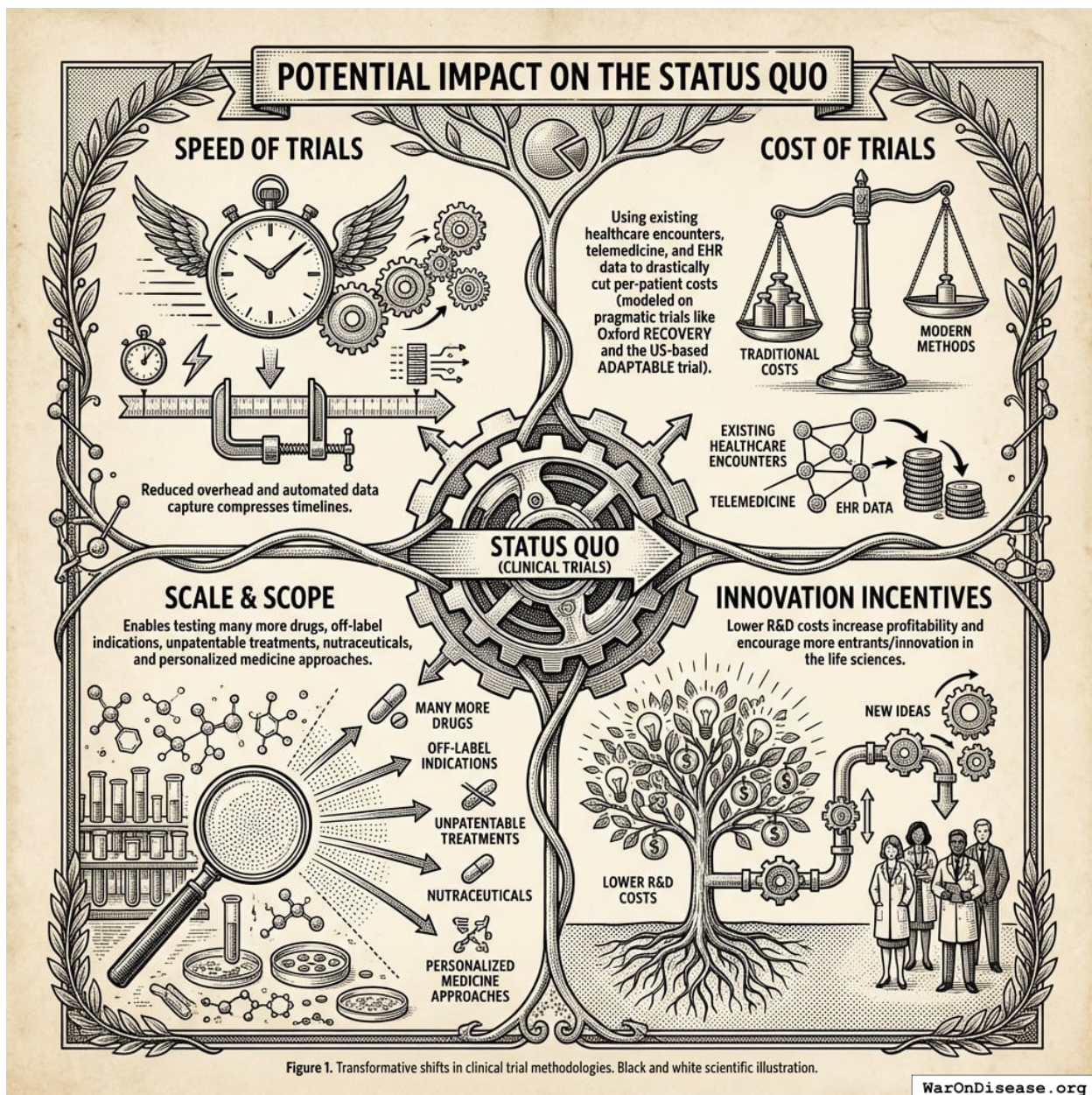


Figure 6: An infographic illustrating how technological optimizations in clinical trials lead to a virtuous cycle of reduced costs, faster timelines, and increased innovation.

3 Addressing Key Concerns

3.1 Why This Differs from Failed Megaprojects

Large-scale interventions face legitimate skepticism. The development economics literature documents numerous failures: infrastructure megaprojects that exceed budgets by 50-100%, foreign aid programs with negative or negligible returns, and “grand challenges” that fail to materialize promised benefits.

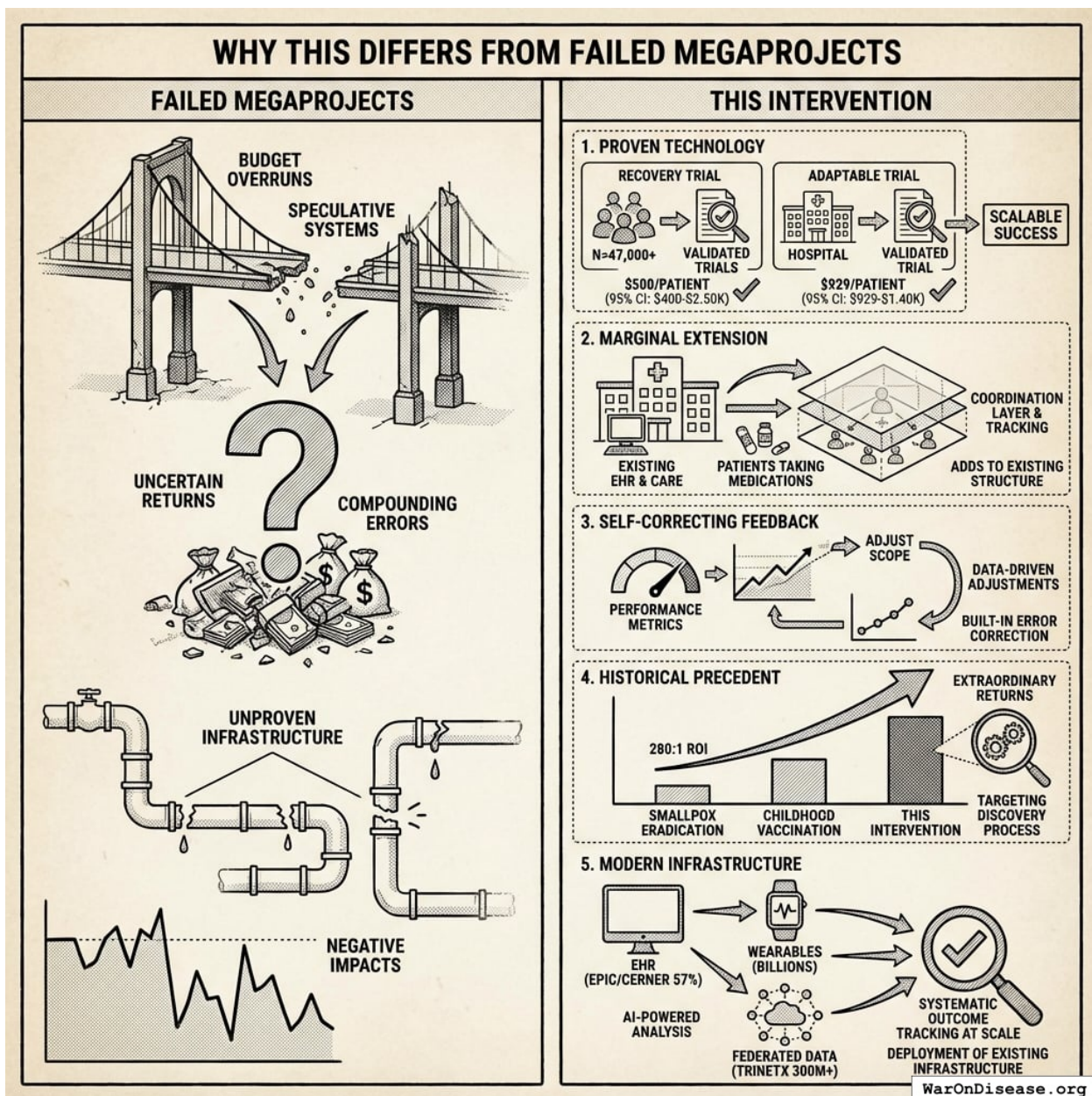


Figure 7: A comparison between the risks of traditional megaprojects and the advantages of the proposed intervention, highlighting the integration of existing infrastructure and feedback loops.

This intervention differs in four critical ways:

- 1. Proven Technology:** Unlike speculative moonshots, pragmatic trials using existing EHR infrastructure have been validated. The RECOVERY trial enrolled 47,000+ patients at \$500 (95% CI: \$400-\$2.50K)/patient. ADAPTABLE achieved \$929 (95% CI: \$929-\$1.40K)/patient in routine US healthcare settings. This isn't "we hope this works" - it's "we've proven this works, now scale it."
- 2. Marginal Extension, Not Novel System:** The framework extends existing clinical trial infrastructure rather than replacing it. Hospitals already have EHRs. Patients already take

medications. We’re adding a coordination layer and outcome tracking, not building from scratch.

3. **Self-Correcting Feedback:** Unlike infrastructure projects where failures compound, the protocol has built-in error correction. If a treatment doesn’t work, the data shows it. If costs exceed projections, we can adjust scope. The system generates its own performance metrics.
4. **Historical Precedent:** Smallpox eradication (280:1 ROI) and childhood vaccination programs demonstrate that systematic health interventions can achieve extraordinary returns. The difference: those targeted specific diseases. This targets the discovery process itself, potentially even higher leverage.
5. **Modern Infrastructure Makes This Possible Now:** The convergence of electronic health records (Epic/Cerner covering 57% of US hospitals), consumer wearables (billions of devices tracking health metrics), federated data networks (TriNetX queries 300M+ patient records without moving data), and AI-powered analysis enables systematic outcome tracking at scale that wasn’t feasible even a decade ago. This isn’t speculation about future technology - it’s deployment of existing, proven infrastructure.

3.2 Historical Validation: Pre-1962 Physician-Led Efficacy Testing

The decentralized physician-led efficacy model is not theoretical. It operated successfully from 1883 to 1960, providing 77 years of empirical validation.

How the pre-1962 system worked:

From 1883 to 1960, 144k physicians across America tested drug efficacy on real patients in routine clinical practice. The Journal of the American Medical Association (JAMA) compiled observational reports, leading medical experts peer-reviewed the aggregated data, and effective treatments received endorsement. This decentralized approach successfully identified antibiotics, vaccines, and countless surgical techniques.

Cost comparison demonstrates dramatic efficiency:

Era	Cost per Drug (2024 USD)	System
Pre-1962	\$24.7M (95% CI: \$19.5M-\$30M)	Decentralized physician-led efficacy testing
Post-1962	\$2.60B (95% CI: \$1.50B-\$4B)	Centralized pharmaceutical company trials
Cost Increase	105x (95% CI: 90.6x-119x)	Regulatory mandate, not drug complexity

The cost explosion began exactly when efficacy testing was centralized within pharmaceutical companies. This wasn’t a natural evolution of drug development or increasing drug complexity. The same types of compounds (small molecules, biologics) that cost \$24.7M (95% CI: \$19.5M-\$30M) to develop in 1960 now cost \$2.60B (95% CI: \$1.50B-\$4B).

The thalidomide success story: Thalidomide is often cited as justification for the 1962 amendments, but the US *already* blocked thalidomide under existing 1938 safety regulations. The FDA’s

Frances Kelsey refused approval based on inadequate safety data, not efficacy requirements. The 1962 amendments added *efficacy* proof requirements, not additional safety testing.

Implications: The dFDA model returns to decentralized physician-led efficacy testing but with modern automation (electronic health records, AI-assisted analysis, real-time data aggregation), targeting the same 50-95% cost reductions that the pre-1962 system achieved.

3.3 Why “Eventually Avoidable” Matters

A critical assumption in this analysis is that 92.6% (95% CI: 50%-98%) of disease deaths are “eventually avoidable” - meaning they could be prevented with sufficient biomedical research over time.

WHY “EVENTUALLY AVOIDABLE” MATTERS

A critical assumption: 92.6% (95% CI: 50%-98%) of disease deaths are “eventually avoidable” with sufficient biomedical research.

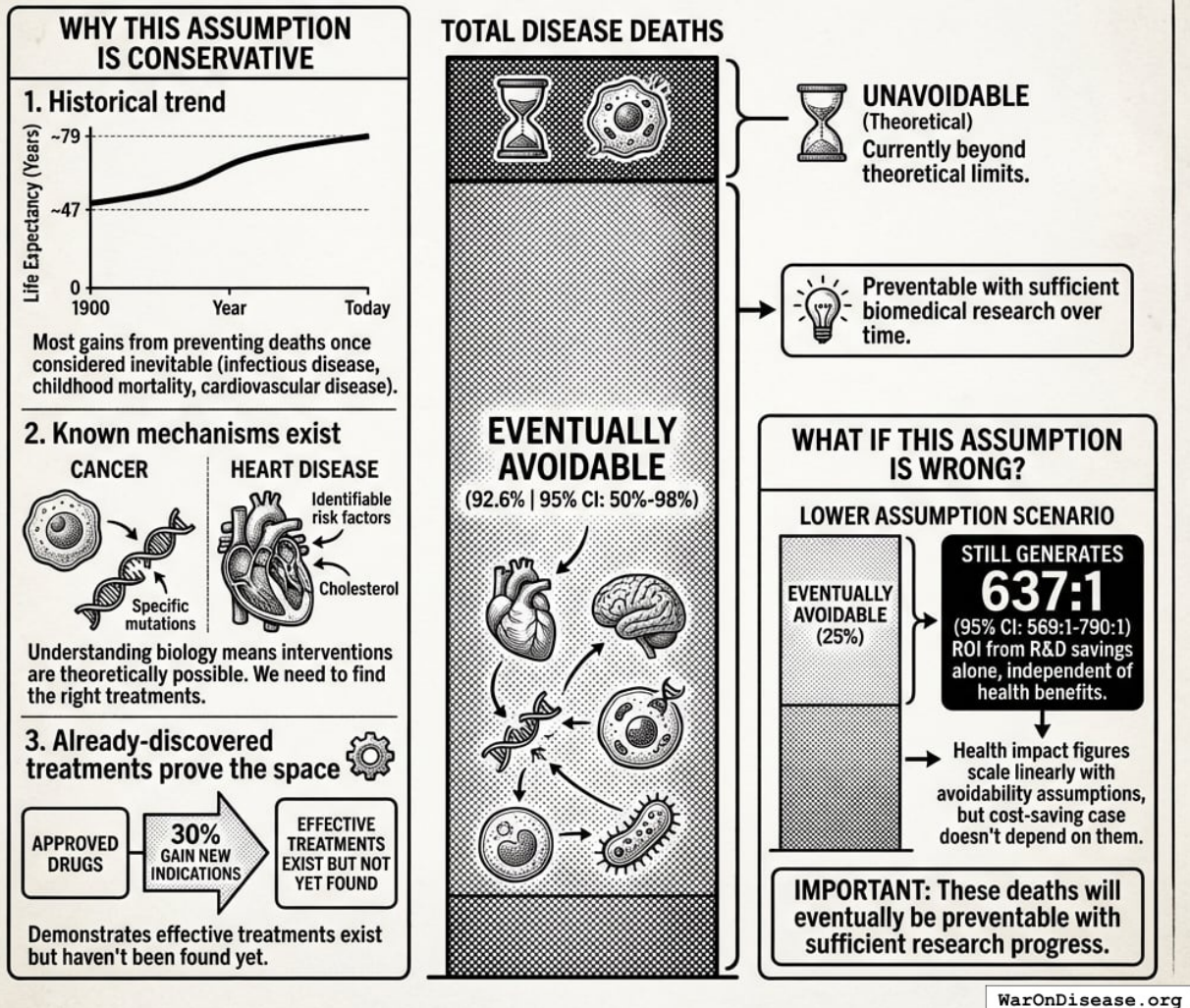


Figure 8: A visual comparison of historical life expectancy gains alongside a sensitivity analysis showing that high ROI remains even if the percentage of avoidable deaths is significantly lower than estimated.

Why this assumption is conservative:

- 1. Historical trend:** In 1900, life expectancy was ~47 years. Today it's ~79. Most of that gain came from preventing deaths that were once considered inevitable (infectious disease, childhood mortality, cardiovascular disease).
- 2. Known mechanisms exist:** For most major disease categories, we understand enough biology to know that interventions are theoretically possible. Cancer is caused by specific mutations. Heart disease has identifiable risk factors. The question is finding the right treatments, not

whether treatments can exist.

3. **Already-discovered treatments prove the space:** 30% of approved drugs gain new indications, demonstrating that effective treatments exist but haven't been found yet.

What if this assumption is wrong?

Even if only 25% of deaths are eventually avoidable (half our estimate), the framework still generates 637:1 (95% CI: 569:1-790:1) ROI from R&D savings alone, independent of health benefits. These deaths will eventually be preventable with sufficient research progress. The health impact figures scale linearly with avoidability assumptions, but the cost-saving case doesn't depend on them.

3.4 Trial Funding Scenario

This analysis models a scenario with \$21.8B/year allocated to pragmatic clinical trials. At \$929 (95% CI: \$97-\$3K)/patient, this funds approximately 23.4M patients/year (95% CI: 9.44M patients/year-96.8M patients/year) patient-years annually.

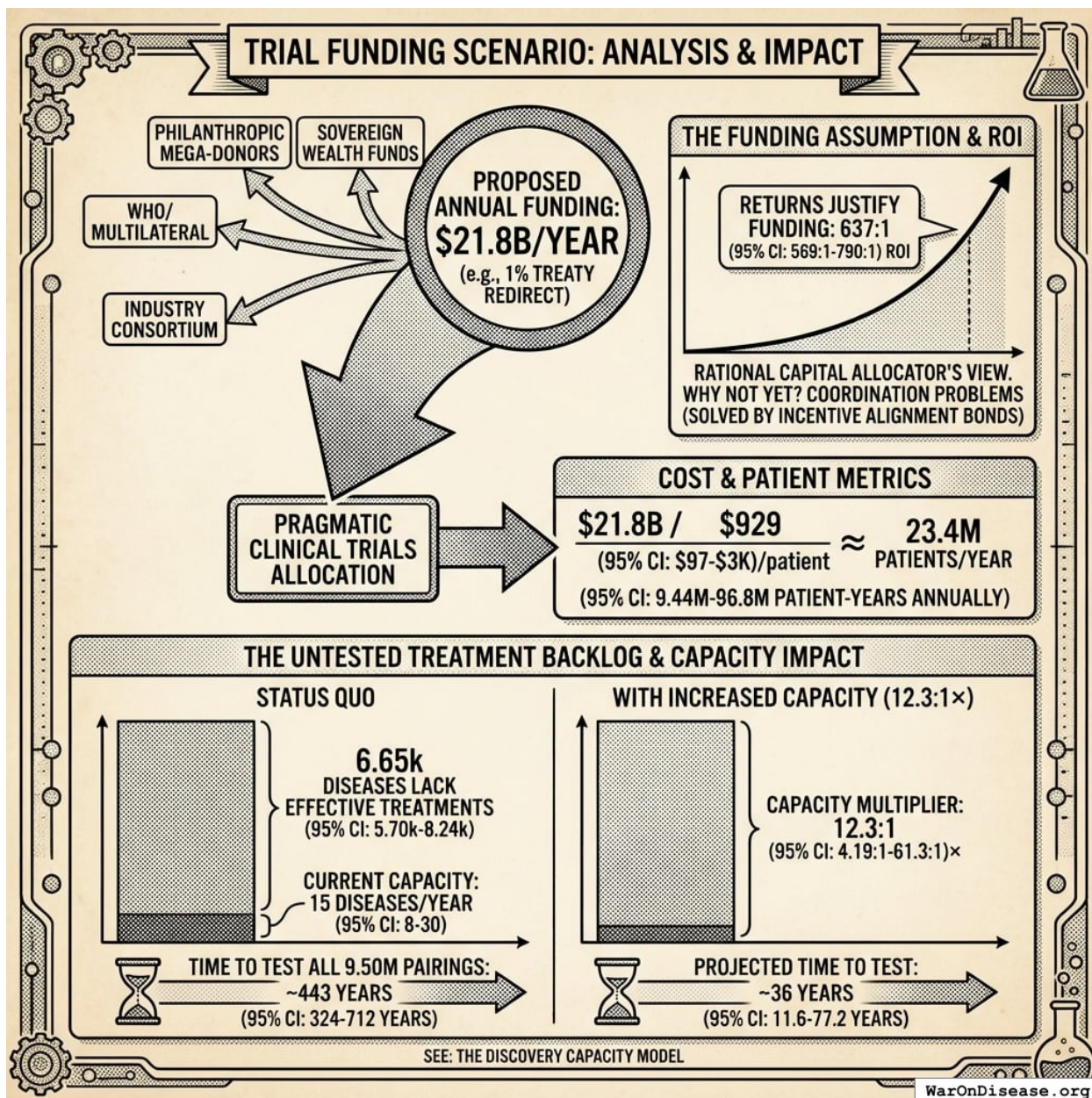


Figure 9: A comparison showing the impact of increased trial funding on the disease treatment backlog, contrasting a 443-year timeline under current capacity with a 36-year timeline under the proposed model.

Tip

On the Funding Assumption: This analysis demonstrates what becomes possible when the funding constraint is removed. The \$21.8B/year figure comes from a proposed [1% Treaty](#) redirecting military spending - but this is one mechanism among many:

- **Philanthropic mega-donors:** A single Gates Foundation-scale commitment could fund the protocol infrastructure and initial years

- **Sovereign wealth funds:** Norway’s \$1.4T fund or similar could view this as humanity-scale infrastructure
- **WHO/multilateral coordination:** Comparable to GAVI or the Global Fund
- **Industry consortium:** Pharma collectively spends \$60B (95% CI: \$50B-\$75B)/year on trials; even 10% redirection exceeds this threshold

The **returns justify the funding**, not vice versa. At 637:1 (95% CI: 569:1-790:1) ROI, any rational capital allocator would fund this if they believed the analysis. The question is not “where does the money come from?” but “why hasn’t this happened yet?” (Answer: coordination problems that mechanisms like the [Incentive Alignment Bonds](#) are designed to solve.)

Trial Capacity Impact: See The Discovery Capacity Model for the full comparison of status quo vs. dFDA trial capacity metrics.

The Untested Treatment Backlog:

Approximately 6.65k diseases (95% CI: 5.70k diseases-8.24k diseases) lack effective treatments. At current trial capacity (15 diseases/year (95% CI: 8 diseases/year-30 diseases/year)), systematically testing all 9.50M combinations plausible pairings would take ~443 years (95% CI: 324 years-712 years). With 12.3x (95% CI: 4.19x-61.3x) capacity, this drops to ~36 years (95% CI: 11.6 years-77.2 years).

4 Framework Costs (ROM Estimates)

Protocol Infrastructure Costs

The dFDA is protocol infrastructure - open standards and APIs that existing clinical trial systems adopt. This is analogous to HTTP enabling any browser to access any website, or FHIR enabling health data interoperability.

- **Upfront protocol/API build:** \$15–25M
- **Annual protocol operations:** \$5–12M
- **Integration onboarding fund:** \$20-50M (one-time, to support Epic/Cerner/Medable adoption)
- **Total initiative:** ~\$40-75M upfront, \$5-12M annual

Ecosystem Integration: The protocol leverages existing infrastructure - DCT platforms (Medable, Science 37), EHR systems (Epic, Cerner), academic medical centers, and health apps. These organizations have invested billions in infrastructure that becomes more valuable through protocol interoperability.

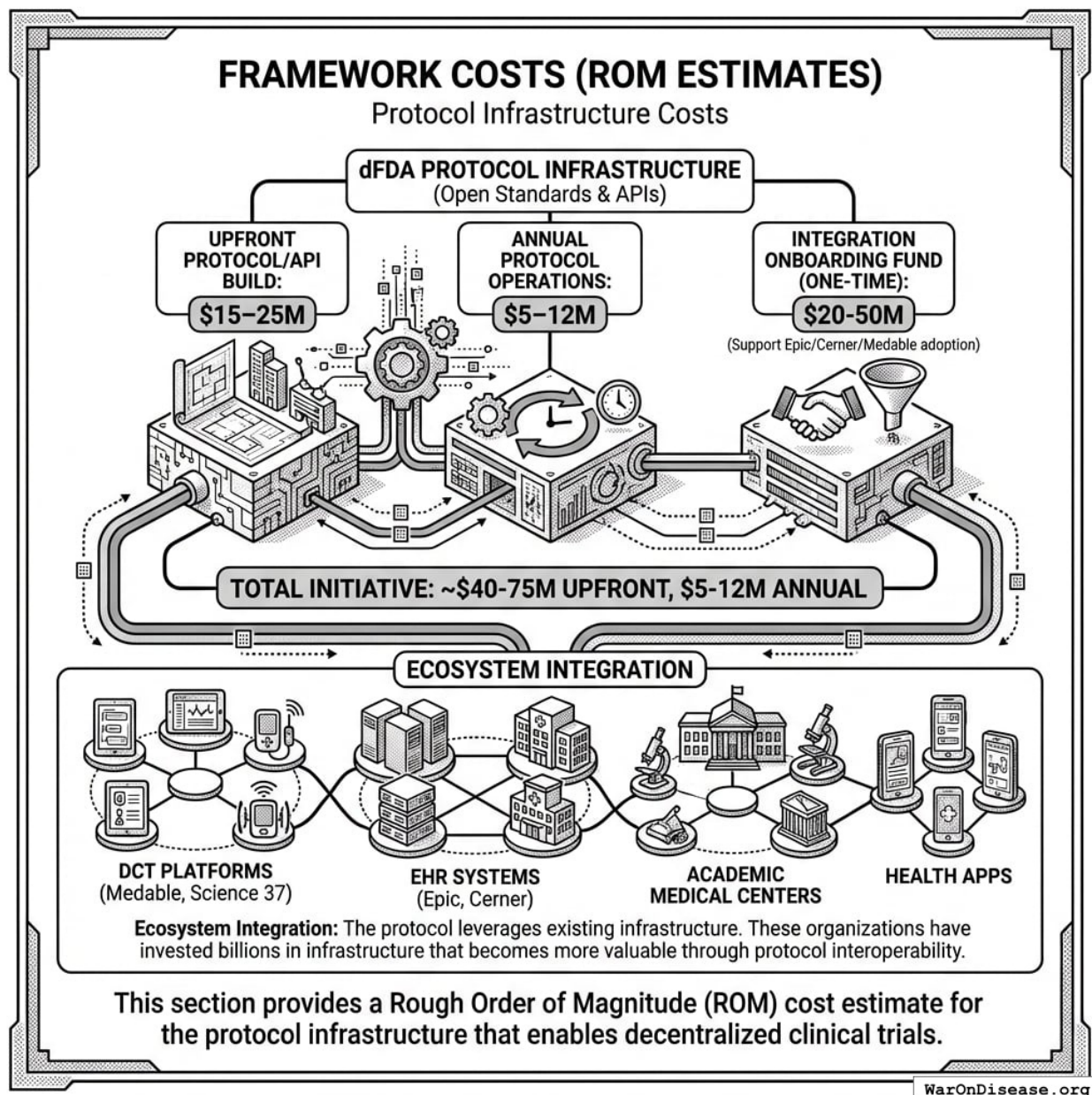


Figure 10: A visual representation of the dFDA protocol ecosystem showing how a central interoperability layer connects EHR systems, clinical trial platforms, and medical centers, paired with a breakdown of upfront versus recurring costs.

This section provides a **Rough Order of Magnitude (ROM)** cost estimate for the protocol infrastructure that enables decentralized clinical trials.

4.1 Upfront Build Costs (30 Months)

1. Core Engineering & Development Effort:

- *Basis:* ~75 FTEs 2.5 years \$200k/FTE/year

- *Activities:* Detailed design, Core framework development (API, storage, mapping/validation, auth), reference frontend, initial plugin interfaces, testing, documentation, initial deployment.

The engineering cost is calculated as:

$$C_{\text{engineering}} = N_{\text{FTEs}} \times T \times C_{\text{FTE}} = 75 \times 2.5 \times \$200\text{k} = \$37.5\text{M}$$

Where $N_{\text{FTEs}} = 75$ is the number of full-time equivalents, $T = 2.5$ years is the development timeline, and $C_{\text{FTE}} = \$200\text{k}$ per FTE per year.

- **Estimated ROM:** \$35 - \$40M

2. Infrastructure Setup & Initial Cloud Costs:

- *Activities:* Establishing cloud accounts, VPCs, Kubernetes cluster (EKS) setup, database provisioning (RDS/TimescaleDB), S3 buckets, CI/CD pipeline setup, initial IaC development (Terraform).
- *Costs:* Includes initial compute/storage during development/testing, potential small upfront reservations.
- **Estimated ROM:** \$1 - \$3 Million

3. Software Licenses & Tooling (Initial):

- *Examples:* Potential costs for monitoring tools (Datadog), security scanners (Snyk), specialized libraries, collaboration tools if not already covered.
- **Estimated ROM:** \$0.5 - \$1 Million

4. Compliance, Legal & Security (Initial Setup):

- *Activities:* Initial HIPAA/GDPR compliance assessment, policy development, security architecture review, legal consultation for data sharing frameworks.
- **Estimated ROM:** \$1 - \$2 Million

The total upfront cost is the sum of all components:

$$C_0 = C_{\text{engineering}} + C_{\text{infrastructure}} + C_{\text{software}} + C_{\text{compliance}}$$

Where:

- $C_{\text{engineering}} = \$35 - \$40$ million (Core Engineering & Development)
- $C_{\text{infrastructure}} = \$1 - \$3$ million (Infrastructure Setup & Initial Cloud Costs)
- $C_{\text{software}} = \$0.5 - \1 million (Software Licenses & Tooling)
- $C_{\text{compliance}} = \$1 - \$2$ million (Compliance, Legal & Security)

Total Estimated Upfront Cost (ROM): \$37.5 - \$46M

*Note: This ROM estimate focuses **only on the Core framework build effort and associated setup**. It represents the foundational first step. A full global implementation requires significant additional investment in broader initiatives to achieve goals of global integration, legal harmonization, and massive scale. These crucial, follow-on costs are estimated separately in the [Scenario Based ROM Estimates for Broader Initiative Costs](#) section below and include:*

- *Global EHR/Data Source Integration Effort*: Building/buying connectors for *thousands* of systems worldwide.
- *Large-Scale Plugin Development*: Funding the ecosystem of data importers, analysis tools, and visualization plugins.
- *International Legal/Regulatory Harmonization*: Major diplomatic and legal efforts to create a global standard.
- *Global Rollout & Adoption*: Costs associated with driving adoption and providing training worldwide.
- *Massive-Scale Infrastructure*: Scaling hardware and cloud resources beyond initial targets to support millions of users.

The following sections provide ROM estimates for both the ongoing operational costs of the Core framework and for these essential broader initiatives.

4.2 Ecosystem Participants

The protocol creates value by enabling interoperability among existing clinical trial infrastructure. These organizations represent the ecosystem that would adopt and benefit from the protocol:

Participant	Current Investment	How They Integrate	What They Gain
DCT Platforms (Medable, Science 37, Thread)	\$500M+ collectively	Adopt outcome reporting standards	Interoperability, larger patient pools, regulatory credibility
EHR Systems (Epic, Cerner)	Billions in infrastructure	Enable federated queries	New revenue from research queries, competitive differentiation
Pharma Sponsors	\$60B (95% CI: \$50B-\$75B)/year on trials	Submit trials via compliant systems	Lower costs, faster enrollment, real-world evidence
Academic Medical Centers	Research infrastructure	Contribute federated data nodes	Research funding, publication opportunities
Health Apps (Apple Health, Fitbit)	Consumer health data	Report patient outcomes to protocol	User engagement, clinical validation

Note

Why This Matters: Companies like Medable (\$521M raised), Science 37, Epic, and Cerner have already built the infrastructure for patient recruitment, data collection, and trial management. The protocol doesn't replicate this work - it makes their existing investments more valuable by enabling data to flow across systems.

4.3 Annual Operational Costs (5M MAU Target Scale)

1. Cloud Infrastructure Costs (AWS):

- *Components:* EKS cluster, RDS/TimescaleDB hosting, S3 storage & requests, SQS messaging, API Gateway usage, Data Transfer (egress), CloudWatch logging/monitoring.
- *Basis:* Highly dependent on actual usage patterns, data retrieval frequency, processing intensity. Assumes optimized resource usage.
- **Estimated ROM:** \$5 - \$15 Million / year (Very sensitive to scale and usage patterns)

2. Ongoing Engineering, Maintenance & Operations:

- *Team Size:* Assume ~20 FTEs (SREs, DevOps, Core Maintainers, Security).
- *Basis:* 20 FTEs * \$200k/FTE/year

The ongoing engineering cost is calculated as:

$$C_{\text{engineering}}^{\text{ops}} = N_{\text{FTEs}}^{\text{ops}} \times C_{\text{FTE}} = 20 \times \$200\text{k} = \$4\text{M/year}$$

Where $N_{\text{FTEs}}^{\text{ops}} = 20$ is the number of FTEs for ongoing operations.

- **Estimated ROM:** \$4 - \$6 Million / year

3. Software Licenses & Tooling (Ongoing):

- *Examples:* Monitoring (Datadog/New Relic), Error Tracking (Sentry), Security Tools, potential DB license/support costs at scale.
- **Estimated ROM:** \$0.5 - \$1.5 Million / year

4. Compliance & Auditing (Ongoing):

- *Activities:* Regular security audits (penetration tests, compliance checks), maintaining certifications, legal reviews.
- **Estimated ROM:** \$0.5 - \$1 Million / year

5. Support (User & Developer):

- *Activities:* Tier 1/2 support for protocol participants and third-party plugin developers.
- **Estimated ROM:** \$1 - \$3 Million / year (Scales with user base)

The total annual operational cost is the sum of all components:

$$C_{\text{op}} = C_{\text{cloud}} + C_{\text{engineering}} + C_{\text{software}} + C_{\text{compliance}} + C_{\text{support}}$$

Where:

- $C_{\text{cloud}} = \$5 - \15 million/year (Cloud Infrastructure Costs)
- $C_{\text{engineering}} = \$4 - \$6$ million/year (Ongoing Engineering, Maintenance & Operations)
- $C_{\text{software}} = \$0.5 - \1.5 million/year (Software Licenses & Tooling)
- $C_{\text{compliance}} = \$0.5 - \$1$ million/year (Compliance & Auditing)
- $C_{\text{support}} = \$1 - \3 million/year (Support)

Total Estimated Annual Operations (Platform Only, ROM): \$11 - \$26.5 Million / year

4.3.1 Marginal Cost Analysis per User

The **5M MAU** target is an illustrative milestone used for these initial ROM estimates, not the ultimate goal for the framework, which is to support hundreds of millions or billions of users. At this initial scale, you can analyze the cost on a per-user basis.

- **Average Cost Range Per User (at 5M MAU):**

- Based on the total annual operational cost range of **\$11M - \$26.5M**, the average cost per user is:

$$\frac{\$11,000,000 \text{ to } \$26,500,000}{5,000,000 \text{ users}} = \$2.20 \text{ to } \$5.30 \text{ per user per year}$$

- **Marginal Cost Per Additional User:**

- As federated data infrastructure, the framework has high fixed costs (protocol development, core engineering) but very low variable costs. Therefore, the **marginal cost** of supporting one additional participant is expected to be a small fraction of the average cost, likely **pennies per year**. This cost will decrease further as the protocol achieves greater economies of scale, making it exceptionally efficient at supporting a global participant base.

(Note: The underlying cloud infrastructure cost (\$5M-\$15M/year) is a top-down ROM estimate. A more granular, bottom-up analysis based on projected per-user storage, data transfer, and compute would provide further support for these figures and is a key area for future refinement of this model.)

Note on Participant Financial Contributions:

This cost estimate **covers building the protocol infrastructure, not paying patients for trial participation**. Trial participation costs would be handled separately through funding mechanisms (government grants, foundation funding, or sponsor payments). The protocol coordinates information exchange but doesn't move money around directly.

This estimate excludes costs for governance structure and plugin development.

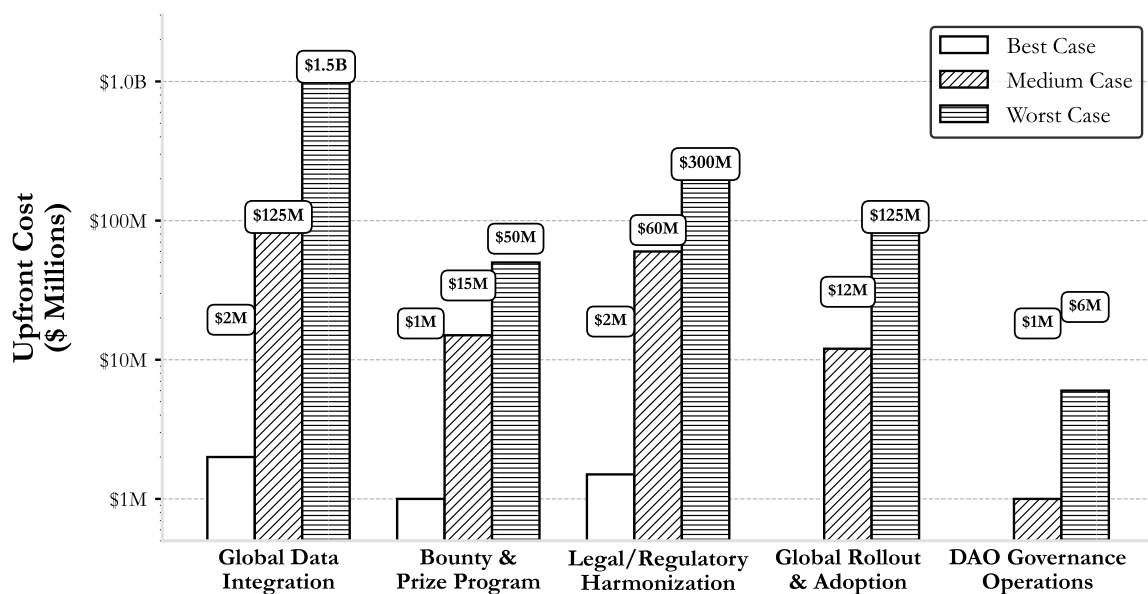
4.4 Scenario Based ROM Estimates for Broader Initiative Costs

This table presents point estimates for each scenario, with the overall range of possibilities captured by comparing the Best, Medium, and Worst Case columns.

Component	Best Case (Upfront / Annual)	Medium Case (Upfront / Annual)	Worst Case (Upfront / Annual)	Key Assumptions & Variables Driving Range
Global Data Integration	\$2M / ~\$0	\$125M / \$10M	\$1.5B / \$150M	Success of AI/automation, standards adoption, #systems, vendor cooperation.
Bounty & Prize Program (Act SEC. 204(i))	\$1M (Prizes) / ~\$0	\$15M (Bounties) / \$2M	\$50M (Major Bounties) / \$10M	Success of organic ecosystem growth vs. need to incentivize critical plugin/tool development via bounties.

Component	Best Case (Upfront / Annual)	Medium Case (Upfront / Annual)	Worst Case (Upfront / Annual)	Key Assumptions & Variables Driving Range
Legal/Regulatory Harmonization	\$1.5M / ~\$0	\$60M / \$3M	\$300M / \$30M	Effectiveness of AI legal tools, political will, complexity of global law.
Global Rollout & Adoption	~\$0 / ~\$0	\$12M / \$3M	\$125M / \$30M	Need for training/support beyond protocol adoption, user interface complexity.
Governance Operations	~\$0 / ~\$0	~\$1M / \$0.3M	~\$6M / \$1M	Automation level, need for audits, grants, core support staff.
— TOTAL —	~\$4.5M / ~\$0	~\$213M / ~\$18.3M	~\$1.98B+ / ~\$221M+	Represents total initiative cost excluding Core framework build/ops.

Costs for a Decentralized Drug Assessment Framework Upfront Investment by Scenario



Total Initiative Cost (excluding Core framework): Best Case \$4.5M | Medium Case \$213M | Worst Case \$1.98B

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4.4.1 Interpretation

Even when pursuing efficient strategies, the potential cost for the full initiative for a decentralized framework (beyond the Core framework) varies dramatically based on real-world execution challenges. The Medium Case suggests upfront costs in the low hundreds of millions and annual costs in the low tens of millions, while the Worst Case pushes towards multi-billion dollar upfront figures and annual costs in the hundreds of millions, dominated by integration, plugin funding, and legal costs

if automation and community efforts fall short.

4.4.2 Summary

Based on the detailed technical specification, a ROM estimate suggests:

- **Initial Core framework Build (~2.5 years): ~\$37.5 - \$46M**
- **Annual Core framework Operations (at ~5M MAU scale): ~\$11 - \$26.5 Million**
(These framework operational costs are distinct from the financial flows of patient contributions and the NIH Trial Participation Cost Discount Fund, and also exclude plugin ecosystem costs not covered by protocol bounties)

The core protocol infrastructure build costs tens of millions; the broader global initiative (integration, legal frameworks, rollout) accounts for the larger cost estimates detailed in the scenario table above.

5 Benefit Analysis - Quantifying the Savings

This section quantifies the potential societal benefits of an infrastructure for a decentralized framework for drug assessment, focusing primarily on R&D cost savings and health outcome improvements.

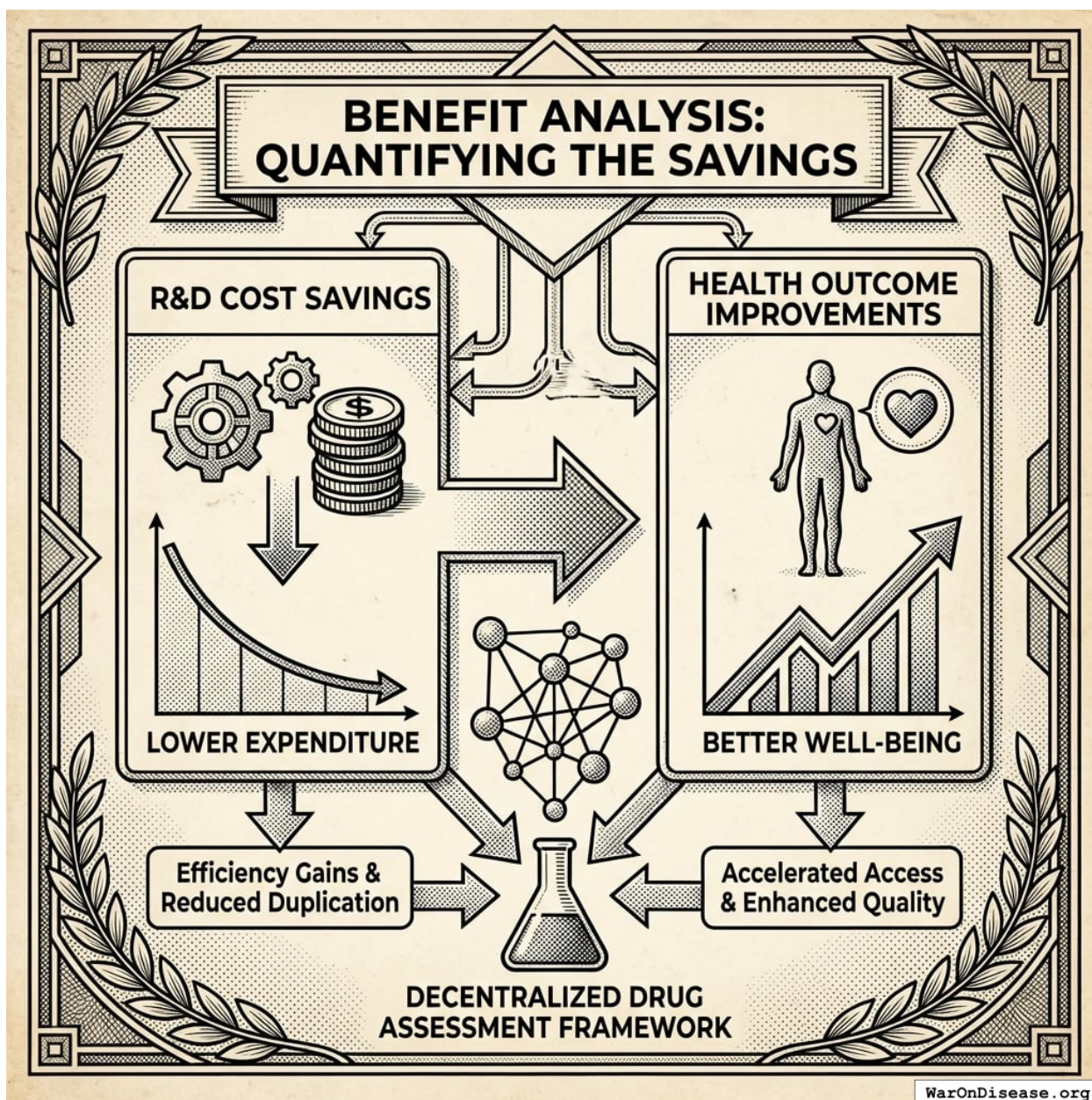


Figure 11: A conceptual model showing how a decentralized drug assessment framework leads to societal benefits such as R&D cost savings and improved health outcomes.

5.1 Market Size and Impact

The global pharmaceutical and medical device R&D market is vast. Annual global spending on clinical trials is approximately \$60B (95% CI: \$50B-\$75B). Much of this spending is made dramatically more efficient through protocol standardization. If such a framework enables even a fraction of these trials to use pragmatic designs, the economic impact will be substantial.

- **Current Average Costs:** Estimates suggest \$2.60B (95% CI: \$1.50B-\$4B) to bring a new drug from discovery through FDA approval, spread across ~10 years.
- **Clinical Trial Phase Breakdown:**

- Phase I: \$2 - \$5 million/trial (smaller scale).
- Phase II: \$10 - \$50 million/trial (depending on disease area).
- Phase III: \$100M - \$500M/trial (large patient populations).
- **Per-Patient Phase III Costs:** Often \$41K (95% CI: \$20K-\$120K) per patient (site fees, overhead, staff, monitoring, data management).

5.2 Decentralized Trial Costs Modeled on Pragmatic Trials

- **Oxford RECOVERY:** Achieved ~\$500 (95% CI: \$400-\$2.50K) per patient. Key strategies included:
 1. Embedding trial protocols within routine hospital care.
 2. Minimizing overhead by leveraging existing staff/resources and electronic data capture.
 3. Focused, pragmatic trial designs.
- **Systematic Review Evidence:** A systematic review of 64 embedded pragmatic clinical trials found a **median cost per patient of \$97 (95% CI: \$19-\$478)**⁸⁵. This confirms that low-cost execution is a replicable property of the pragmatic design, not an anomaly of any single trial.
- **ADAPTABLE Trial (PCORnet):** The US-based [ADAPTABLE trial](#) (\$14M (95% CI: \$14M-\$20M) / 15.1k patients = **\$929 (95% CI: \$929-\$1.40K)/patient**) provides a more representative benchmark for pragmatic trial costs in typical healthcare settings without emergency conditions.
- **dFDA Cost Projection:** Our projections use \$929 (95% CI: \$97-\$3K)/patient based on ADAPTABLE. Confidence interval (\$500-\$3,000) captures range from RECOVERY-like efficiency to complex chronic disease trials.

Input: Pragmatic Trial Cost Distribution

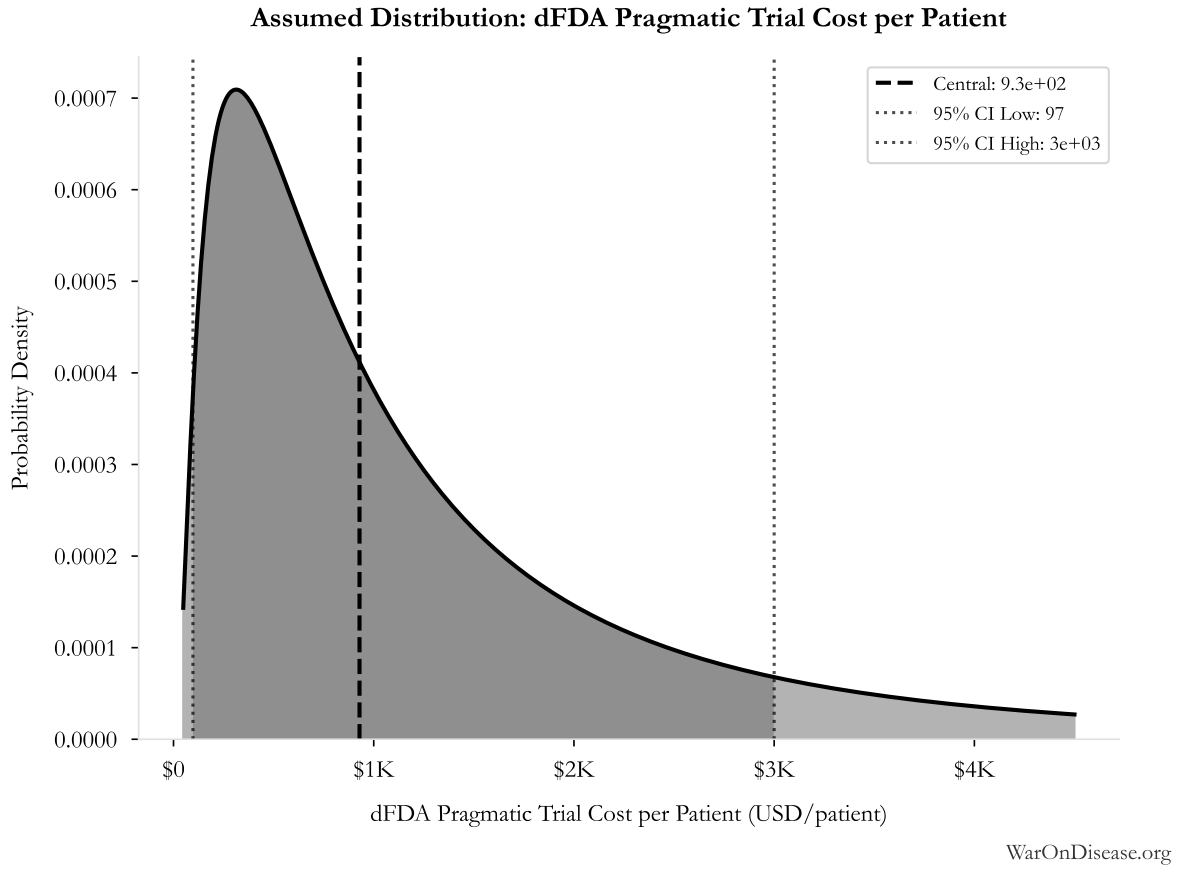


Figure 12: Probability Distribution: dFDA Pragmatic Trial Cost per Patient

This chart shows the assumed probability distribution for this parameter. The shaded region represents the 95% confidence interval where we expect the true value to fall.

- **Extrapolation to New System:**

- A well-integrated global framework could achieve \$929 (95% CI: \$97-\$3K) per patient in many cases, especially for pragmatic or observational designs.
- Up to **~44.1x (95% CI: 39.4x-89.1x) cost reduction** is achievable by comparing pragmatic trial costs (\$929 (95% CI: \$97-\$3K)) against traditional costs of \$41K (95% CI: \$20K-\$120K).

The cost reduction factor:

$$\begin{aligned}
 & k_{reduce} \\
 &= \frac{Cost_{P3,pt}}{Cost_{pragmatic,pt}} \\
 &= \frac{\$41K}{\$929} \\
 &= 44.1
 \end{aligned}$$

The percentage reduction:

$$\begin{aligned}
& \text{Reduce}_{pct} \\
&= 1 - \frac{\text{Cost}_{pragmatic,pt}}{\text{Cost}_{P3,pt}} \\
&= 1 - \frac{\$929}{\$41K} \\
&= 97.7\%
\end{aligned}$$

i Note

Scope of Cost Reduction: This reduction applies to trials amenable to pragmatic design - approximately 70% of Phase III trial volume by patient count (chronic disease management, comparative effectiveness, dose optimization). First-in-human studies, novel mechanism trials, and high-risk interventions retain traditional controlled protocols. The confidence interval (\$500-\$3,000/patient) captures this heterogeneity: simple comparative studies approach RECOVERY-level efficiency while complex trials remain closer to traditional costs. The headline 97.7% (95% CI: 97.5%-98.9%) figure represents the **weighted average** across the addressable trial market, not a claim that every trial achieves this reduction.

5.3 Overall Savings

1. By Reducing Per-Patient Costs

- If a trial with 5,000 participants costs \$929 (95% CI: \$97-\$3K)/patient, total cost is ~\$6 million, versus \$200 - \$600 million under traditional models.
- This magnitude of savings can drastically reduce the total cost of clinical development.

For a trial with x participants, the total cost savings is:

$$S_{\text{trial}}(x) = (c_t - c_d) \cdot x$$

Where:

- c_t is the traditional cost per patient (\$41K (95% CI: \$20K-\$120K))
- c_d is the decentralized cost per patient (\$929 (95% CI: \$97-\$3K))

For a trial with $x = 5,000$ participants, savings are approximately:

$$(\text{Traditional} - \text{Pragmatic}) \times 5,000 \approx \$194\text{M per trial}$$

2. Volume of Trials & Speed

- Faster, cheaper trials allow more drug candidates, off-label uses, nutraceuticals, and personalized dosing strategies to be tested.
- Shorter development cycles reduce carrying costs and risk, further increasing ROI for sponsors.

3. Regulatory Savings

- A unified protocol standard with automated data audits cuts bureaucratic duplication across multiple countries, drastically lowering compliance costs.

4. Transparent, Standardized Environment

- The transparent nature of the protocol creates a standardized operating environment. Outcome data is visible across the ecosystem, enabling sponsors to benchmark their trial designs and optimize operational costs, further driving down R&D expenditure beyond the technical efficiencies of pragmatic trials.

5.4 Economic Value of Earlier Access to Treatments

- Faster approvals and access to effective treatments can save lives and improve quality of life.
- **Value of a Statistical Life (VSL):** U.S. agencies use ~\$10M (95% CI: \$5M-\$15M) per life saved.
- **QALY Framework:** Standard willingness-to-pay is **\$100,000–\$150K (95% CI: \$100K-\$199K) per QALY gained.**
- These benefits are additive to direct cost savings and can be substantial depending on the scale of acceleration.

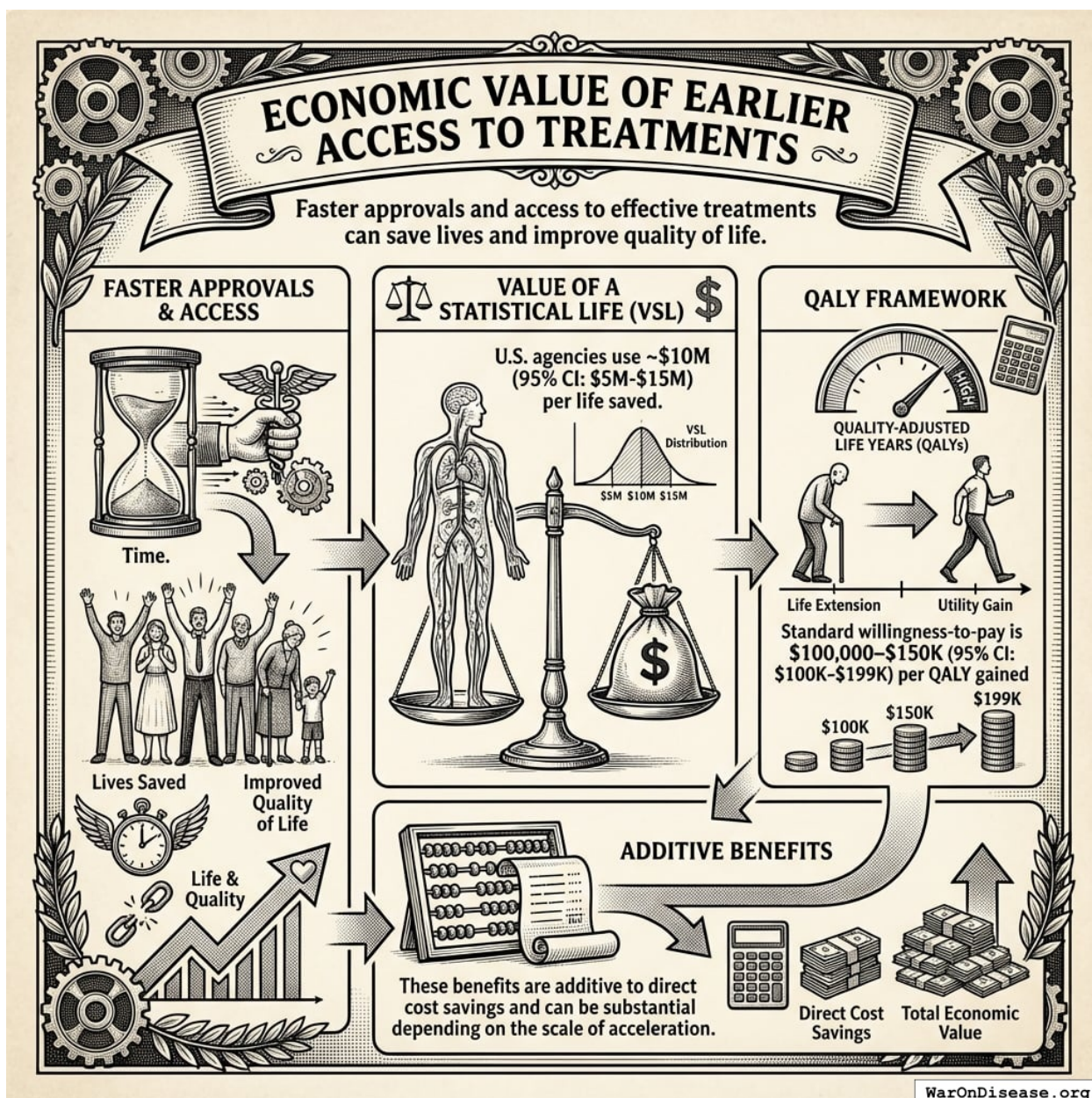


Figure 13: A conceptual breakdown showing how accelerated treatment access creates economic value by combining the Value of a Statistical Life (VSL) for lives saved and Quality-Adjusted Life Year (QALY) metrics for improved health outcomes.

5.5 Gross R&D Savings from Implementing a Decentralized Framework

- **Parameter:** Percentage reduction in addressable clinical trial costs due to a decentralized framework for drug assessment.
- **Central Estimate:** 97.7% (95% CI: 97.5%-98.9%) (44.1x (95% CI: 39.4x-89.1x))
- **Source/Rationale:**
 - Decentralized Clinical Trials (DCTs) demonstrate significant cost reductions¹³⁸ through reduced site management, travel, and streamlined data collection.

- **Empirical evidence:** ADAPTABLE trial achieved \$929 (95% CI: \$929-\$1.40K)/patient in routine US settings. Harvard meta-analysis of 108 pragmatic trials found median cost of \$97 (95% CI: \$19-\$478)/patient.
- **Our estimate:** \$929 (95% CI: \$97-\$3K)/patient (vs. \$41K (95% CI: \$20K-\$120K) traditional). This deliberately uses ADAPTABLE as a conservative baseline; actual costs may be lower.
- **Confidence interval** captures uncertainty from complex chronic disease trials to highly efficient EHR-integrated designs.

The annual gross R&D savings can be calculated as:

$$S_{\text{annual}} = \alpha \cdot R_d$$

Where:

- $\alpha \in [0, 1]$ is the cost reduction percentage (as decimal)
- $R_d = \$60B$ (95% CI: \$50B-\$75B) annual global clinical trial spending

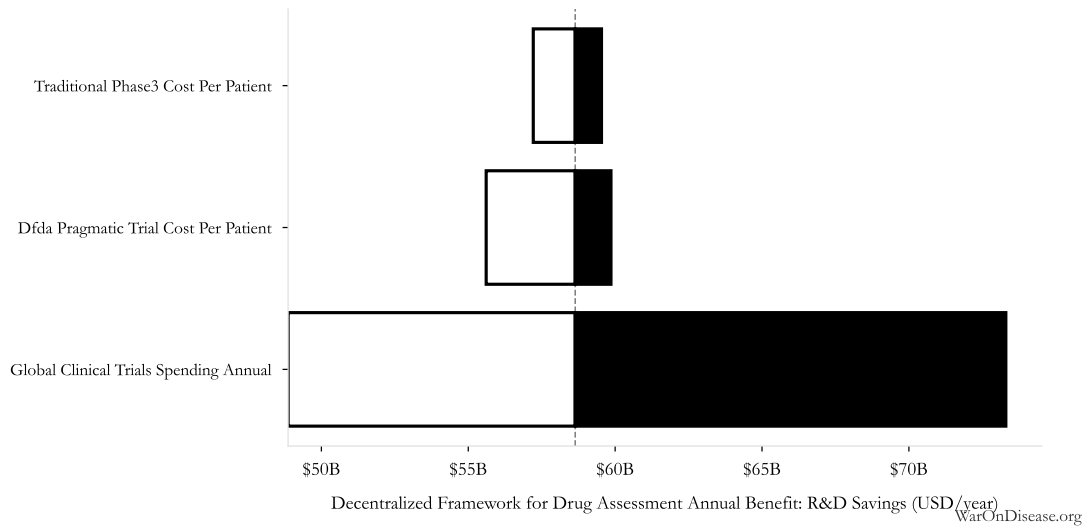
Base Case Calculation:

Using 97.7% (95% CI: 97.5%-98.9%) cost reduction (pragmatic trial costs of \$929 (95% CI: \$97-\$3K) vs traditional \$41K (95% CI: \$20K-\$120K)):

$$\begin{aligned}
& \textit{Benefit}_{RD,ann} \\
&= \textit{Spending}_{trials} \times \textit{Reduce}_{pct} \\
&= \$60B \times 97.7\% \\
&= \$58.6B \\
&\text{where } \textit{Reduce}_{pct} \\
&= 1 - \frac{\textit{Cost}_{pragmatic,pt}}{\textit{Cost}_{P3,pt}} \\
&= 1 - \frac{\$929}{\$41K} \\
&= 97.7\%
\end{aligned}$$

Uncertainty Analysis - R&D Savings:

Sensitivity Analysis: Decentralized Framework for Drug Assessment Annual Benefit: R&D Savings



Monte Carlo Analysis: Decentralized Framework for Drug Assessment Annual Benefit: R&D Savings

Distribution of Outcomes Probability of Exceeding Value

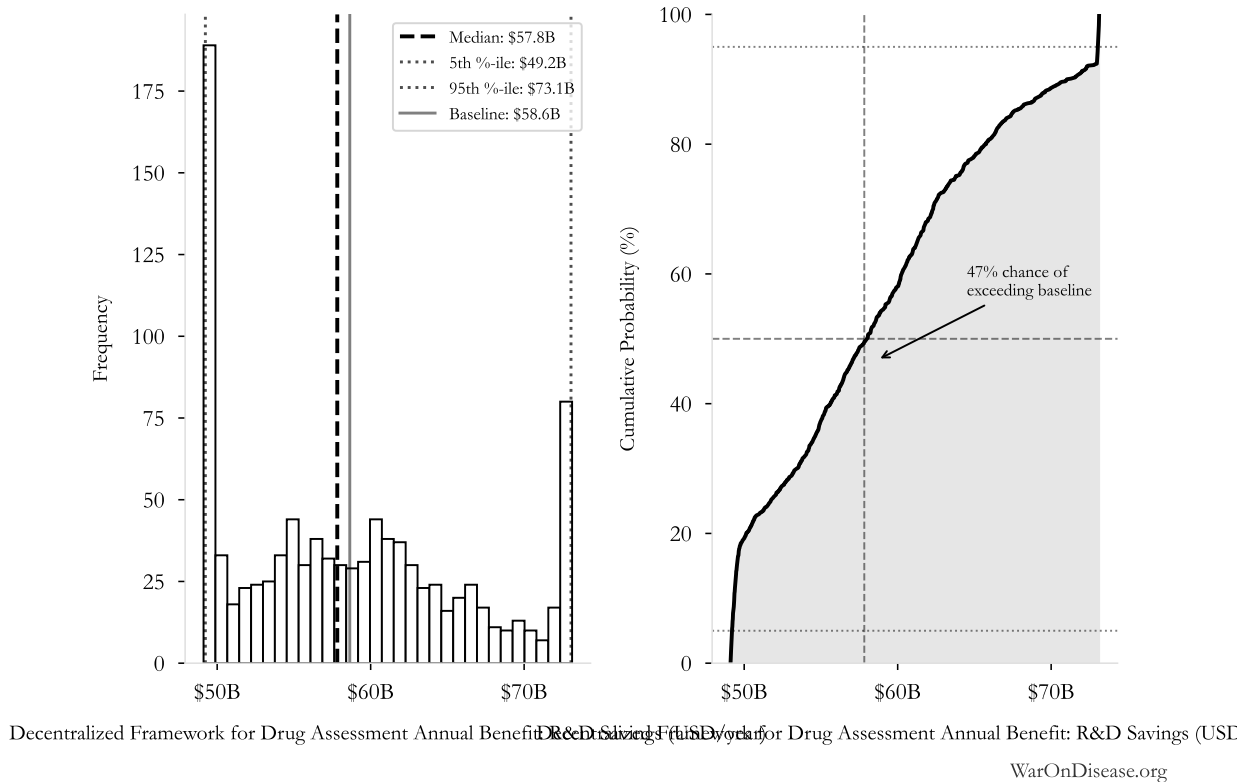


Figure 14: Monte Carlo Distribution: Decentralized Framework for Drug Assessment Annual Benefit: R&D Savings (10,000 simulations)

Simulation Results Summary: Decentralized Framework for Drug Assessment Annual Benefit: R&D Savings

Statistic	Value
Baseline (deterministic)	\$58.6B
Mean (expected value)	\$58.8B
Median (50th percentile)	\$57.8B
Standard Deviation	\$7.66B
90% Confidence Interval	[\$49.2B, \$73.1B]

The histogram shows the distribution of Decentralized Framework for Drug Assessment Annual Benefit: R&D Savings across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.

5.5.1 Key Sources

- [138](#)
- [139](#)
- [140](#)

5.6 Post-Safety Efficacy Lag Elimination

A primary health benefit of a decentralized framework for drug assessment comes from eliminating the “efficacy lag”, the 8.2 years (95% CI: 4.85 years-11.5 years) Phase II/III delay between Phase I safety verification and final approval. **Critical: This does NOT eliminate safety testing.** Phase I safety testing (2.3 years) is preserved.

5.6.1 The Efficacy Lag Problem

A [comprehensive quantitative analysis](#) of post-safety efficacy lag costs (1962-2024) found:

- **Total Deaths:** 416M deaths (95% CI: 225M deaths-630M deaths) eventually avoidable deaths over 8.2 years (95% CI: 4.85 years-11.5 years) efficacy lag (1962-2024)
- **Total DALYs:** 7.94B DALYs (95% CI: 4.43B DALYs-12.1B DALYs) Disability-Adjusted Life Years lost
- **Total Timeline Shift:** One-time 8.2 years (95% CI: 4.85 years-11.5 years) acceleration in disease eradication

The analysis shows that for every 1 unit of harm the FDA prevents through safety testing, it generates **3.07k:1 (95% CI: 2.88k:1-3.12k:1) units** of harm through efficacy delay (Type II vs. Type I error ratio).

Input: Efficacy Lag Duration Distribution

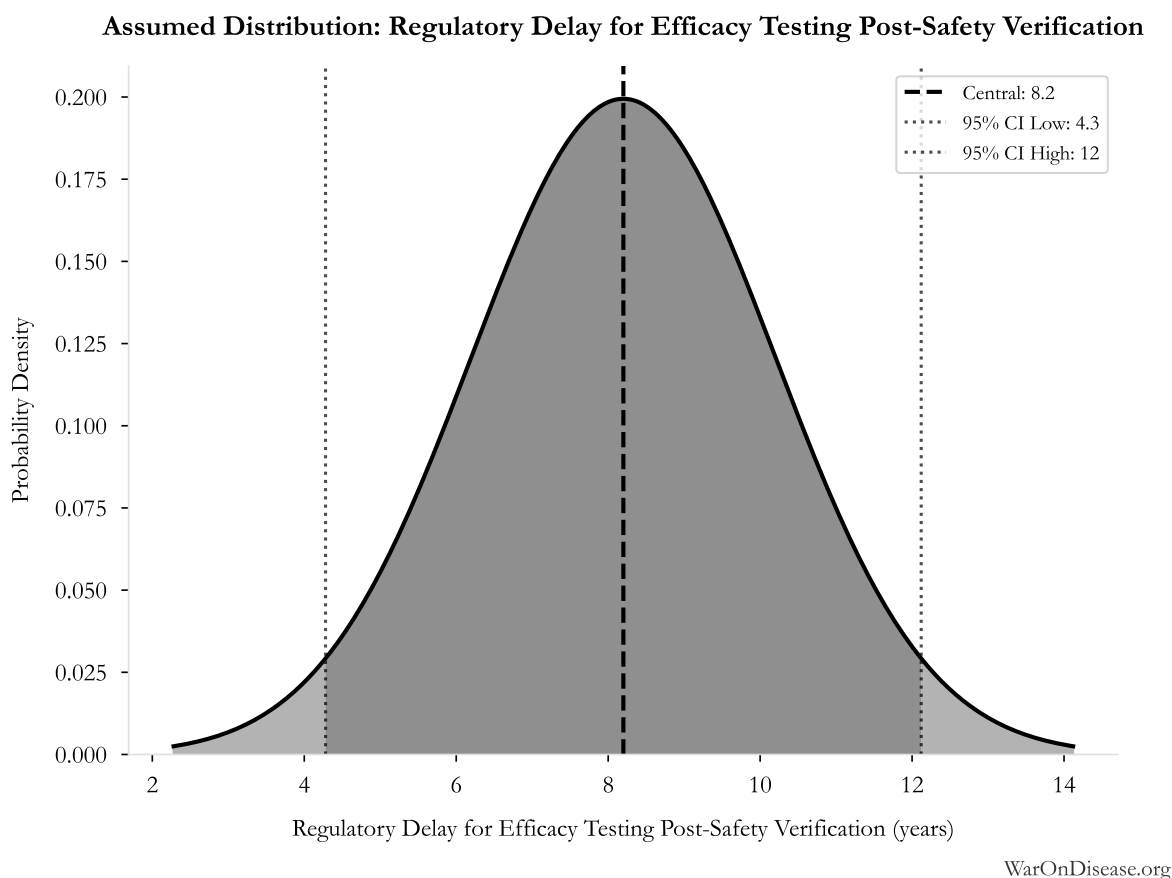


Figure 15: Probability Distribution: Regulatory Delay for Efficacy Testing Post-Safety Verification

This chart shows the assumed probability distribution for this parameter. The shaded region represents the 95% confidence interval where we expect the true value to fall.

5.6.2 How a Decentralized Framework Eliminates the Efficacy Lag

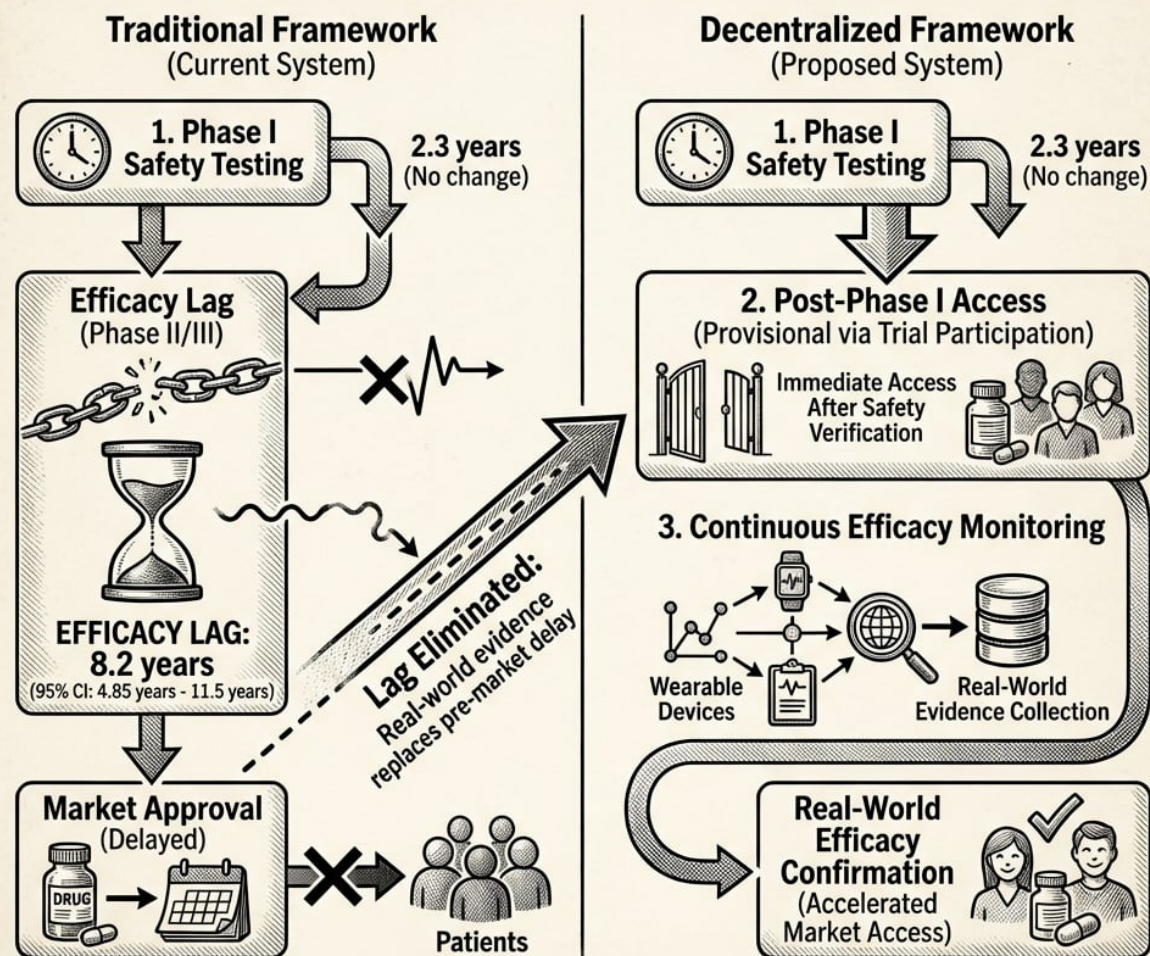
Such a framework provides **provisional access post-Phase I** via trial participation:

1. **Phase I Safety Testing:** Maintained at 2.3 years (no change)
2. **Post-Phase I Access:** Patients can access drugs through trial participation immediately after safety verification
3. **Continuous Efficacy Monitoring:** Real-world evidence replaces the 8.2 years (95% CI: 4.85 years-11.5 years) pre-market efficacy delay

This eliminates the post-safety efficacy lag (the Phase II/III portion, while preserving Phase I safety testing) by enabling real-world evidence collection during trials.

How a Decentralized Framework Eliminates the Efficacy Lag

Such a framework provides **provisional access post-Phase I** via trial participation.



This eliminates the post-safety efficacy lag (the Phase II/III portion, while preserving Phase I safety testing) by enabling real-world evidence collection during trials.

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Figure 16: A timeline comparison showing how a decentralized framework replaces the traditional multi-year efficacy lag with immediate post-Phase I access and continuous real-world evidence monitoring.

5.6.3 Quantified Benefits (One-Time Timeline Shift)

The elimination of the post-safety efficacy lag by such a framework achieves a one-time 8.2 years (95% CI: 4.85 years-11.5 years) timeline acceleration:

- **Total DALYs Averted:** 7.94B DALYs (95% CI: 4.43B DALYs-12.1B DALYs) (total one-time impact from 8.2 years (95% CI: 4.85 years-11.5 years) timeline shift)
- **Total Economic Value:** \$1.19 quadrillion (95% CI: \$443T-\$2.41 quadrillion) (total one-time benefit from timeline shift)

- **Deaths Prevented:** 416M deaths (95% CI: 225M deaths-630M deaths) (total over the 8.2 years (95% CI: 4.85 years-11.5 years) period)

$$DALY_{s_{lag}} = YLL_{lag} + YLD_{lag} = 7.07B + 873M = 7.94B$$

$$\text{where } YLL_{lag}$$

$$\begin{aligned} &= Deaths_{s_{lag}} \times (LE_{global} - Age_{death, delay}) \\ &= 416M \times (79 - 62) \\ &= 7.07B \end{aligned}$$

$$\text{where } Deaths_{s_{lag}}$$

$$\begin{aligned} &= T_{lag} \times Deaths_{disease, daily} \times 338 \\ &= 8.2 \times 150,000 \times 338 \\ &= 416M \end{aligned}$$

$$\text{where } YLD_{lag}$$

$$\begin{aligned} &= Deaths_{s_{lag}} \times T_{suffering} \times DW_{chronic} \\ &= 416M \times 6 \times 0.35 \\ &= 873M \end{aligned}$$

$$\text{where } Deaths_{s_{lag}}$$

$$\begin{aligned} &= T_{lag} \times Deaths_{disease, daily} \times 338 \\ &= 8.2 \times 150,000 \times 338 \\ &= 416M \end{aligned}$$

$$\begin{aligned}
& Value_{lag} \\
&= DALY_{s_{lag}} \times Value_{QALY} \\
&= 7.94B \times \$150K \\
&= \$1190T
\end{aligned}$$

$$\text{where } DALY_{s_{lag}} = YLL_{lag} + YLD_{lag} = 7.07B + 873M = 7.94B$$

$$\begin{aligned}
& \text{where } YLL_{lag} \\
&= Deaths_{lag} \times (LE_{global} - Age_{death, delay}) \\
&= 416M \times (79 - 62) \\
&= 7.07B
\end{aligned}$$

$$\begin{aligned}
& \text{where } Deaths_{lag} \\
&= T_{lag} \times Deaths_{disease, daily} \times 338 \\
&= 8.2 \times 150,000 \times 338 \\
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& Deaths_{lag} \\
&= T_{lag} \times Deaths_{disease, daily} \times 338 \\
&= 8.2 \times 150,000 \times 338 \\
&= 416M
\end{aligned}$$

5.6.4 Efficacy Lag Elimination - Uncertainty Analysis

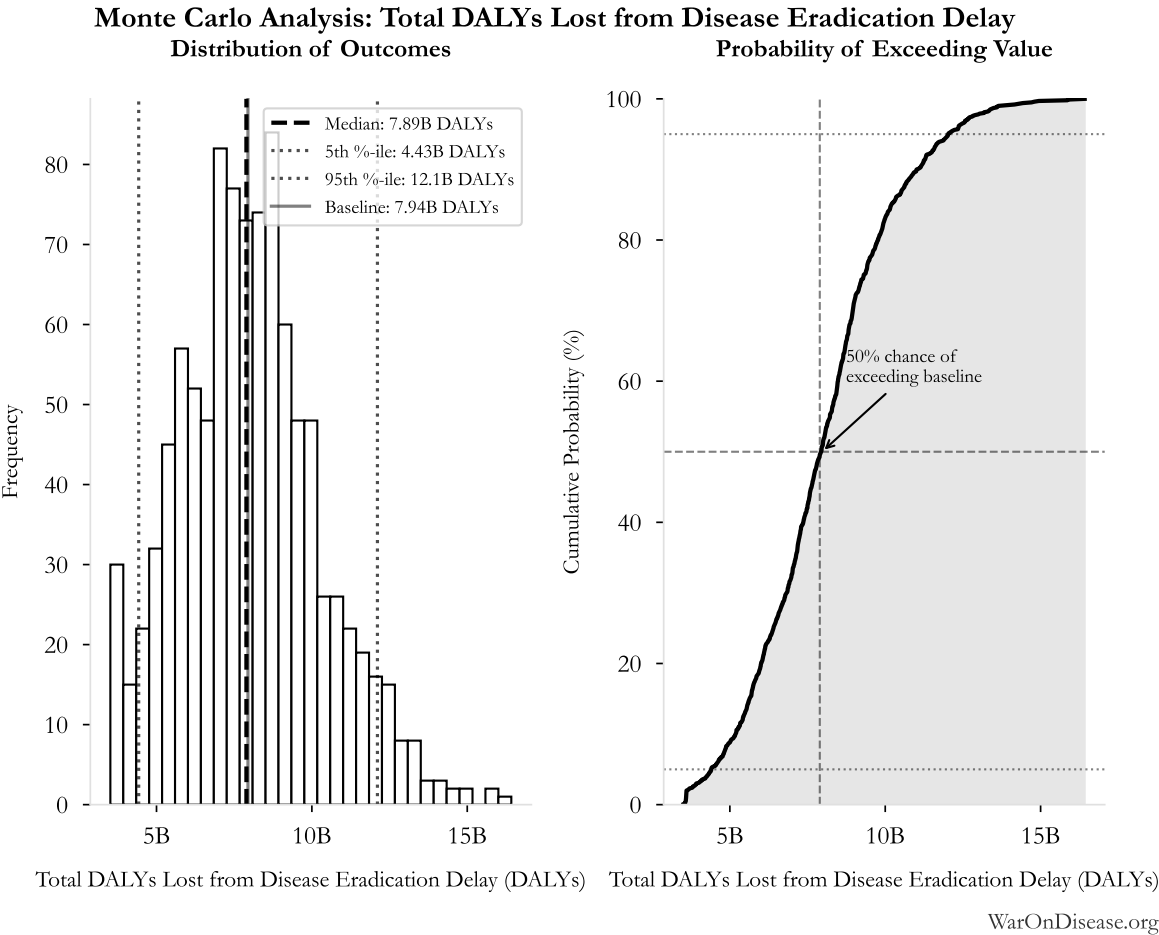


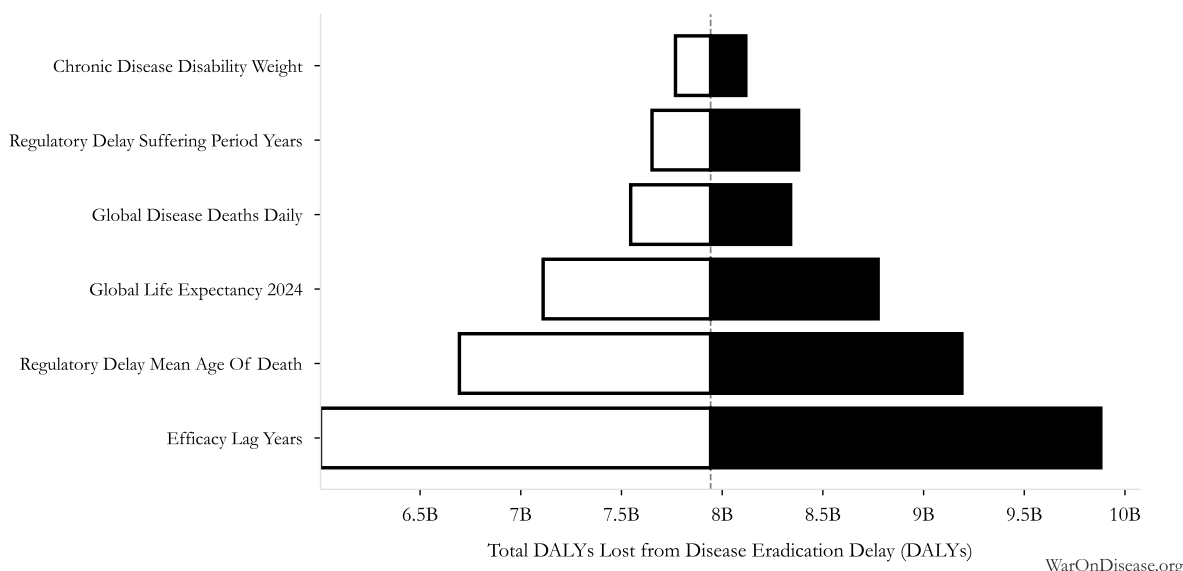
Figure 17: Monte Carlo Distribution: Total DALYs Lost from Disease Eradication Delay (10,000 simulations)

Simulation Results Summary: Total DALYs Lost from Disease Eradication Delay

Statistic	Value
Baseline (deterministic)	7.94B
Mean (expected value)	8.05B
Median (50th percentile)	7.89B
Standard Deviation	2.31B
90% Confidence Interval	[4.43B, 12.1B]

The histogram shows the distribution of Total DALYs Lost from Disease Eradication Delay across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.

Sensitivity Analysis: Total DALYs Lost from Disease Eradication Delay



This represents the health benefits from eliminating the post-safety efficacy lag **only**. The efficacy lag (8.2 years (95% CI: 4.85 years-11.5 years)) is one of two components of the total timeline shift:

1. **Efficacy lag elimination:** 8.2 years (95% CI: 4.85 years-11.5 years) (this section)
2. **Trial capacity/discovery acceleration:** 204 years (95% CI: 123 years-350 years) (from 12.3x (95% CI: 4.19x-61.3x) more trials)

The **combined total** timeline shift is 212 years (95% CI: 135 years-355 years), yielding 10.7B deaths (95% CI: 7.39B deaths-16.2B deaths) and 565B DALYs (95% CI: 361B DALYs-877B DALYs). See [Key Findings](#) for the combined totals.

For detailed methodology and assumptions on efficacy lag, see [The Human Cost of Regulatory Latency](#).

5.7 Safety and Risk Management

Common concern: Won't faster trials with lower costs compromise safety?

SAFETY AND RISK MANAGEMENT

Common concern: Won't faster trials with lower costs compromise safety?

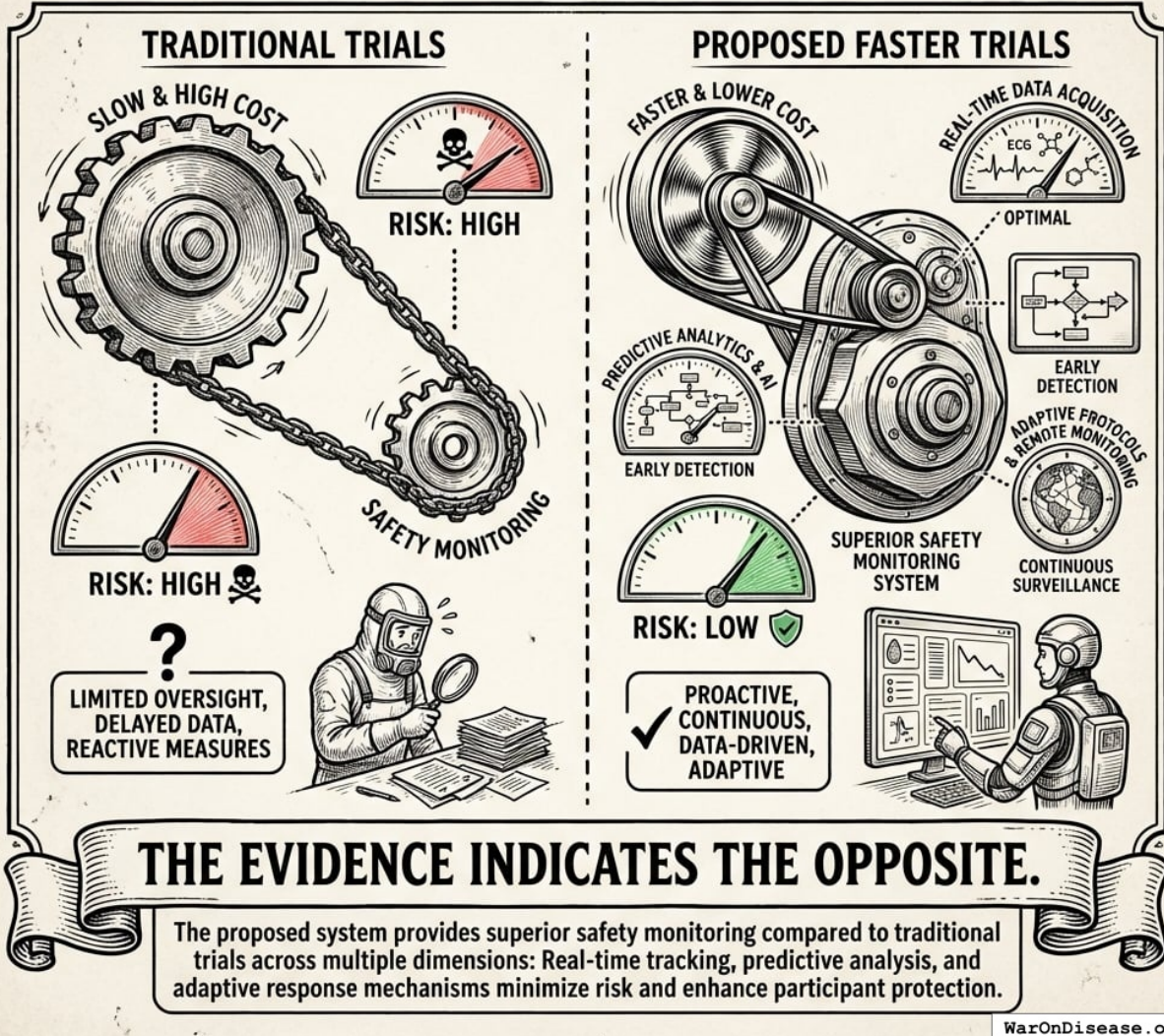


Figure 18: A comparison showing how the proposed system enhances safety monitoring across multiple dimensions compared to traditional clinical trial methods.

The evidence indicates the opposite. The proposed system provides superior safety monitoring compared to traditional trials across multiple dimensions.

5.7.1 Current System Limitations: Dangerously Blind to Real-World Harms

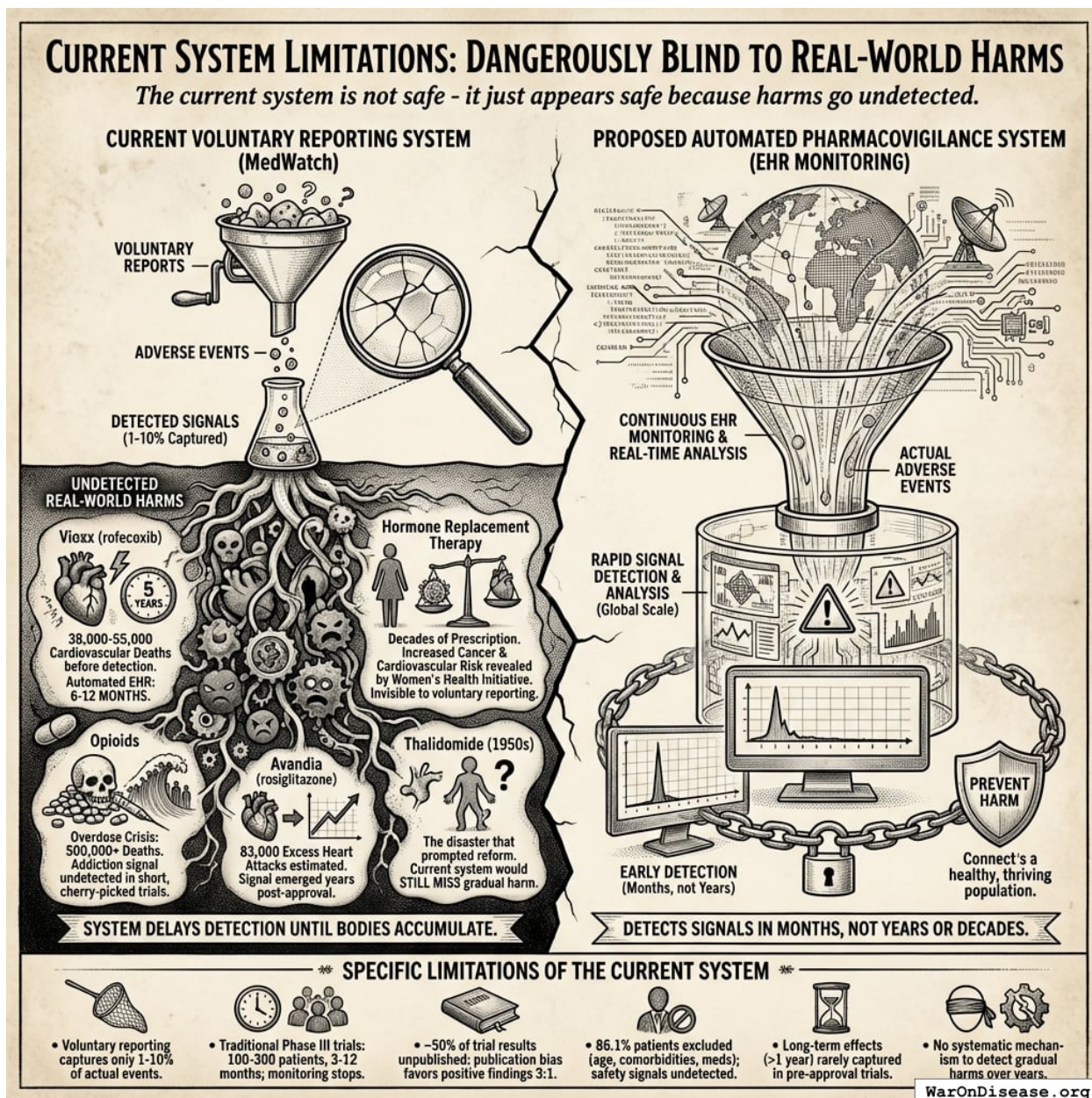


Figure 19: Comparison of detection timelines for major drug safety disasters, contrasting the years of delay under voluntary reporting systems with the estimated months required for detection via automated EHR monitoring.

The current system is not safe - it just appears safe because harms go undetected. The FDA's voluntary adverse event reporting system (MedWatch) captures only **1-10% of actual adverse events**. Long-term harms that develop gradually over years - the most insidious and deadly kind - are virtually invisible:

- **Vioxx** (rofecoxib): Caused 38,000-55,000 cardiovascular deaths over 5 years before detection through voluntary reporting. With automated EHR pharmacovigilance, the elevated MI risk would have been detected within 6-12 months.
- **Hormone Replacement Therapy**: Prescribed for decades before the Women's Health Initiative revealed increased cancer and cardiovascular risk - risks invisible to voluntary reporting
- **Opioids**: The overdose crisis killed 500,000+ Americans; the addiction signal was undetectable in short trials with cherry-picked populations
- **Avandia** (rosiglitazone): 83,000 excess heart attacks estimated before restrictions; signal emerged years post-approval
- **Thalidomide** (1950s): The disaster that prompted regulatory reform - yet the current system would **still miss** a thalidomide-like harm if it manifested gradually rather than as obvious birth defects

The current "safety" system doesn't prevent harm - it **delays detection** until bodies accumulate. A global automated pharmacovigilance system with continuous EHR monitoring would detect these signals in months, not years or decades.

Specific limitations of the current system:

- Voluntary adverse event reporting captures only 1-10% of actual events
- Traditional Phase III trials test 100-300 patients for 3-12 months, then monitoring stops
- Approximately 50% of trial results go unpublished, with publication bias favoring positive findings 3:1
- 86.1% of patients excluded due to age, comorbidities, or medications - safety signals in these populations go undetected
- Long-term effects (>1 year) rarely captured in pre-approval trials
- No systematic mechanism to detect gradual harms that develop over years

5.7.2 Proposed System Safety Advantages

1. **Preserved Phase I Safety Testing:** Rigorous Phase I safety testing (~2.3 years) is maintained. What changes is eliminating the 8.2 years (95% CI: 4.85 years-11.5 years) efficacy delay *after* safety is verified.

PROPOSED SYSTEM SAFETY ADVANTAGES

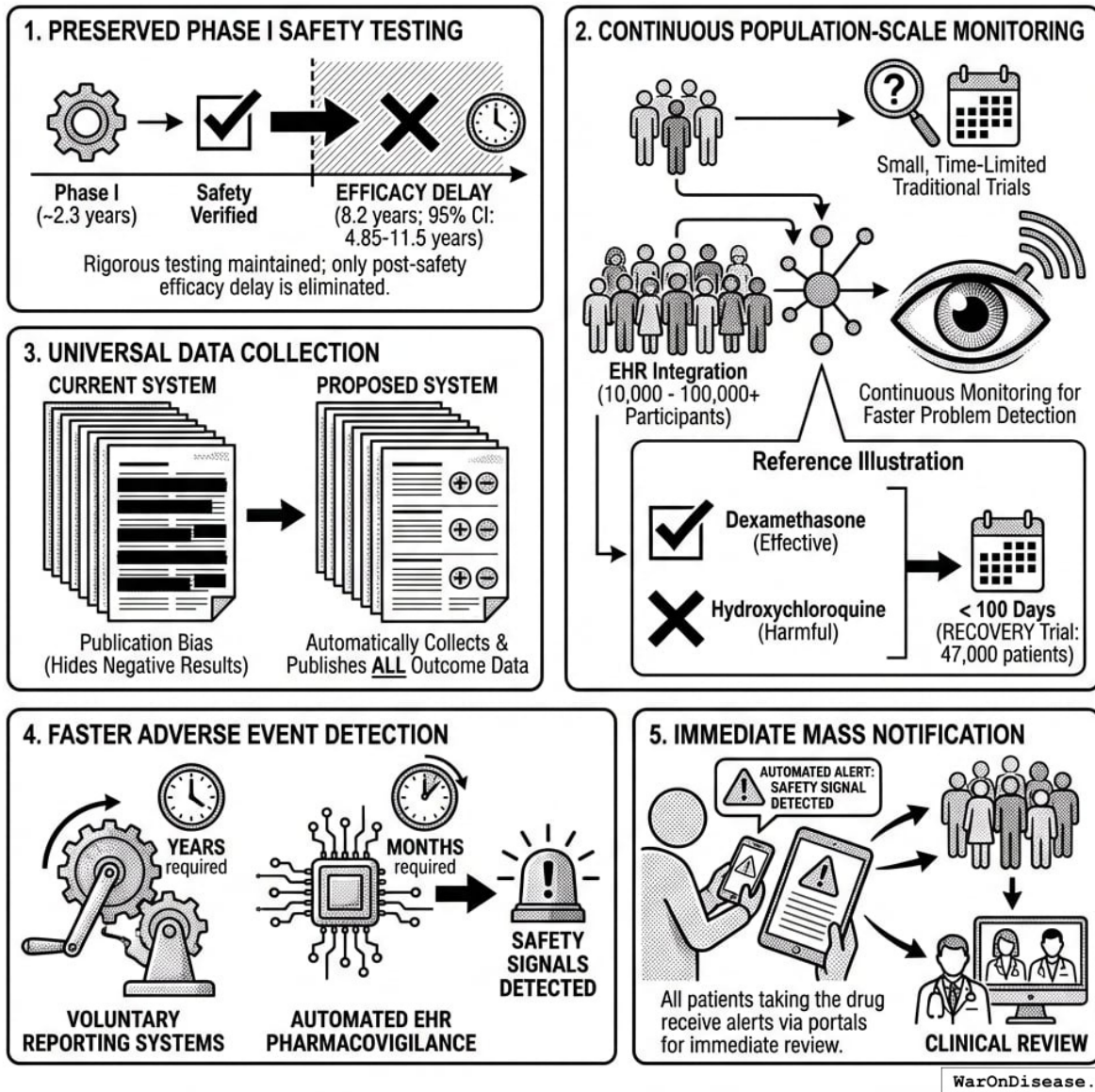


Figure 20: A comparison between the traditional clinical trial system and the proposed system, highlighting differences in safety monitoring scale, detection speed, and the elimination of efficacy delays.

- 2. Continuous Population-Scale Monitoring:** Pragmatic trials with 10,000-100,000+ participants monitored continuously through EHR integration detect safety problems faster than small, time-limited traditional trials. The RECOVERY trial identified both effective treatments (dexamethasone) and harmful ones (hydroxychloroquine) in under 100 days with 47,000 patients.
- 3. Universal Data Collection:** The system automatically collects and publishes outcome data on all treatments, eliminating the publication bias that currently hides negative results.

4. **Faster Adverse Event Detection:** Automated EHR pharmacovigilance detects safety signals in months rather than the years required by voluntary reporting systems.
5. **Immediate Mass Notification:** When safety signals are detected, all patients taking the drug receive automated alerts through patient portals, enabling immediate clinical review.

5.7.3 Detection Timeline Comparison

The following table quantifies how automated EHR monitoring would have changed outcomes for major drug safety disasters:

Drug	Harm	Voluntary Reporting Detection	Projected EHR Detection	Deaths During Delay
Vioxx	Cardiovascular events	5 years (1999-2004)	6-12 months	38,000-55,000
Avandia	Heart attacks	8 years post-approval	12-18 months	~83,000 excess events
Opioids	Addiction/over-dose	Decades	2-3 years	500,000+ deaths
HRT	Cancer/cardiovascular	40+ years	3-5 years	Unknown (millions affected)

Why the difference? Automated EHR surveillance compares treated patients to matched controls continuously. The Vioxx cardiovascular signal would trigger statistical alerts after ~5,000 prescriptions (1.2% elevated event rate vs. expected background). Voluntary reporting requires doctors to notice, remember, and file paperwork, capturing only 1-10% of events.

5.7.4 Comparative Safety Surveillance

Safety Dimension	Traditional Trials	Pragmatic Trials + EHR Monitoring
Sample size	100-300 patients	10,000-100,000+ patients
Patient selection	86.1% excluded	All volunteers (real-world populations)
Monitoring duration	3-12 months (then stops)	Continuous via EHR (indefinite)
Publication rate	~50% unpublished	100% automatically published
Adverse event detection	Voluntary reporting (1-10% capture)	Automated surveillance (100% capture)

5.7.5 Pooled Liability Insurance

The framework includes pooled liability coverage for sponsors, reducing individual company risk while ensuring patient compensation for adverse events. This removes a major barrier to trial participation for smaller sponsors while maintaining accountability.

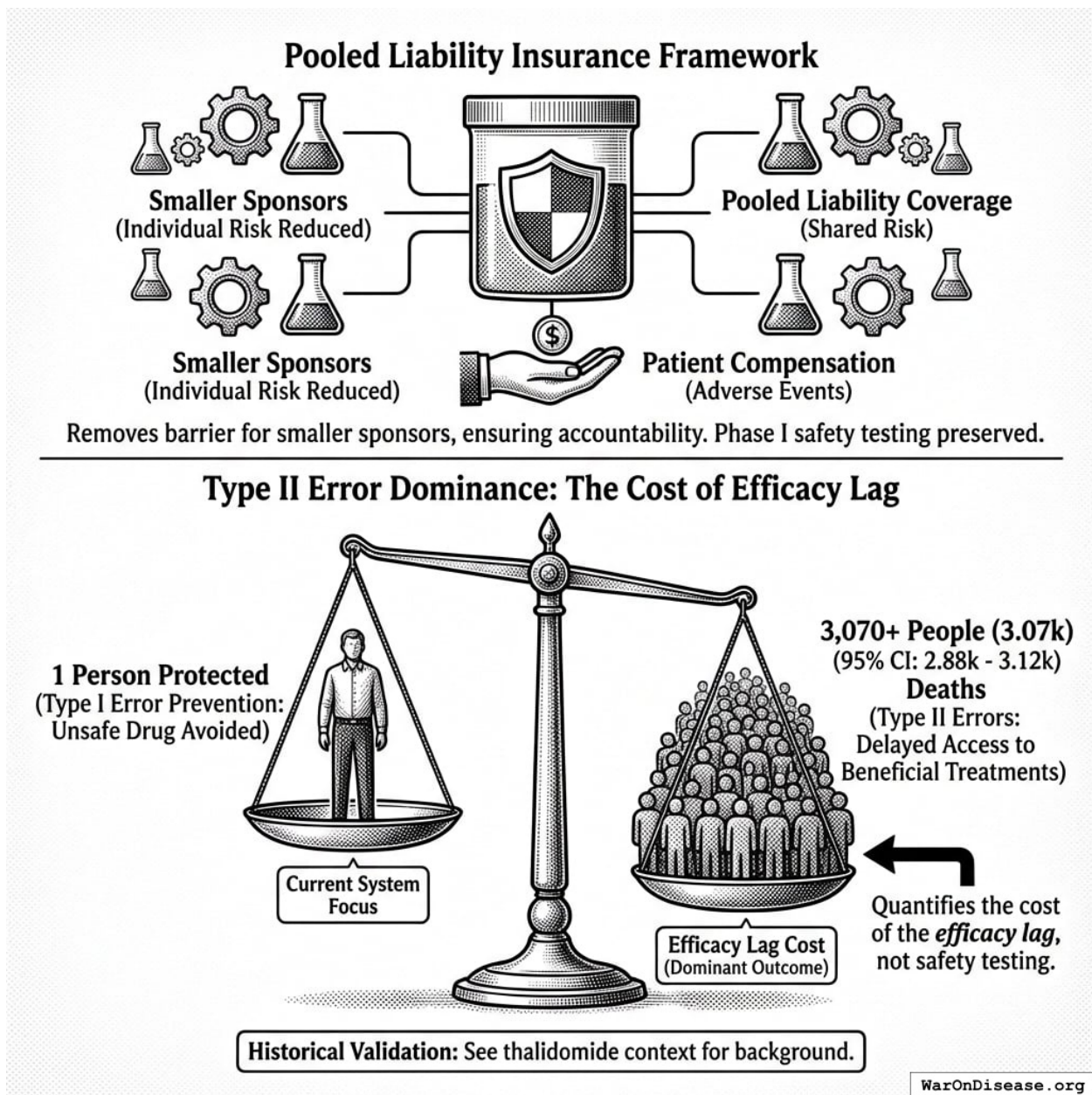


Figure 21: A comparison of Type I and Type II errors in drug regulation, illustrating the 3,070:1 ratio of deaths from delayed treatment access versus deaths prevented from unsafe drugs.

Type II Error Dominance: For every person protected from an unsafe drug (Type I error prevention), 3.07k:1 (95% CI: 2.88k:1-3.12k:1) people die from delayed access to beneficial treatments (Type II errors). The current system prevents harm from unsafe drugs, but causes 3.07k:1 (95% CI: 2.88k:1-3.12k:1) more deaths through delays. Phase I safety testing is preserved in this framework; the ratio quantifies the cost of the *efficacy lag*, not safety testing. (See [Historical Validation](#) for the thalidomide context.)

6 ROI Analysis for a Decentralized Framework

6.1 Monte Carlo Distributions

Monte Carlo Analysis: ROI from Decentralized Framework for Drug Assessment R&D Savings Only

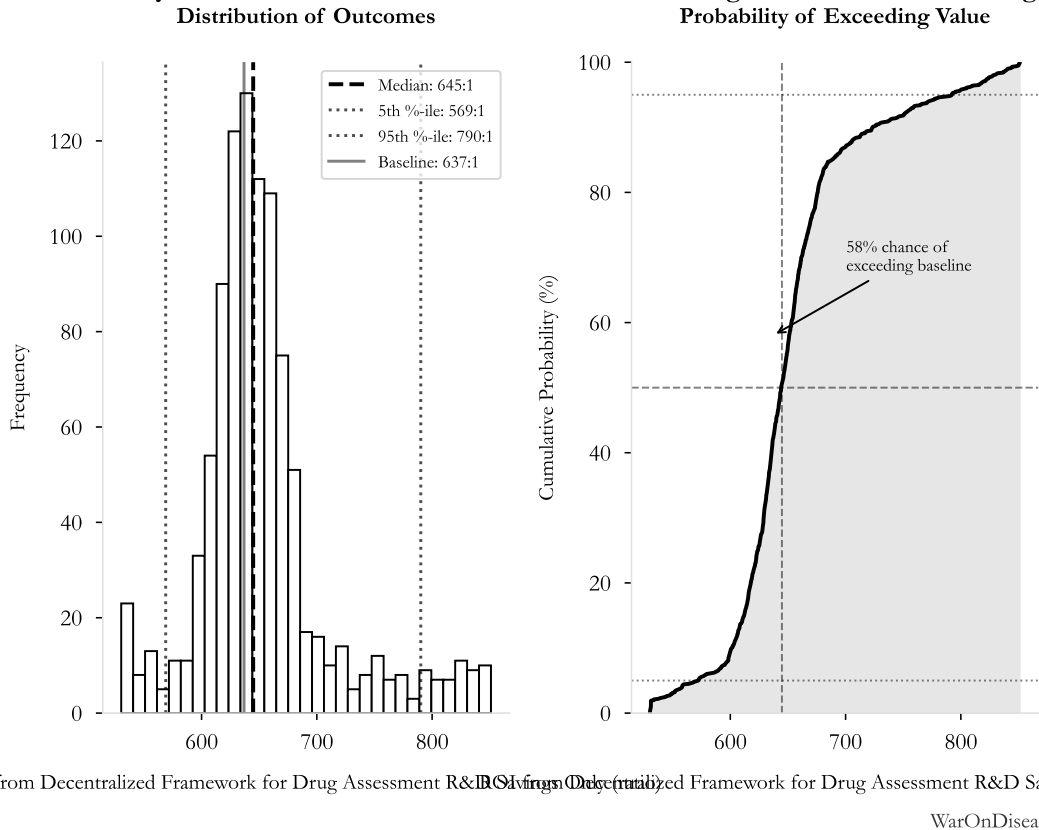


Figure 22: Monte Carlo Distribution: ROI from Decentralized Framework for Drug Assessment R&D Savings Only (10,000 simulations)

Simulation Results Summary: ROI from Decentralized Framework for Drug Assessment R&D Savings Only

Statistic	Value
Baseline (deterministic)	637:1
Mean (expected value)	653:1
Median (50th percentile)	645:1
Standard Deviation	58.4:1
90% Confidence Interval	[569:1, 790:1]

The histogram shows the distribution of ROI from Decentralized Framework for Drug Assessment R&D Savings Only across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.

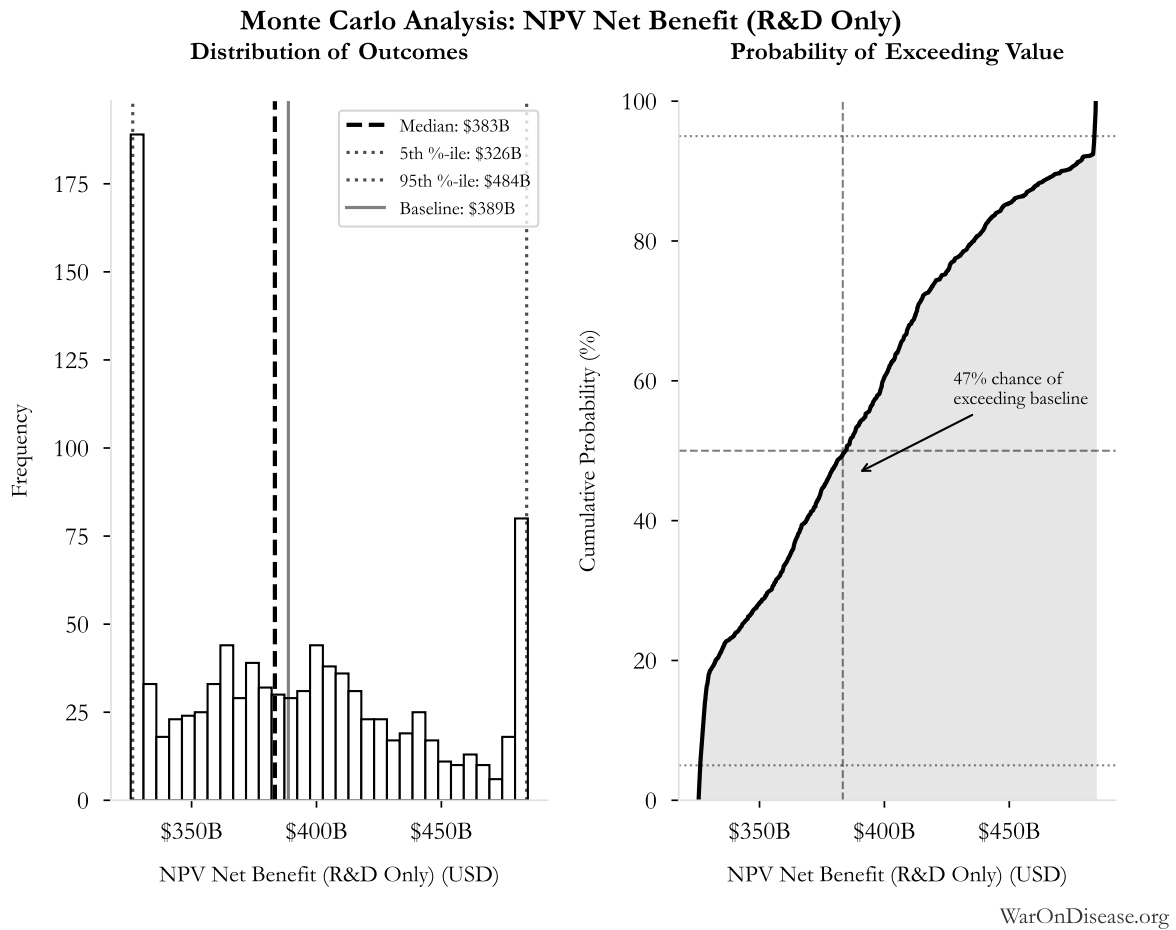


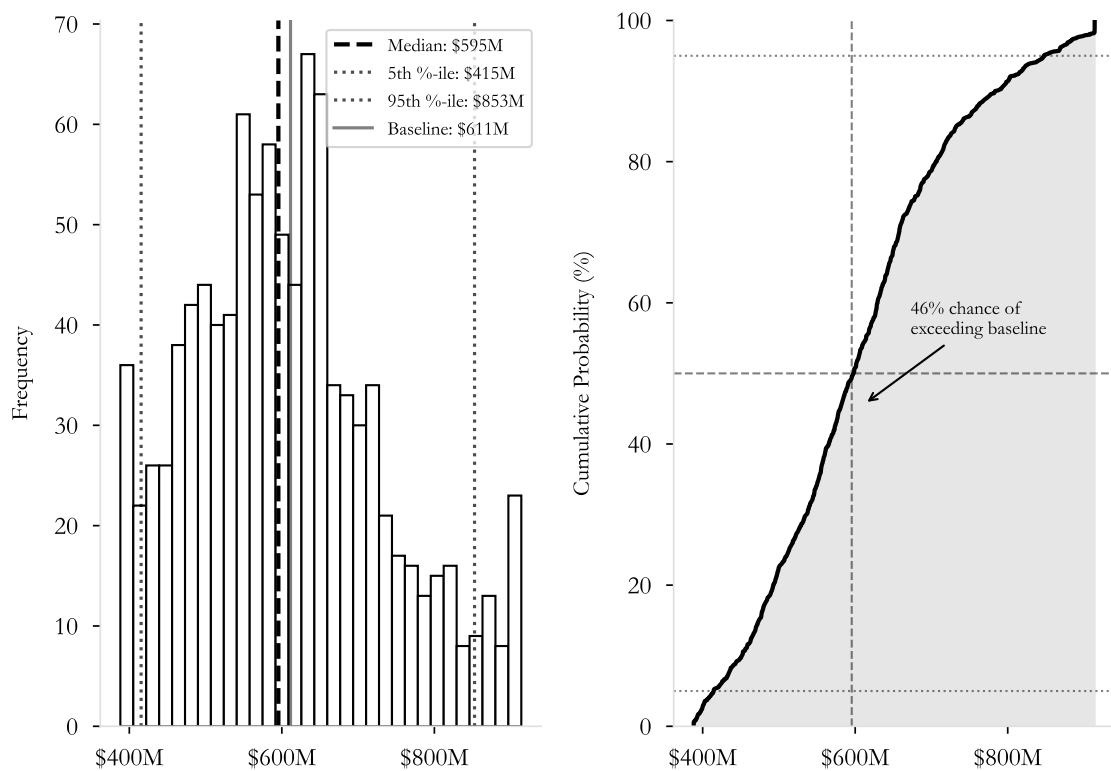
Figure 23: Monte Carlo Distribution: NPV Net Benefit (R&D Only) (10,000 simulations)

Simulation Results Summary: NPV Net Benefit (R&D Only)

Statistic	Value
Baseline (deterministic)	\$389B
Mean (expected value)	\$390B
Median (50th percentile)	\$383B
Standard Deviation	\$50.7B
90% Confidence Interval	[\$326B, \$484B]

The histogram shows the distribution of NPV Net Benefit (R&D Only) across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.

Monte Carlo Analysis: Decentralized Framework for Drug Assessment Total NPV Cost



Decentralized Framework for Drug Assessment Total NPV Cost (USD) Decentralized Framework for Drug Assessment Total NPV Cost (USD),
WarOnDisease.org

Figure 24: Monte Carlo Distribution: Decentralized Framework for Drug Assessment Total NPV Cost (10,000 simulations)

Simulation Results Summary: Decentralized Framework for Drug Assessment Total NPV Cost

Statistic	Value
Baseline (deterministic)	\$611M
Mean (expected value)	\$609M
Median (50th percentile)	\$595M
Standard Deviation	\$127M
90% Confidence Interval	[\$415M, \$853M]

The histogram shows the distribution of Decentralized Framework for Drug Assessment Total NPV Cost across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.

Exceedance Probability: ROI from Decentralized Framework for Drug Assessment R&D Savings Only

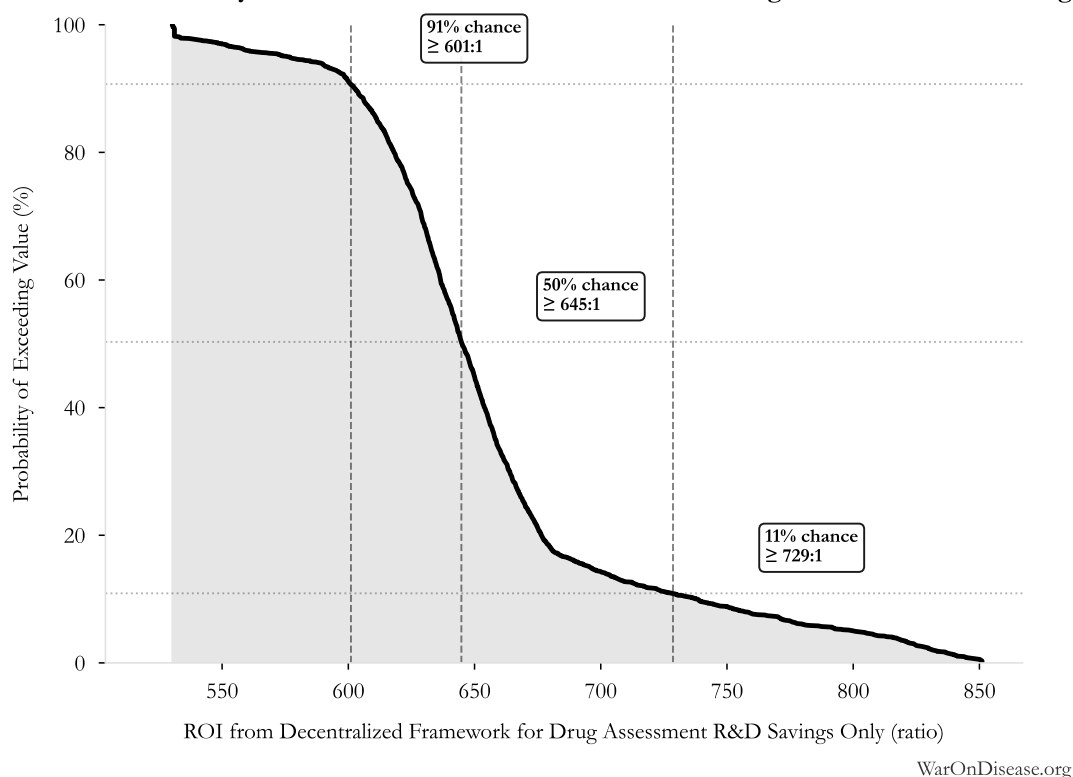


Figure 25: Probability of Exceeding Threshold: ROI from Decentralized Framework for Drug Assessment R&D Savings Only

This exceedance probability chart shows the likelihood that ROI from Decentralized Framework for Drug Assessment R&D Savings Only will exceed any given threshold. Higher curves indicate more favorable outcomes with greater certainty.

7 Research Acceleration Mechanism

The 12.3x (95% CI: 4.19x-61.3x) research acceleration transforms our ability to explore the vast therapeutic space where undiscovered cures already exist.

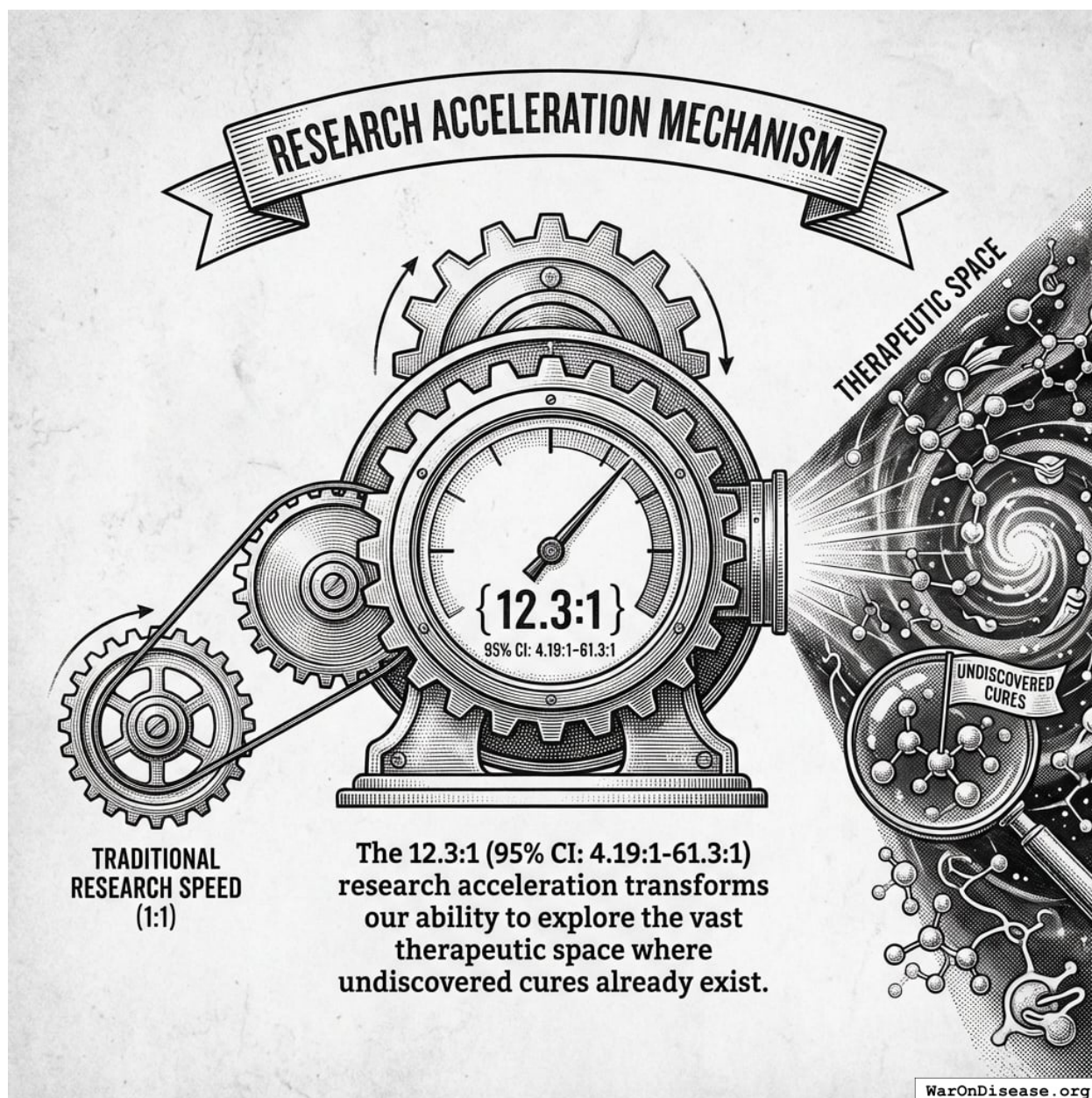


Figure 26: A visual comparison showing the scale of research acceleration, contrasting standard discovery progress with the 12.3x multiplier and its 95% confidence interval.

7.1 The Unexplored Therapeutic Frontier

The fundamental problem isn't that cures are hard to discover. It's that we're barely looking:

- **9.50M combinations** plausible drug-disease pairings exist (9.50k compounds (95% CI: 7.00k compounds-12.0k compounds) safe × 1.00k diseases (95% CI: 800 diseases-1.20k diseases))
- **Only 0.342% (95% CI: 0.21%-0.514%)** of these combinations have been tested - **99.7% (95% CI: 99.5%-99.8%)** remains unexplored
- **Only 12%** of the human interactome has ever been targeted by drugs
- **30%** of approved drugs gain new indications, proving undiscovered uses exist

$$Ratio_{explore} = \frac{N_{tested}}{N_{combos}} = \frac{32,500}{9.5M} = 0.342\%$$

$$\begin{aligned} & \text{where } N_{combos} \\ &= N_{safe} \times N_{diseases,trial} \\ &= 9,500 \times 1,000 \\ &= 9.5M \end{aligned}$$

The cures likely already exist among tested-safe compounds. We just haven't looked. See [The Untapped Therapeutic Frontier](#) for detailed analysis of this exploration gap, and The Discovery Capacity Model for the quantitative framework showing how 12.3x (95% CI: 4.19x-61.3x) trial capacity produces the 212 years (95% CI: 135 years-355 years) timeline shift.

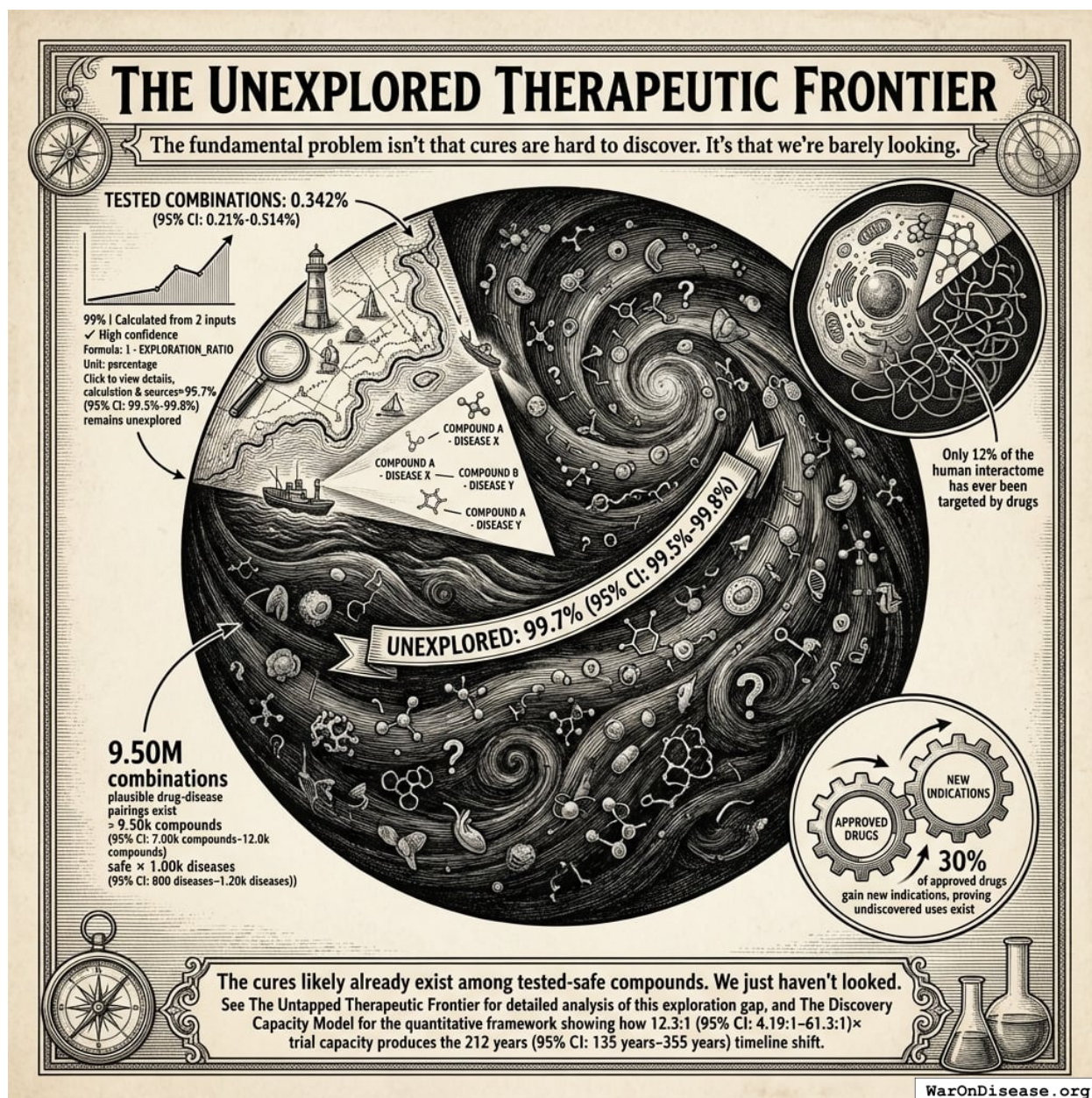


Figure 27: A visualization of the therapeutic exploration gap, contrasting the 9.5 million potential drug-disease combinations with the tiny 0.3% sliver that has been clinically tested.

7.2 Addressing the Returns Question: Diminishing, Linear, or Compounding?

A common objection is that “more trials won’t produce proportionally more cures” - the diminishing returns hypothesis. This deserves serious consideration, but the evidence suggests the opposite may be true.

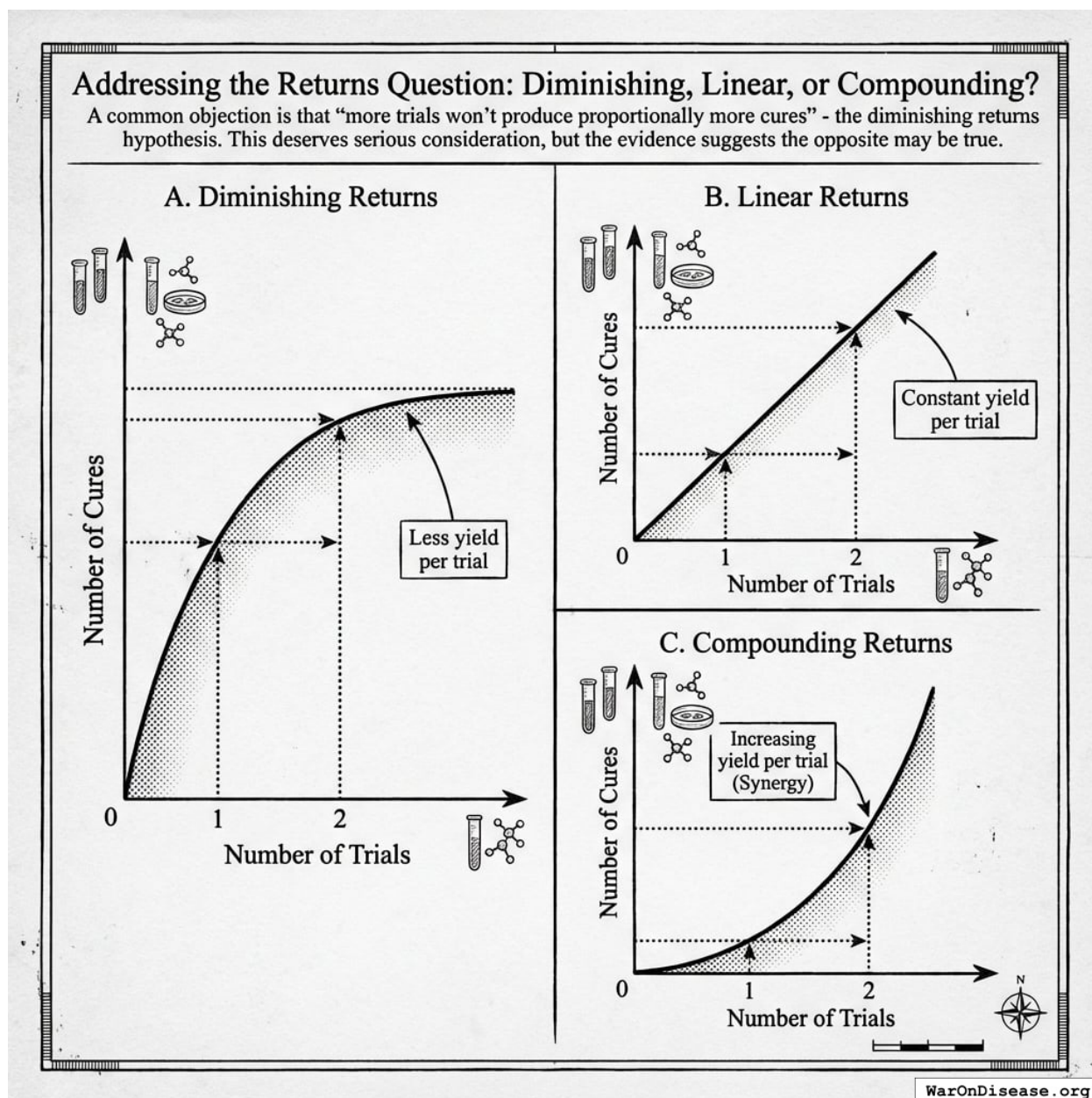


Figure 28: A comparison of three growth trajectories (diminishing, linear, and compounding) showing the relationship between the number of trials and the resulting output of cures.

7.2.1 Why Diminishing Returns Is Unlikely (We Haven’t Started Looking)

The diminishing returns objection assumes we’ve exhausted low-hanging fruit. But we’ve barely begun:

1. **Single compounds alone:** 9.50M combinations possible combinations of known safe compounds × diseases. At current trial capacity, systematically testing these would take **2.88k years** (95% CI: 2.45k years-3.43k years). We won’t finish until the year 5000+.

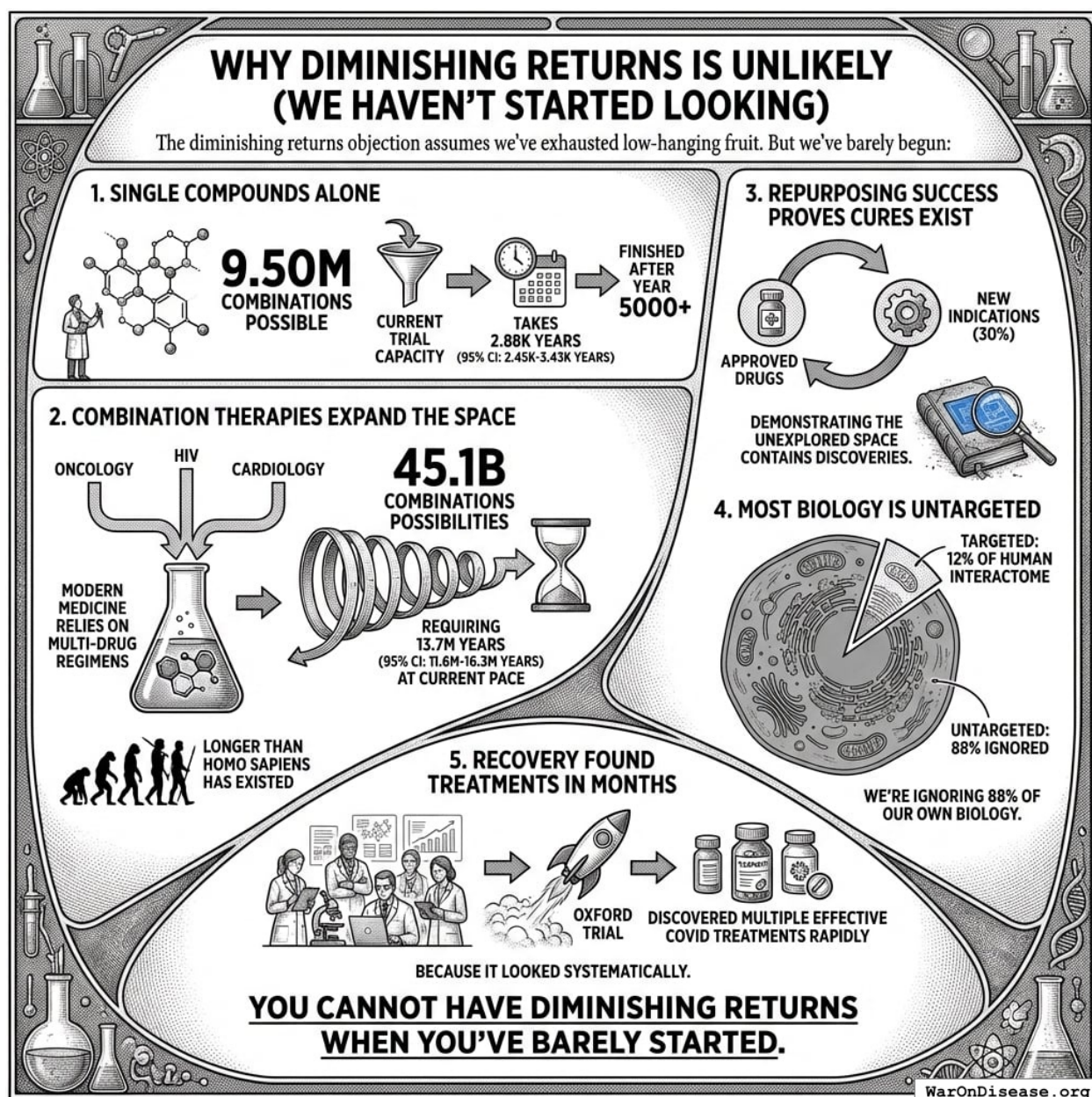


Figure 29: A comparison showing the massive scale of unexplored medical combinations and biological targets versus the tiny fraction currently understood, highlighting the vast remaining potential for discovery.

$$\begin{aligned}
& T_{\text{explore,safe}} \\
&= \frac{N_{\text{combos}}}{\text{Trials}_{\text{ann,curr}}} \\
&= \frac{9.5M}{3,300} \\
&= 2,880
\end{aligned}$$

$$\begin{aligned}
& \text{where } N_{\text{combos}} \\
&= N_{\text{safe}} \times N_{\text{diseases,trial}} \\
&= 9,500 \times 1,000 \\
&= 9.5M
\end{aligned}$$

2. **Combination therapies expand the space:** Modern medicine relies on multi-drug regimens (oncology, HIV, cardiology). Pairwise combinations of safe compounds create **45.1B combinations** possibilities, requiring **13.7M years (95% CI: 11.6M years-16.3M years)** at current pace - longer than *Homo sapiens* has existed.
3. **Repurposing success proves cures exist:** 30% of approved drugs gain new indications, demonstrating the unexplored space contains discoveries.
4. **Most biology is untargeted:** Only 12% of the human interactome has been targeted. We're ignoring 88% of our own biology.
5. **RECOVERY found treatments in months:** The Oxford trial discovered multiple effective COVID treatments rapidly *because it looked systematically*.

You cannot have diminishing returns when you've barely started.

7.2.2 Mathematical Framework: When Would Diminishing Returns Dominate?

We can formalize the competing models to identify when diminishing returns would actually matter.

Model 1: Linear (Baseline)

$$T_{\text{discovered}} = k_0 \cdot N_{\text{trials}}$$

Where k_0 is the constant discovery rate (effective treatments per trial). This assumes the therapeutic space is sampled uniformly at random.

Model 2: Diminishing Returns (Pessimistic)

As we exhaust the therapeutic space, the hit rate decreases:

$$k_{\text{dim}}(s) = k_0 \cdot (1 - s)$$

Where $s = S_{\text{explored}}/S_{\text{total}}$ is the fraction of therapeutic space already tested. At current exploration ($s < 0.01$), this gives $k_{\text{dim}} \approx 0.99 \cdot k_0$, virtually identical to linear.

Model 3: Learning/Compounding (Optimistic)

Each trial improves our biological models, increasing future hit rates:

$$k_{learn}(n) = k_0 \cdot (1 + \alpha \cdot \ln(1 + n))$$

Where α is the learning coefficient and n is cumulative trials completed. Even modest learning ($\alpha = 0.1$) with 100,000 trials yields $k_{learn} \approx 2.15 \cdot k_0$.

Model 4: Combined (Realistic)

Both effects operate simultaneously:

$$k_{combined}(s, n) = k_0 \cdot (1 - s) \cdot (1 + \alpha \cdot \ln(1 + n))$$

The Crossover Point: When Does Depletion Dominate Learning?

Diminishing returns dominates when the depletion factor exceeds the learning factor. Solving for the critical exploration fraction:

$$s_{crossover} = 1 - \frac{1}{1 + \alpha \cdot \ln(1 + n)}$$

Learning Coefficient (α)	Trials Completed (n)	Crossover Exploration ($s_{crossover}$)
0.05 (weak)	100,000	37%
0.10 (modest)	100,000	53%
0.15 (strong)	100,000	63%

Interpretation: Even with weak learning effects, diminishing returns only dominates after exploring 37%+ of therapeutic space. With modest learning, the crossover occurs at 53%+ exploration.

Timeline to Crossover:

At current exploration of 0.342% (95% CI: 0.21%-0.514%) (<1%), reaching the 53% crossover would require ~1,500 years at current pace or ~125 years with dFDA. For combination therapies (45.1B combinations combinations), reaching 53% exploration would take millions of years.

Conclusion: For any plausible planning horizon, learning effects dominate. Diminishing returns is a theoretical concern for civilizations operating on multi-century timescales, not a practical constraint for the next 100+ years of medical research.

7.2.3 The Conservative Default: Linear Assumption

Given genuine uncertainty about whether returns are diminishing or compounding, our analysis assumes a **linear relationship** between trial capacity and treatment discoveries. This is the conservative choice because:

1. **It's the neutral prior:** Without strong evidence for either diminishing or compounding returns, linearity is the least assumptive model

2. **It may underestimate benefits:** If platform technologies and learning effects produce compounding returns, our projections are conservative
3. **It's empirically defensible:** The RECOVERY trial's success (multiple treatments found with increased search) is consistent with linear or better returns
4. **It avoids both failure modes:** Assuming diminishing returns would justify inaction; assuming compounding returns might overstate benefits. Linearity is the responsible middle ground

Bottom line: Even under the conservative linear assumption, 12.3x (95% CI: 4.19x-61.3x) more trials produces 12.3x (95% CI: 4.19x-61.3x) more discoveries from a space that is 99%+ unexplored. The expected value calculation remains overwhelmingly positive.

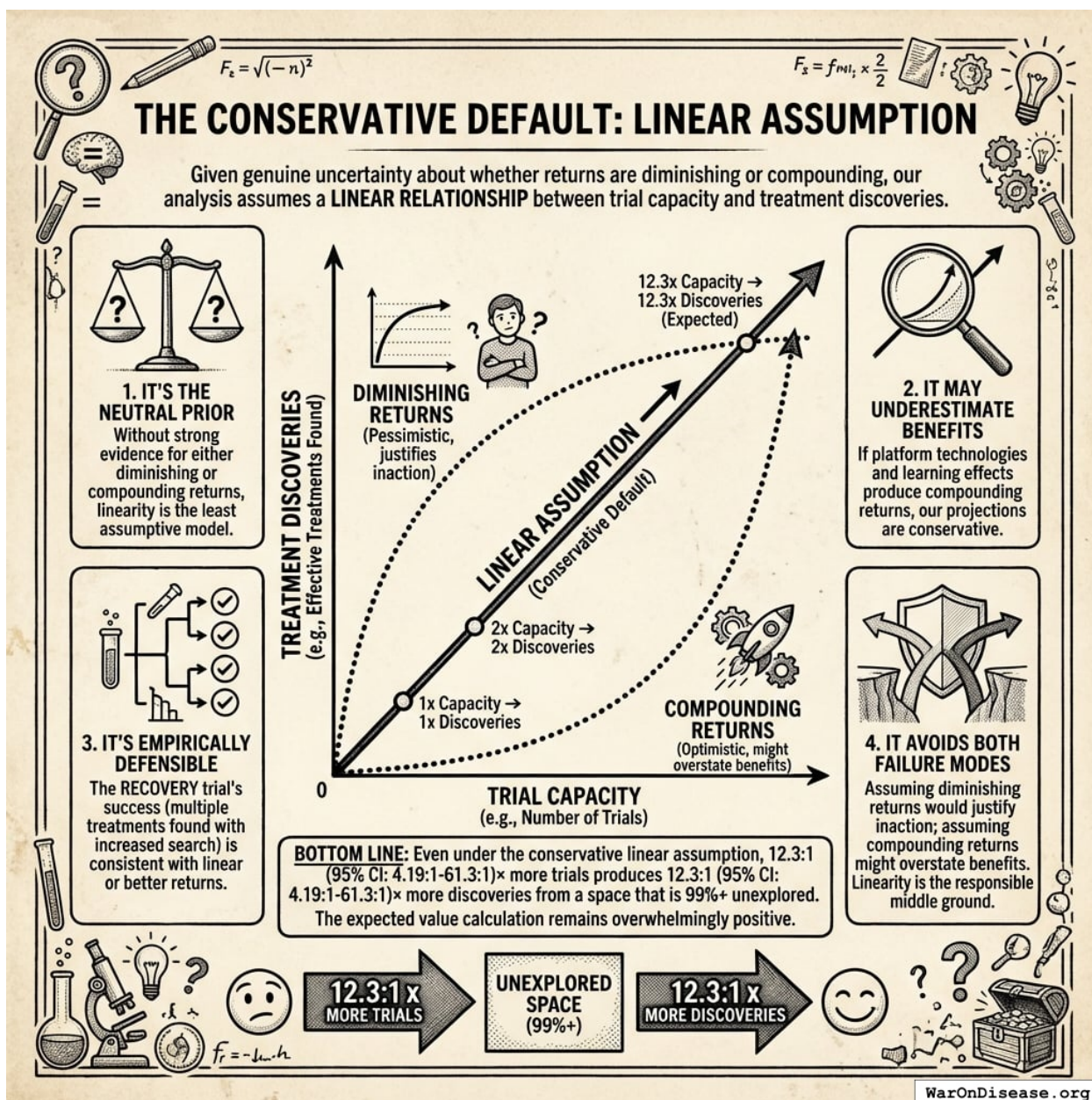


Figure 30: A comparison of three models of trial discovery: diminishing, linear, and compounding returns, illustrating the linear assumption as the conservative middle ground.

8 Data Sources and Methodological Notes

1. Cost of Current Drug Development:

- Tufts Center for the Study of Drug Development often cited for \$1.0 - \$2.6 billion/drug.
- Journal articles and industry reports (IQVIA, Deloitte) also highlight \$2+ billion figures.
- Oxford RECOVERY trial: \$500 (95% CI: \$400-\$2.50K)/patient (exceptional NHS/COVID conditions). ADAPTABLE trial: \$929 (95% CI: \$929-\$1.40K)/patient (typical US pragmatic trial). Our projections use \$929 (95% CI: \$97-\$3K)/patient based

on ADAPTABLE; confidence interval captures uncertainty.

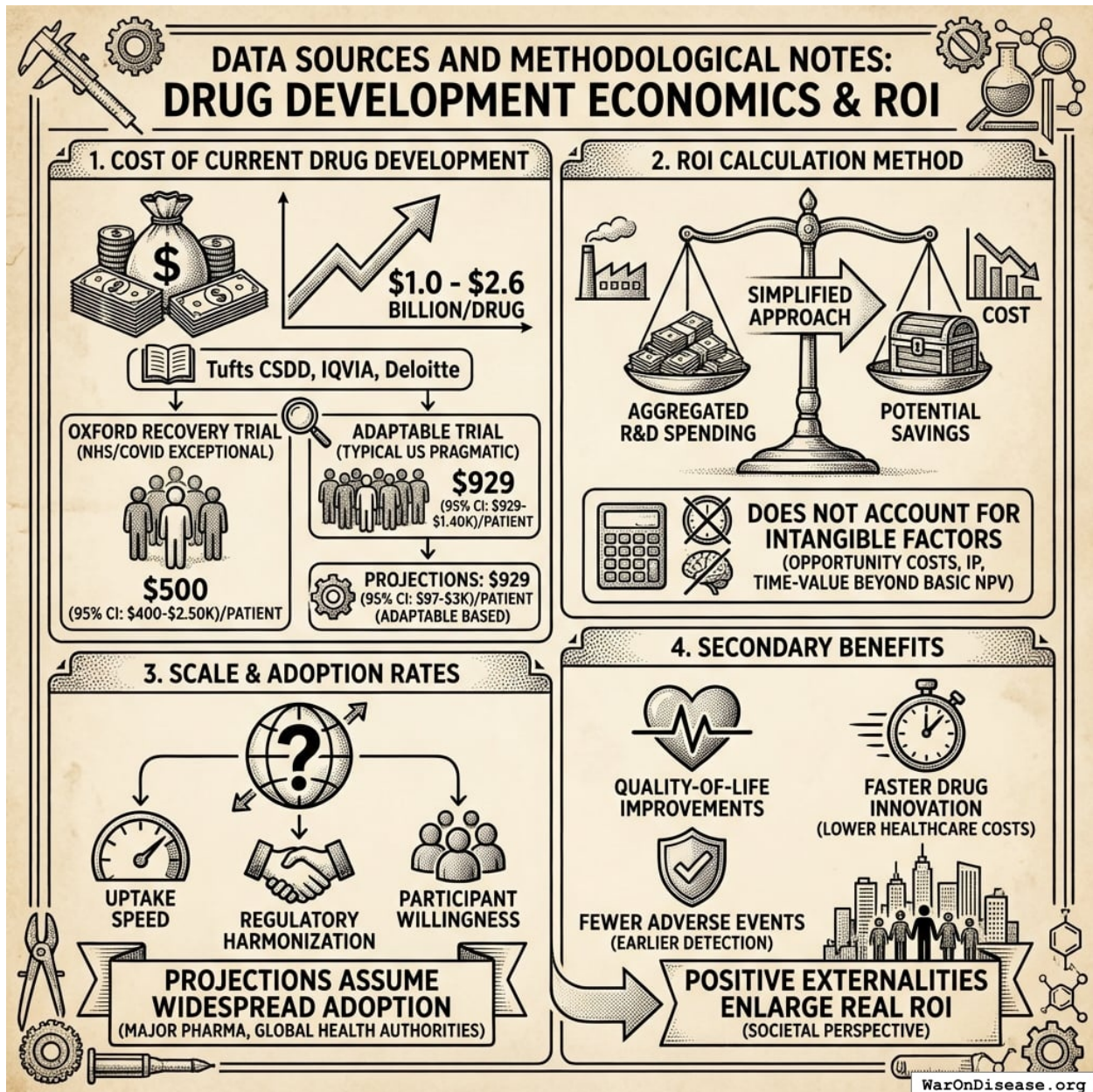


Figure 31: A comparison of traditional multi-billion dollar drug development cost estimates versus the significantly lower per-patient costs observed in pragmatic clinical trials.

2. ROI Calculation Method:

- Simplified approach comparing aggregated R&D spending to potential savings.
- Does not account for intangible factors (opportunity costs, IP complexities, time-value of money) beyond a basic Net Present Value (NPV) perspective.

3. Scale & Adoption Rates:

- The largest uncertainties revolve around uptake speed, regulatory harmonization, and

participant willingness.

- Projections assume widespread adoption by major pharmaceutical companies and global health authorities.

4. Secondary Benefits:

- Quality-of-life improvements, lower healthcare costs from faster drug innovation, and potentially fewer adverse events from earlier detection.
- These are positive externalities that can significantly enlarge real ROI from a societal perspective.

9 Conclusion

Transforming the FDA’s centralized regulatory approach into a global, decentralized model offers the potential to reduce clinical trial costs by a factor of 44.1x (95% CI: 39.4x-89.1x), accelerate approval timelines, and expand therapeutic coverage to neglected diseases. The 10-year NPV total cost is \$611M (95% CI: \$415M-\$853M) (upfront plus discounted annual operations), generating \$389B (95% CI: \$326B-\$484B) in net R&D savings. Given that the pharmaceutical industry collectively spends \$60B (95% CI: \$50B-\$75B) annually on clinical trials, a 97.7% (95% CI: 97.5%-98.9%) reduction yields an ROI of **637:1 (95% CI: 569:1-790:1)** at scale.

Beyond direct savings, the effects on medical progress are substantial: expanded therapeutic exploration, real-time treatment effectiveness rankings, and research on off-patent treatments that currently lack commercial incentives. With appropriate privacy protections and international coordination, this framework enables evidence-based personalized medicine at global scale.

9.1 Disclaimer

All figures in this document are estimates based on publicly available information, industry benchmarks, and simplifying assumptions. Real-world costs, savings, and ROI will vary greatly depending on the scope of implementation, the speed of adoption, regulatory cooperation, and numerous other factors. Nonetheless, this high-level exercise illustrates the substantial potential gains from a global, decentralized, continuously learning clinical trial and regulatory ecosystem.

10 Verification: Complete Derivation Chains

For economist verification, this section provides complete derivation chains for all headline figures. Each metric traces back to primary data sources.

10.1 1. Trial Capacity Multiplier Derivation (12.3x (95% CI: 4.19x-61.3x))

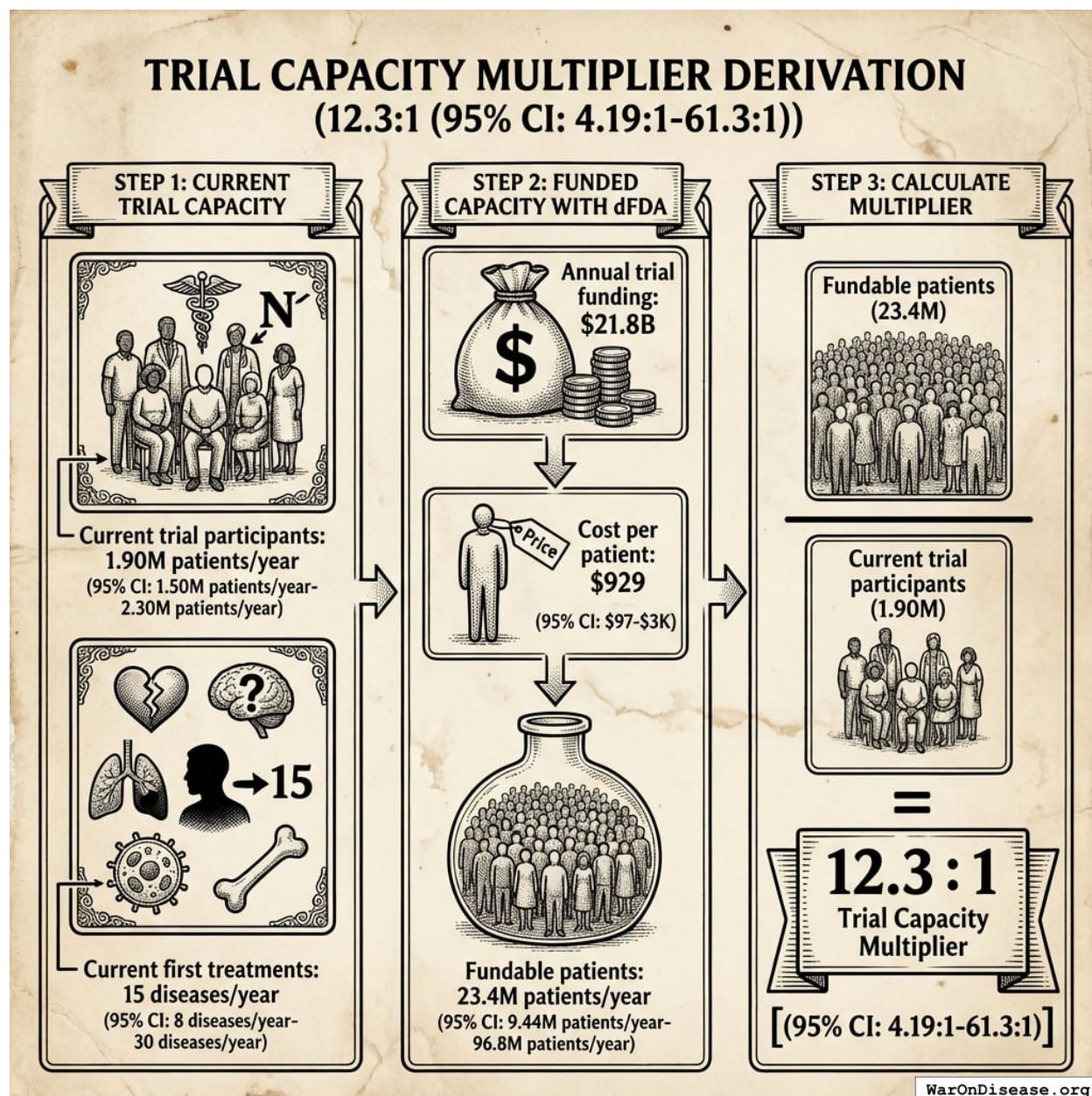


Figure 32: A comparison showing the massive scale-up from current clinical trial capacity (1.9M patients) to the projected capacity under dFDA (23.4M patients), highlighting the 12.3x multiplier effect.

Step 1: Current trial capacity - Current trial participants: 1.90M patients/year (95% CI: 1.50M patients/year-2.30M patients/year) - Current first treatments: 15 diseases/year (95% CI: 8 diseases/year-30 diseases/year)

Step 2: Funded capacity with dFDA - Annual trial funding: \$21.8B - Cost per patient: \$929 (95% CI: \$97-\$3K) - Fundable patients: 23.4M patients/year (95% CI: 9.44M patients/year-96.8M patients/year)

patients/year)

Step 3: Calculate multiplier

$$k_{capacity} = \frac{N_{fundable,ann}}{Slots_{curr}} = \frac{23.4M}{1.9M} = 12.3$$

$$\begin{aligned} & \text{where } N_{fundable,ann} \\ &= \frac{Subsidies_{trial,ann}}{Cost_{pragmatic,pt}} \\ &= \frac{\$21.7B}{\$929} \\ &= 23.4M \end{aligned}$$

$$\begin{aligned} & \text{where } Subsidies_{trial,ann} \\ &= Treasury_{RD,ann} - OPEX_{dFDA} \\ &= \$21.8B - \$40M \\ &= \$21.7B \end{aligned}$$

$$\begin{aligned} & \text{where } OPEX_{dFDA} \\ &= Cost_{platform} + Cost_{staff} + Cost_{infra} \\ & \quad + Cost_{regulatory} + Cost_{community} \\ &= \$15M + \$10M + \$8M + \$5M + \$2M \\ &= \$40M \end{aligned}$$

$$\begin{aligned} & \text{where } Treasury_{RD,ann} \\ &= Funding_{treaty} - Payout_{bond,ann} - Funding_{political,ann} \\ &= \$27.2B - \$2.72B - \$2.72B \\ &= \$21.8B \end{aligned}$$

$$\begin{aligned} & \text{where } Funding_{treaty} \\ &= Spending_{mil} \times Reduce_{treaty} \\ &= \$2.72T \times 1\% \\ &= \$27.2B \end{aligned}$$

$$\begin{aligned} & \text{where } Payout_{bond,ann} \\ &= Funding_{treaty} \times Pct_{bond} \\ &= \$27.2B \times 10\% \\ &= \$2.72B \end{aligned}$$

$$\begin{aligned} & \text{where } Funding_{treaty} \\ &= Spending_{mil} \times Reduce_{treaty} \\ &= \$2.72T \times 1\% \\ &= \$27.2B \end{aligned}$$

$$\begin{aligned} & \text{where } Funding_{political,ann} \\ &= Funding_{treaty} \times Pct_{political} \\ &= \$27.2B \times 10\% \\ &= \$2.72B \end{aligned}$$

$$\begin{aligned} & \text{where } Funding_{treaty} \\ &= Spending_{mil} \times Reduce_{treaty} \\ &= \$2.72T \times 1\% \end{aligned}$$

10.2 2. Timeline Shift Derivation (212 years (95% CI: 135 years-355 years))

Components: | Component | Value | Source | |—————|——-|———| | Efficacy Lag Elimination | 8.2 years (95% CI: 4.85 years-11.5 years) | FDA drug approval timeline data | | Discovery Acceleration | 204 years (95% CI: 123 years-350 years) | Capacity vs. backlog model | | **Combined Total** | **212 years (95% CI: 135 years-355 years)** | Sum of components |

$$T_{accel,max} = T_{accel} + T_{lag} = 204 + 8.2 = 212$$

$$\begin{aligned} & \text{where } T_{accel} \\ &= T_{first,SQ} \times \left(1 - \frac{1}{k_{capacity}}\right) \\ &= 222 \times \left(1 - \frac{1}{12.3}\right) \\ &= 204 \end{aligned}$$

$$\text{where } T_{first,SQ} = T_{queue,SQ} \times 0.5 = 443 \times 0.5 = 222$$

$$\text{where } T_{queue,SQ} = \frac{N_{untreated}}{Treatments_{new,ann}} = \frac{6,650}{15} = 443$$

$$\text{where } N_{untreated} = N_{rare} \times 0.95 = 7,000 \times 0.95 = 6,650$$

$$\text{where } k_{capacity} = \frac{N_{fundable,ann}}{Slots_{curr}} = \frac{23.4M}{1.9M} = 12.3$$

$$\begin{aligned} & \text{where } N_{fundable,ann} \\ &= \frac{Subsidies_{trial,ann}}{Cost_{pragmatic,pt}} \\ &= \frac{\$21.7B}{\$929} \\ &= 23.4M \end{aligned}$$

$$\begin{aligned} & \text{where } Subsidies_{trial,ann} \\ &= Treasury_{RD,ann} - OPEX_{dFDA} \\ &= \$21.8B - \$40M \\ &= \$21.7B \end{aligned}$$

$$\begin{aligned} & \text{where } OPEX_{dFDA} \\ &= Cost_{platform} + Cost_{staff} + Cost_{infra} \\ & \quad + Cost_{regulatory} + Cost_{community} \\ &= \$15M + \$10M + \$8M + \$5M + \$2M \\ &= \$40M \end{aligned}$$

$$\begin{aligned} & \text{where } Treasury_{RD,ann} \\ &= Funding_{treaty} - Payout_{bond,ann} - Funding_{political,ann} \\ &= \$27.2B - \$2.72B - \$2.72B \\ &= \$21.8B \end{aligned}$$

$$\begin{aligned} & \text{where } Funding_{treaty} \\ &= Spending_{mil} \times Reduce_{treaty} \\ &= \$2.72T \times 1\% \\ &= \$27.2B \end{aligned}$$

$$\begin{aligned} & \text{where } Payout_{bond,ann} \\ &= Funding_{treaty} \times Pct_{bond} \\ &= \$27.2B \times 10\% \\ &= \$2.72B \end{aligned}$$

10.3 3. Lives Saved Derivation (10.7B deaths (95% CI: 7.39B deaths-16.2B deaths))

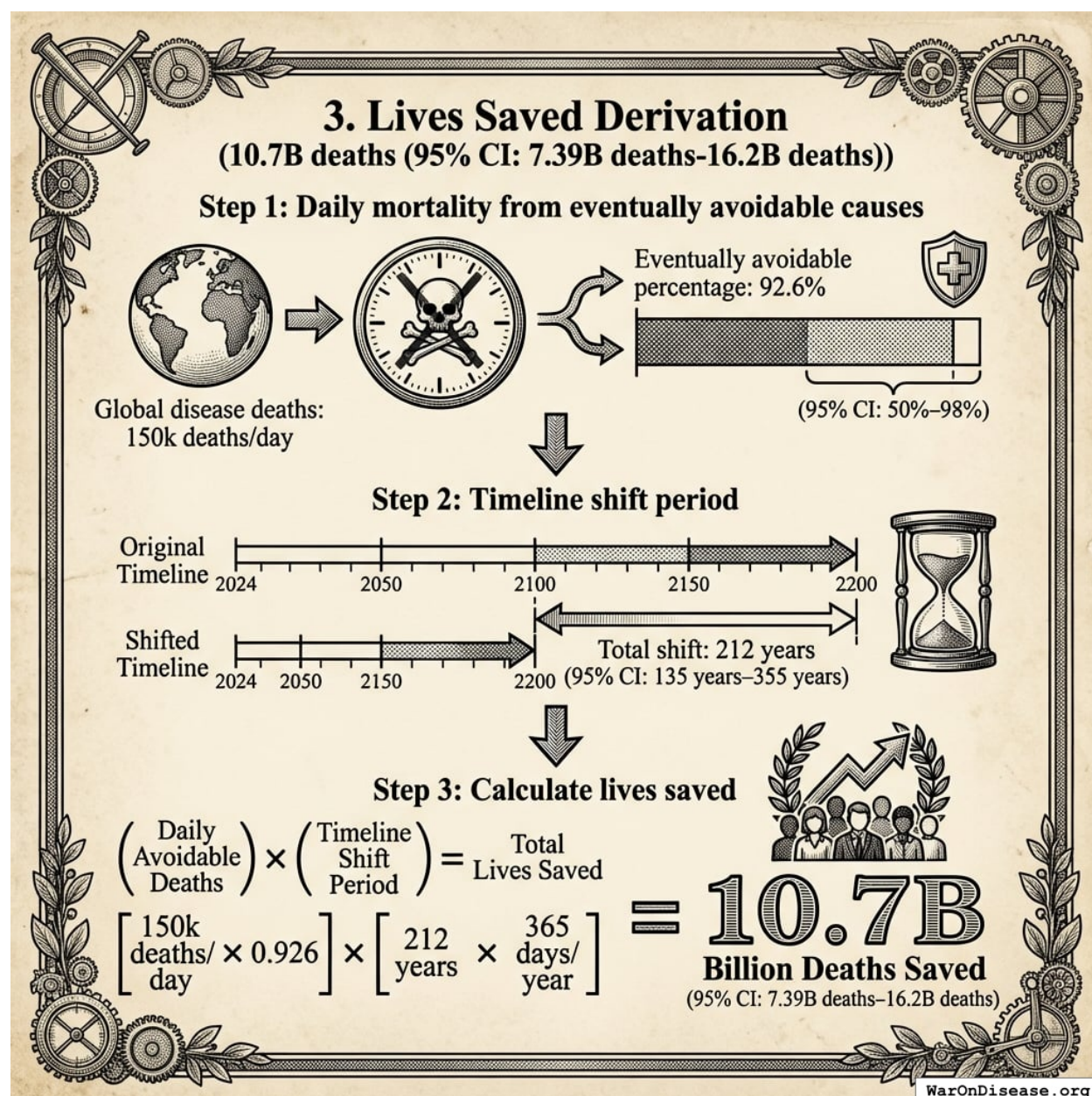


Figure 33: A flowchart illustrating the multi-step derivation of total lives saved by combining global daily mortality, the percentage of avoidable deaths, and the total timeline shift period.

Step 1: Daily mortality from eventually avoidable causes - Global disease deaths: 150k deaths/day/day - Eventually avoidable percentage: 92.6% (95% CI: 50%-98%)

Step 2: Timeline shift period - Total shift: 212 years (95% CI: 135 years-355 years)

Step 3: Calculate lives saved

$$\begin{aligned}
& Lives_{max} \\
&= Deaths_{disease,daily} \times T_{accel,max} \times 338 \\
&= 150,000 \times 212 \times 338 \\
&= 10.7B
\end{aligned}$$

$$\text{where } T_{accel,max} = T_{accel} + T_{lag} = 204 + 8.2 = 212$$

$$\begin{aligned}
& \text{where } T_{accel} \\
&= T_{first,SQ} \times \left(1 - \frac{1}{k_{capacity}}\right) \\
&= 222 \times \left(1 - \frac{1}{12.3}\right) \\
&= 204
\end{aligned}$$

$$\text{where } T_{first,SQ} = T_{queue,SQ} \times 0.5 = 443 \times 0.5 = 222$$

$$\text{where } T_{queue,SQ} = \frac{N_{untreated}}{Treatments_{new,ann}} = \frac{6,650}{15} = 443$$

$$\text{where } N_{untreated} = N_{rare} \times 0.95 = 7,000 \times 0.95 = 6,650$$

$$\text{where } k_{capacity} = \frac{N_{fundable,ann}}{Slots_{curr}} = \frac{23.4M}{1.9M} = 12.3$$

$$\begin{aligned}
& \text{where } N_{fundable,ann} \\
&= \frac{Subsidies_{trial,ann}}{Cost_{pragmatic,pt}} \\
&= \frac{\$21.7B}{\$929} \\
&= 23.4M
\end{aligned}$$

$$\begin{aligned}
& \text{where } Subsidies_{trial,ann} \\
&= Treasury_{RD,ann} - OPEX_{dFDA} \\
&= \$21.8B - \$40M \\
&= \$21.7B
\end{aligned}$$

$$\begin{aligned}
& \text{where } OPEX_{dFDA} \\
&= Cost_{platform} + Cost_{staff} + Cost_{infra} \\
&\quad + Cost_{regulatory} + Cost_{community} \\
&= \$15M + \$10M + \$8M + \$5M + \$2M \\
&= \$40M
\end{aligned}$$

$$\begin{aligned}
& \text{where } Treasury_{RD,ann} \\
&= Funding_{treaty} - Payout_{bond,ann} - Funding_{political,ann} \\
&= \$27.2B - \$2.72B - \$2.72B \\
&= \$21.8B
\end{aligned}$$

$$\begin{aligned}
& \text{where } Funding_{treaty} \\
&= Spending_{mil} \times Reduce_{treaty} \\
&= \$2.72T \times 1\% \\
&= \$27.2B
\end{aligned}$$

10.4 4. Cost per DALY Derivation (\$0.841 (95% CI: \$0.242-\$1.75))

Step 1: Total protocol infrastructure cost (10-year NPV) - NPV total cost: \$611M (95% CI: \$415M-\$853M)

Step 2: DALYs averted - Total DALYs: 565B DALYs (95% CI: 361B DALYs-877B DALYs)

Step 3: Calculate cost per DALY

$$Cost_{direct,DALY} = \frac{NPV_{direct}}{DALY_{s_{max}}} = \frac{\$475B}{565B} = \$0.841$$

$$\begin{aligned} & \text{where } NPV_{direct} \\ &= \frac{T_{queue,dFDA}}{Treasury_{RD,ann} \times r_{discount}} \\ &= \frac{36}{\$21.8B \times 3\%} \\ &= \$475B \end{aligned}$$

$$\begin{aligned} & \text{where } Treasury_{RD,ann} \\ &= Funding_{treaty} - Payout_{bond,ann} - Funding_{political,ann} \\ &= \$27.2B - \$2.72B - \$2.72B \\ &= \$21.8B \end{aligned}$$

$$\begin{aligned} & \text{where } Funding_{treaty} \\ &= Spending_{mil} \times Reduce_{treaty} \\ &= \$2.72T \times 1\% \\ &= \$27.2B \end{aligned}$$

$$\begin{aligned} & \text{where } Payout_{bond,ann} \\ &= Funding_{treaty} \times Pct_{bond} \\ &= \$27.2B \times 10\% \\ &= \$2.72B \end{aligned}$$

$$\begin{aligned} & \text{where } Funding_{treaty} \\ &= Spending_{mil} \times Reduce_{treaty} \\ &= \$2.72T \times 1\% \\ &= \$27.2B \end{aligned}$$

$$\begin{aligned} & \text{where } Funding_{political,ann} \\ &= Funding_{treaty} \times Pct_{political} \\ &= \$27.2B \times 10\% \\ &= \$2.72B \end{aligned}$$

$$\begin{aligned} & \text{where } Funding_{treaty} \\ &= Spending_{mil} \times Reduce_{treaty} \\ &= \$2.72T \times 1\% \\ &= \$27.2B \end{aligned}$$

$$\text{where } T_{queue,dFDA} = \frac{T_{queue,SQ}}{k_{capacity}} = \frac{443}{12.3} = 36$$

$$\text{where } T_{queue,SQ} = \frac{N_{untreated}}{Treatments_{new,ann}} = \frac{6,650}{15} = 443$$

$$\text{where } N_{untreated} = N_{rare} \times 0.95 = 7,000 \times 0.95 = 6,650$$

$$\text{where } k_{capacity} = \frac{N_{fundable,ann}}{Slots_{curr}} = \frac{23.4M}{1.9M} = 12.3$$

Comparison: Malaria bed nets cost \$89 (95% CI: \$78-\$100)/DALY. This framework operates at vastly greater scale while achieving competitive cost-effectiveness.

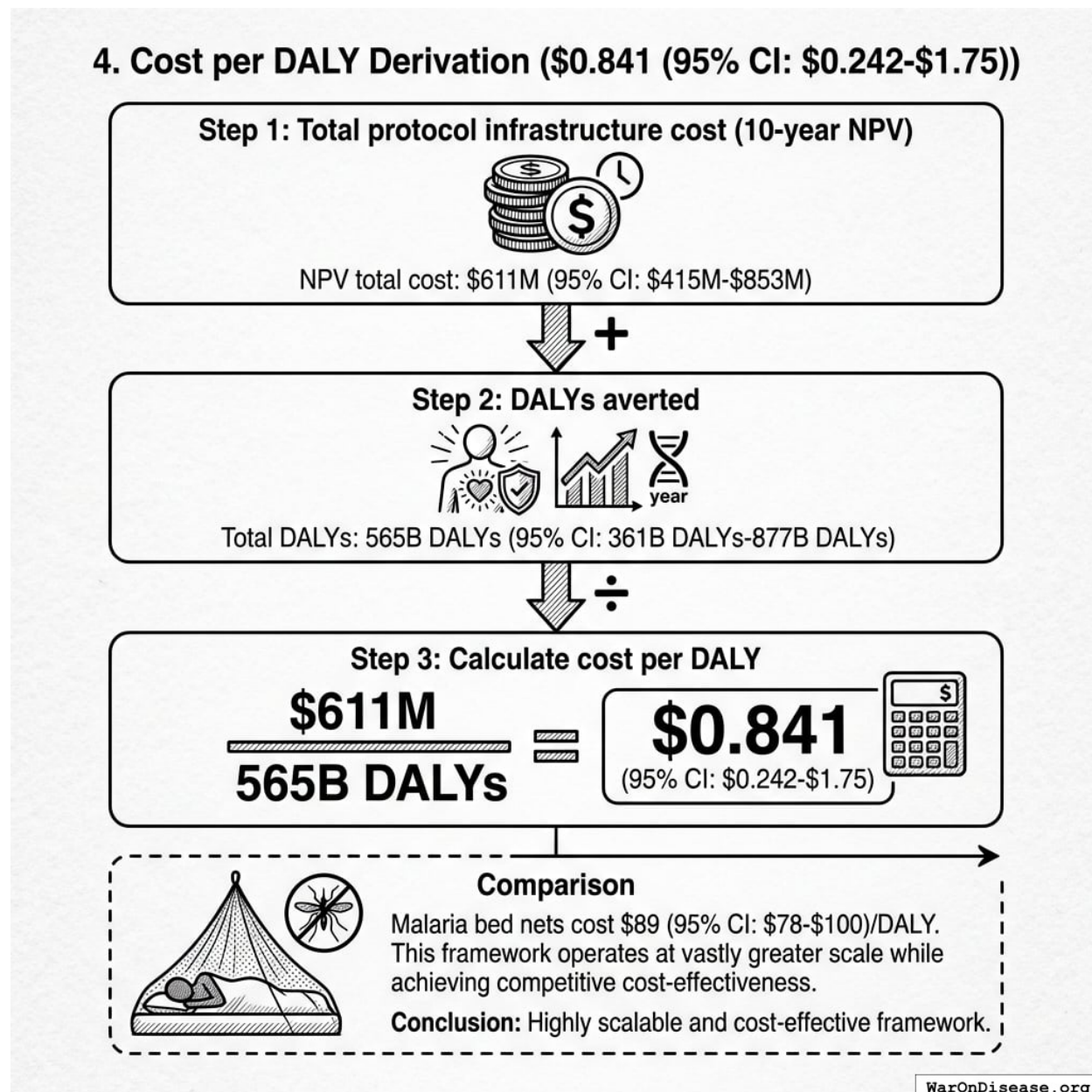


Figure 34: A visual comparison of cost-effectiveness showing the derivation of \$0.84 per DALY alongside the \$89 benchmark for malaria bed nets.

10.5 5. ROI Derivation (637:1 (95% CI: 569:1-790:1))

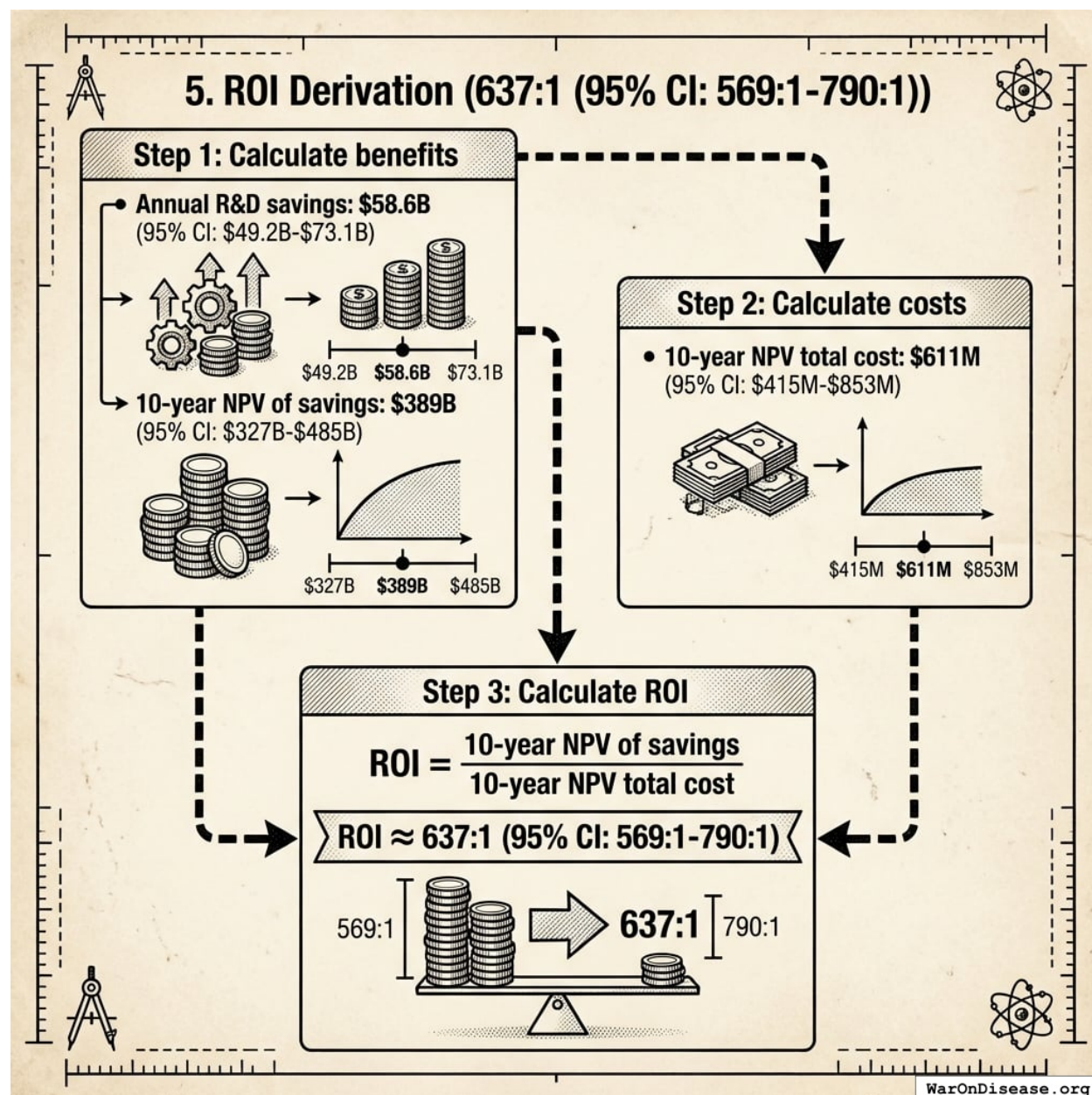


Figure 35: A visual breakdown of the ROI calculation showing the massive scale difference between \$389 billion in 10-year benefits and \$611 million in costs, resulting in a 637:1 ratio.

Step 1: Calculate benefits - Annual R&D savings: \$58.6B (95% CI: \$49.2B-\$73.1B) - 10-year NPV of savings: \$389B (95% CI: \$327B-\$485B)

Step 2: Calculate costs - 10-year NPV total cost: \$611M (95% CI: \$415M-\$853M)

Step 3: Calculate ROI

$$ROI_{RD} = \frac{NPV_{RD}}{Cost_{dFDA,total}} = \frac{\$389B}{\$611M} = 637$$

$$\text{where } NPV_{RD} = \sum_{t=1}^{10} \frac{Savings_{RD,ann} \times \frac{\min(t,5)}{5}}{(1+r)^t}$$

$$\begin{aligned} & \text{where } Savings_{RD,ann} \\ &= Benefit_{RD,ann} - OPEX_{dFDA} \\ &= \$58.6B - \$40M \\ &= \$58.6B \end{aligned}$$

$$\begin{aligned} & \text{where } Benefit_{RD,ann} \\ &= Spending_{trials} \times Reduce_{pct} \\ &= \$60B \times 97.7\% \\ &= \$58.6B \end{aligned}$$

$$\begin{aligned} & \text{where } Reduce_{pct} \\ &= 1 - \frac{Cost_{pragmatic,pt}}{Cost_{P3,pt}} \\ &= 1 - \frac{\$929}{\$41K} \\ &= 97.7\% \end{aligned}$$

$$\begin{aligned} & \text{where } OPEX_{dFDA} \\ &= Cost_{platform} + Cost_{staff} + Cost_{infra} \\ & \quad + Cost_{regulatory} + Cost_{community} \\ &= \$15M + \$10M + \$8M + \$5M + \$2M \\ &= \$40M \end{aligned}$$

$$\begin{aligned} & \text{where } Cost_{dFDA,total} \\ &= PV_{OPEX} + Cost_{upfront,total} \\ &= \$342M + \$270M \\ &= \$611M \end{aligned}$$

$$\begin{aligned} & \text{where } PV_{OPEX} \\ &= \frac{T_{horizon}}{OPEX_{total} \times r_{discount}} \\ &= \frac{10}{\$40M \times 3\%} \\ &= \$342M \end{aligned}$$

$$\begin{aligned} & \text{where } OPEX_{total} \\ &= OPEX_{ann} + OPEX_{DIH,ann} \\ &= \$18.9M + \$21.1M \\ &= \$40M \end{aligned}$$

$$\begin{aligned} & \text{where } Cost_{upfront,total} \\ &= Cost_{upfront} + Cost_{DIH,init} \\ &= \$40M + \$230M \end{aligned}$$

10.6 Verification Summary

Metric	Value	Primary Inputs	Data Sources
Trial Capacity	12.3x (95% CI: 4.19x-61.3x)	Funding, trial costs	ADAPTABLE trial, ClinicalTrials.gov
Timeline Shift	212 years (95% CI: 135 years-355 years)	Efficacy lag, backlog model	FDA approval data, disease registry
Lives Saved	10.7B deaths (95% CI: 7.39B deaths-16.2B deaths)	Mortality rates, timeline	WHO GBD, mortality statistics
Cost/DALY	\$0.841 (95% CI: \$0.242-\$1.75)	NPV costs, DALYs	ROM estimates, DALY calculations
ROI	637:1 (95% CI: 569:1-790:1)	Costs, savings	NPV analysis with 5-year ramp

All parameters, confidence intervals, and Monte Carlo distributions are documented in [Parameters and Calculations](#).

11 Key Analytical Assumptions

This analysis rests on several core assumptions that should be made explicit for academic transparency:

11.1 Linear Scaling Assumption

Assumption: Each additional dollar of trial funding produces proportional additional discoveries.

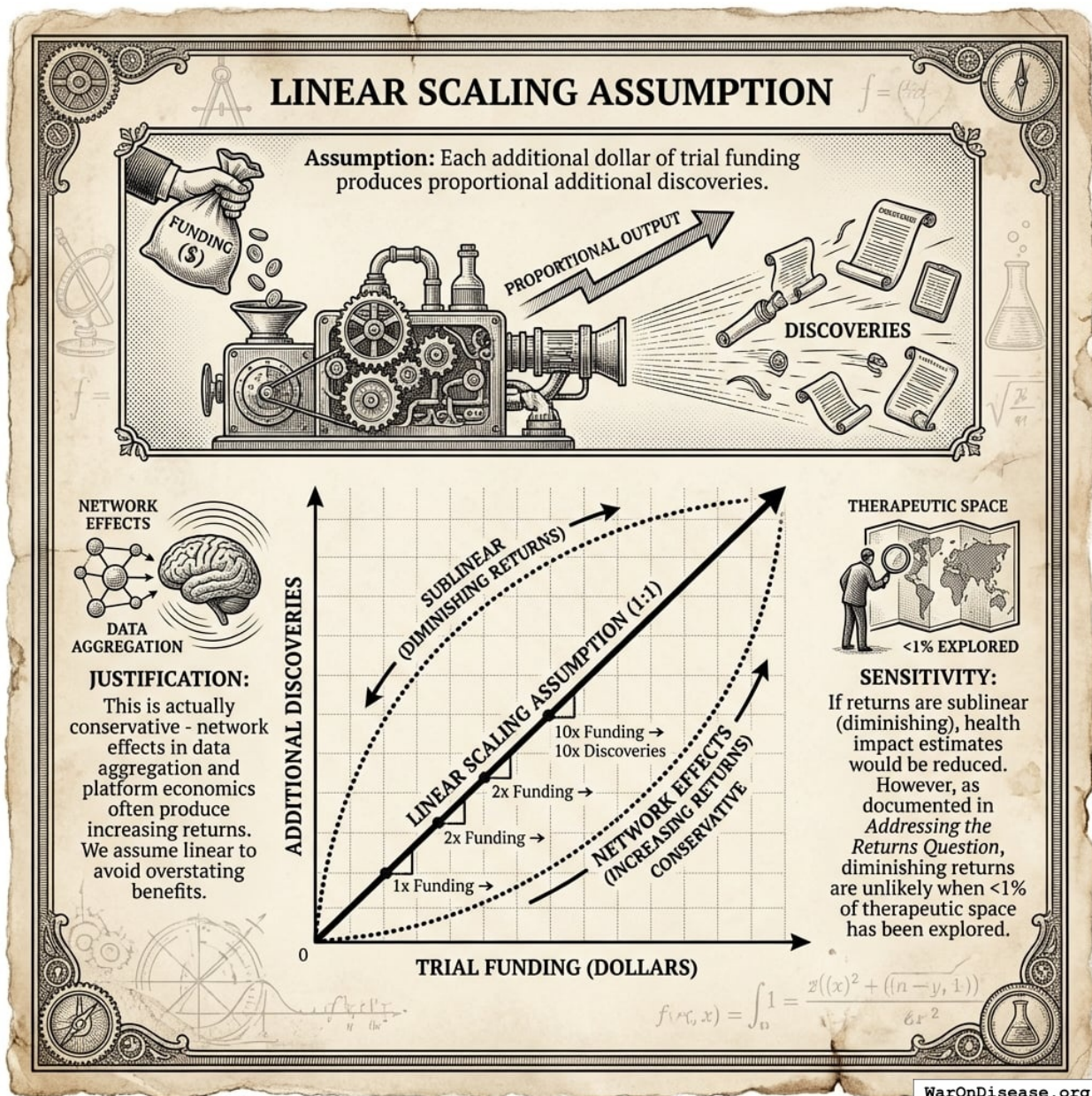


Figure 36: A comparison of linear scaling against diminishing and compounding returns, illustrating why a linear assumption is a conservative model for discovery output relative to funding.

Justification: This is actually conservative - network effects in data aggregation and platform economics often produce increasing returns. We assume linear to avoid overstating benefits.

Sensitivity: If returns are sublinear (diminishing), health impact estimates would be reduced. However, as documented in [Addressing the Returns Question](#), diminishing returns are unlikely when <1% of therapeutic space has been explored.

11.2 Adoption Rate Assumptions

Assumption: Framework adoption follows a 5-year ramp (20%, 40%, 60%, 80%, 100%) before reaching full capacity.

Justification: Based on historical technology adoption curves in healthcare (EHR adoption, telemedicine during COVID). The ramp is built into NPV calculations.

Sensitivity: Slower adoption delays benefits but doesn't change eventual steady-state impact. NPV is reduced with slower adoption due to discounting.

11.3 Cost Reduction Assumptions

Assumption: Pragmatic trials cost \$929 (95% CI: \$97-\$3K)/patient versus \$41K (95% CI: \$20K-\$120K)/patient for traditional trials.

Justification: Based on ADAPTABLE trial (\$929 (95% CI: \$929-\$1.40K)/patient) and systematic review of 64 pragmatic trials (median \$97 (95% CI: \$19-\$478)/patient). RECOVERY achieved \$500 (95% CI: \$400-\$2.50K)/patient under exceptional NHS/COVID conditions.

Sensitivity: The tornado diagrams show ROI remains strongly positive even at 30% cost reduction (vs. baseline 97.7% (95% CI: 97.5%-98.9%)).

11.4 Eventually Avoidable Mortality Assumption

Assumption: 92.6% (95% CI: 50%-98%) of disease deaths are eventually avoidable with sufficient biomedical research.

Justification: Historical trend shows ~70% reduction in age-adjusted mortality since 1900. Most major disease categories have known biological mechanisms amenable to intervention. See [Why “Eventually Avoidable” Matters](#).

Sensitivity: Health impact scales linearly with this assumption. At 25% avoidability (half the estimate), health benefits are halved. R&D savings are unaffected.

11.5 Counterfactual Baseline Specification

This cost-effectiveness analysis uses the **status quo** as the baseline counterfactual: current clinical trial infrastructure continues operating at current efficiency (\$41K (95% CI: \$20K-\$120K)/patient) and capacity (1.90M patients/year (95% CI: 1.50M patients/year-2.30M patients/year) participants/year). Under this baseline, the \$21.8B/year allocated to pragmatic trials would not exist.

Why status quo is the appropriate baseline:

1. **No comparable interventions exist:** There is no competing proposal that would achieve similar trial cost reductions at scale
2. **Historical trend supports it:** Trial costs have increased, not decreased, over the past 50 years (105x (95% CI: 90.6x-119x) since 1962)
3. **Incremental improvements are marginal:** Ongoing digitization efforts (DCT platforms, EHR integration) produce 10-20% efficiency gains, not the 97.7% (95% CI: 97.5%-98.9%) reduction from pragmatic trial design

Alternative counterfactual scenarios:

1. **Organic efficiency improvement:** Clinical trial costs decrease 2-3% annually through technology adoption. Under this scenario, the marginal impact of dFDA is reduced by the amount of improvement that would occur anyway. At 3%/year organic improvement over 10 years, approximately 26% of the cost reduction would occur regardless, reducing dFDA's attributable benefit to ~74% of projections.
2. **Alternative government priorities:** Funds are allocated to other health investments (NIH grants, hospital infrastructure, insurance subsidies). Each alternative use would require separate cost-benefit analysis. However, none of these alternatives address the core trial cost problem; they operate within the existing high-cost paradigm.
3. **Return to taxpayers:** Funds are returned via tax cuts, enabling private consumption and investment. Under this scenario, the opportunity cost equals the weighted average return on private capital (approximately 3% annually). The dFDA ROI of 637:1 (95% CI: 569:1-790:1) substantially exceeds this threshold.

Methodological note: The analysis uses the neutral status quo baseline to avoid biasing results in either direction. Sensitivity analysis (tornado diagrams) demonstrates robustness across baseline assumptions.

The analysis uses the neutral status quo baseline to avoid biasing results in either direction.

11.6 Methodology Validation Against Accepted Benchmarks

This analysis uses standard health economics methodology identical to that employed by EPA, DOT, GiveWell, NICE, WHO-CHOICE, and CBO:

Our Method	Equivalent Standard	Institution Using It
Value of Statistical Life (\$10M (95% CI: \$5M-\$15M))	VSL for regulatory impact	EPA, DOT, FDA
Cost per DALY (\$0.841 (95% CI: \$0.242-\$1.75)/DALY)	ICER thresholds	GiveWell, NICE, WHO-CHOICE
Monte Carlo uncertainty propagation	Probabilistic sensitivity analysis	ICER, Cochrane, HTA agencies
NPV with discount rate	Standard cost-benefit analysis	CBO, OMB Circular A-94
Long-horizon cumulative impact	Social cost of carbon	EPA, IPCC, Stern Review

The large headline figures reflect the cumulative value of permanent infrastructure, the same methodology used to value smallpox eradication (\$300M program valued at total future lives saved) and climate economics (multi-trillion dollar damage estimates that exceed annual GDP). Our 95% confidence intervals span nearly an order of magnitude, wider than most published health economics studies.

12 Appendix Calculation Frameworks and Detailed Analysis

This appendix provides the detailed models and data used in the cost-benefit analysis.

12.1 Calculation Framework - NPV Methodology

Uses 10-year NPV horizon (standard business practice). See [Verification: Complete Derivation Chains](#) for full methodology.

12.2 Financial Analysis Summary

12.2.1 Health Impact Uncertainty Analysis

The Monte Carlo distributions below show the range of health impact estimates across 10,000 simulations, accounting for uncertainty in timeline shift, mortality rates, and avoidable percentages:

Lives Saved Distribution:

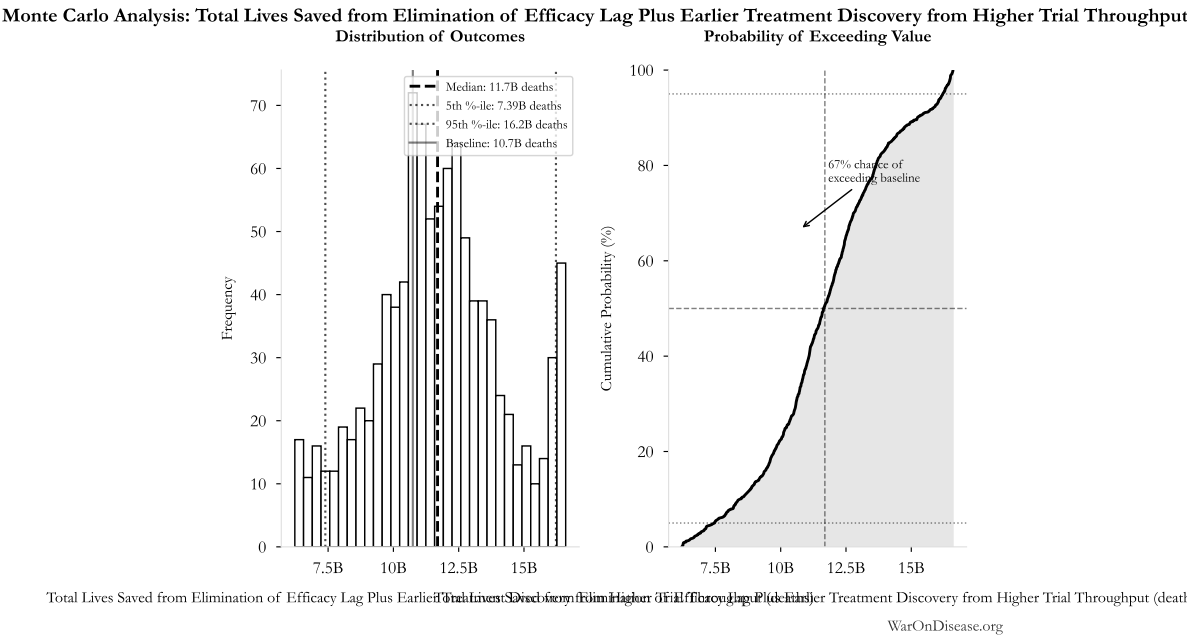


Figure 37: Monte Carlo Distribution: Total Lives Saved from Elimination of Efficacy Lag Plus Earlier Treatment Discovery from Higher Trial Throughput (10,000 simulations)

Simulation Results Summary: Total Lives Saved from Elimination of Efficacy Lag Plus Earlier Treatment Discovery from Higher Trial Throughput

Statistic	Value
Baseline (deterministic)	10.7B
Mean (expected value)	11.7B
Median (50th percentile)	11.7B
Standard Deviation	2.45B

Statistic	Value
90% Confidence Interval	[7.39B, 16.2B]

The histogram shows the distribution of Total Lives Saved from Elimination of Efficacy Lag Plus Earlier Treatment Discovery from Higher Trial Throughput across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.

Economic Value Distribution:

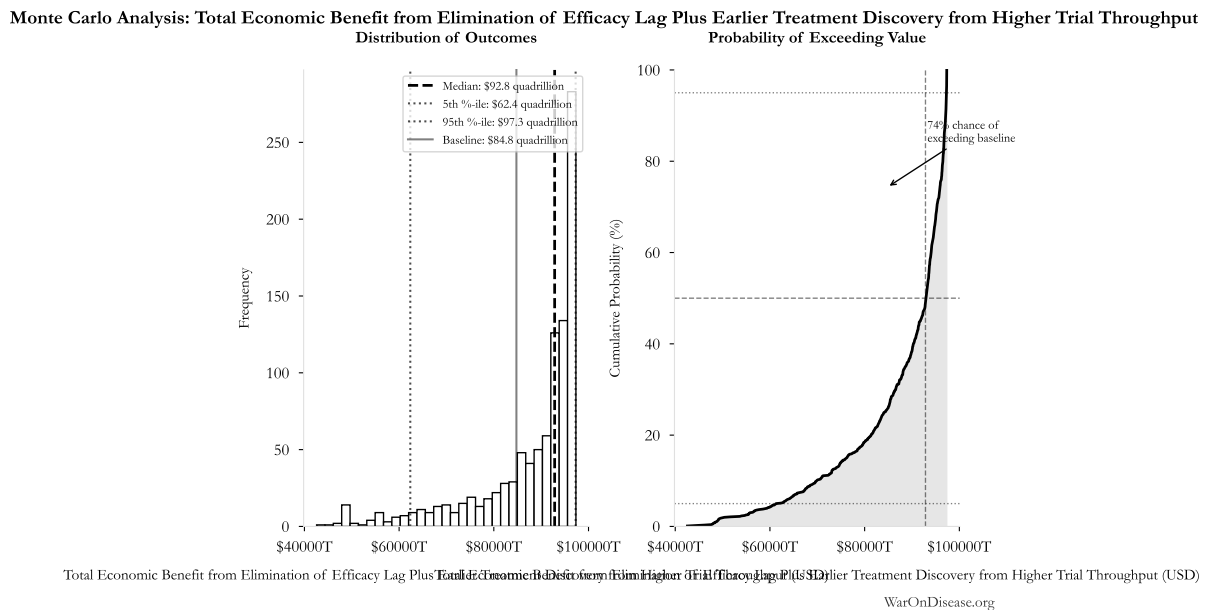


Figure 38: Monte Carlo Distribution: Total Economic Benefit from Elimination of Efficacy Lag Plus Earlier Treatment Discovery from Higher Trial Throughput (10,000 simulations)

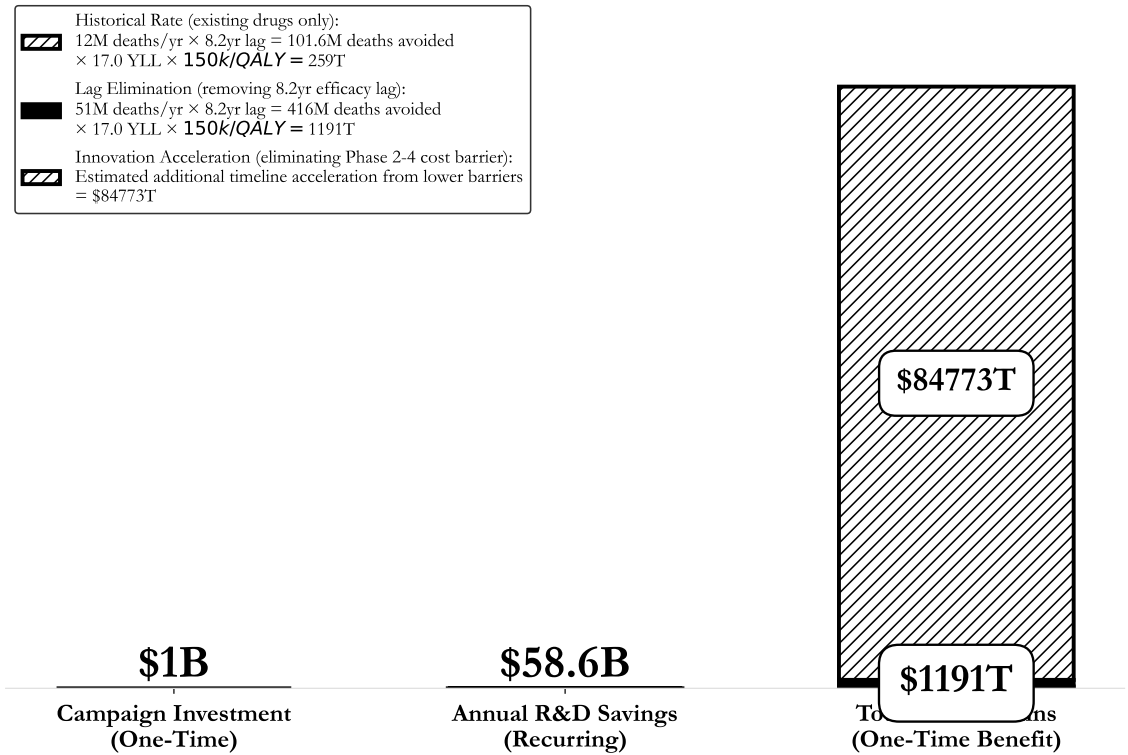
Simulation Results Summary: Total Economic Benefit from Elimination of Efficacy Lag Plus Earlier Treatment Discovery from Higher Trial Throughput

Statistic	Value
Baseline (deterministic)	\$84.8 quadrillion
Mean (expected value)	\$87.8 quadrillion
Median (50th percentile)	\$92.8 quadrillion
Standard Deviation	\$11.5 quadrillion
90% Confidence Interval	[\$62.4 quadrillion, \$97.3 quadrillion]

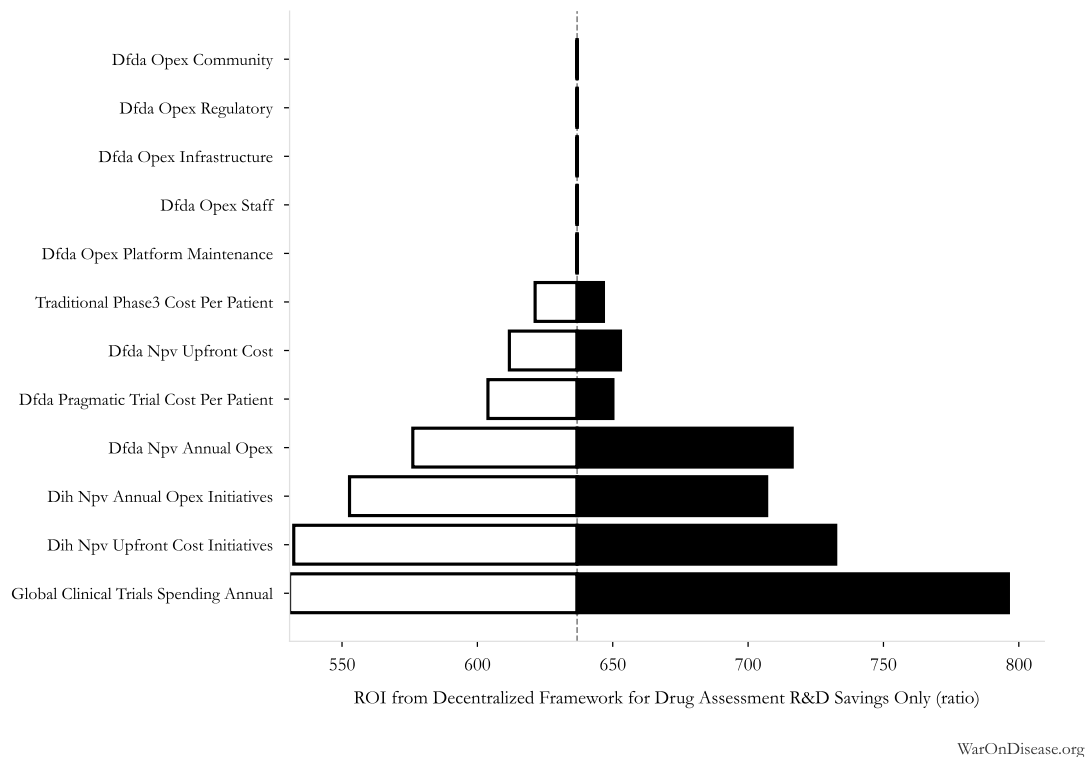
The histogram shows the distribution of Total Economic Benefit from Elimination of Efficacy Lag Plus Earlier Treatment Discovery from Higher Trial Throughput across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.

12.2.2 Financial Visualizations

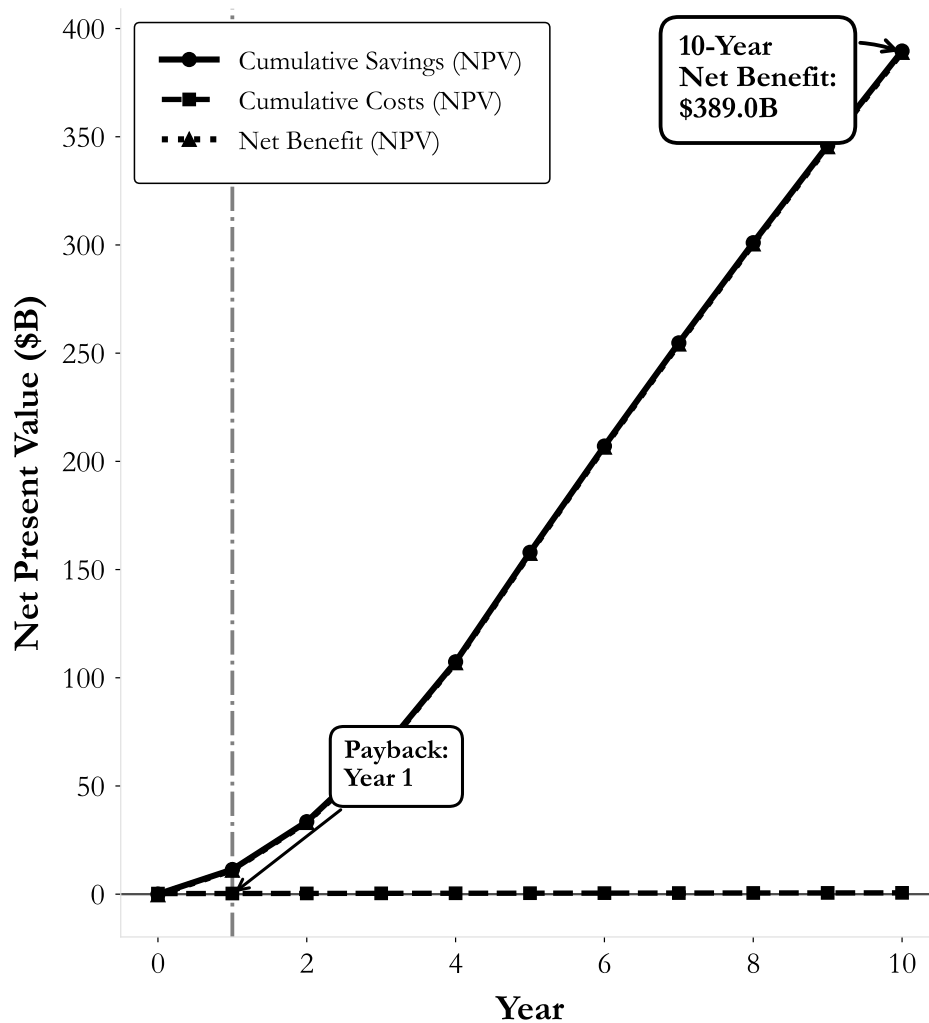
1% Treaty Investment and Returns



Sensitivity Analysis: ROI from Decentralized Framework for Drug Assessment R&D Savings Only



Net Present Value: A Decentralized Drug Assessment Framework 8% Discount, 5-Yr Adoption Ramp



WarOnDisease.org

NPV Analysis Summary:

Total Costs (NPV): \$0.61B
 Total Savings (NPV): \$389.62B
 Net Benefit (NPV): \$389.01B
 ROI: 637:1
 Payback Period: Year 1

12.3 Cost-Utility Framework

We present a cost-utility analysis using the **quality-adjusted life years (QALYs)** and **disability-adjusted life years (DALYs)** metrics. This approach is the US and global standard for evaluating

the value of health interventions¹⁴¹.

- **QALY:** One year of life in perfect health. Gains are calculated as:

$$\text{QALYs Gained} = (Q_1 \times T_1) - (Q_0 \times T_0)$$

Where Q_0/Q_1 = quality of life (0-1) before/after, T_0/T_1 = years of life before/after.

- **Cost-Effectiveness:** A decentralized framework for drug assessment achieves cost-effectiveness through dual pathways:
 1. **R&D Savings:** \$58.6B (95% CI: \$49.2B-\$73.1B)+ annual savings from 97.7% (95% CI: 97.5%-98.9%) trial cost reduction
 2. **Health Gains:** 565B DALYs (95% CI: 361B DALYs-877B DALYs) averted from the full timeline shift (~212 years (95% CI: 135 years-355 years) from 12.3x (95% CI: 4.19x-61.3x) trial capacity + efficacy lag elimination)

This combination creates a **dominant intervention:** simultaneously saves money and improves health outcomes.

- **US Willingness-to-Pay Threshold:** Typically \$100,000–\$150,000 per QALY for interventions that *add* costs. Dominant interventions that both save money and improve health are favorable regardless of this threshold.
- **Sources for Context:**
 - QALY methodology and standards¹⁴¹: “The quality-adjusted life year (QALY) is the academic standard for measuring how well all different kinds of medical treatments lengthen and/or improve patients’ lives...”
 - Health economic evaluation¹⁰⁵: Standard health economic analysis considers cost-effectiveness across intervention types.

12.3.1 QALY Benefit Streams Breakdown

The total DALY impact (565B DALYs (95% CI: 361B DALYs-877B DALYs)) derives from three distinct benefit streams with varying levels of empirical support:

A. Accelerated Development of Existing Pipeline Drugs

Health gains from bringing effective treatments to patients faster through shortened development and approval timelines:

- **Baseline:** Research shows treatment delays significantly increase mortality. Cancer studies indicate approximately 10% increased mortality risk per month of delay.
- **Mechanism:** 12.3x (95% CI: 4.19x-61.3x) trial capacity reduces average development time
- **Confidence:** High. Well-documented costs of delayed access (84,000 life-years lost per year of delay in cancer therapies alone)

B. Improved Preventative Care via Real-World Evidence

Value of using comprehensive data to optimize preventative care and treatment effectiveness:

- **Baseline:** Cancer screenings alone have saved millions of life-years; significant untapped potential remains

- **Mechanism:** Large-scale identification of at-risk populations and real-world effectiveness measurement enables personalized prevention
- **Confidence:** Medium. Preventative care benefits are well-established, but scale of improvement from comprehensive data remains uncertain

C. Enabling Research for Previously Untreatable Diseases

Transformative potential to create viable research pathways for conditions ignored due to high trial costs:

- **Baseline:** 7.00k diseases (95% CI: 6.00k diseases-10.0k diseases)+ rare diseases, 95% lack FDA-approved treatments
- **Mechanism:** Radically lower per-patient costs (\$929 (95% CI: \$97-\$3K) vs \$41K (95% CI: \$20K-\$120K)) make rare disease R&D economically feasible
- **Confidence:** Lower. Economic viability of rare disease research is theoretically sound but empirically unproven at scale

Conservative Approach: Base case uses median estimates across all three streams; sensitivity analysis demonstrates positive returns even when using R&D savings alone (637:1 (95% CI: 569:1-790:1) ROI independent of health benefits).

QALY Benefit Streams Breakdown

Total DALY impact (565B DALYs (95% CI: 361B DALYs-877B DALYs)) from three distinct streams.

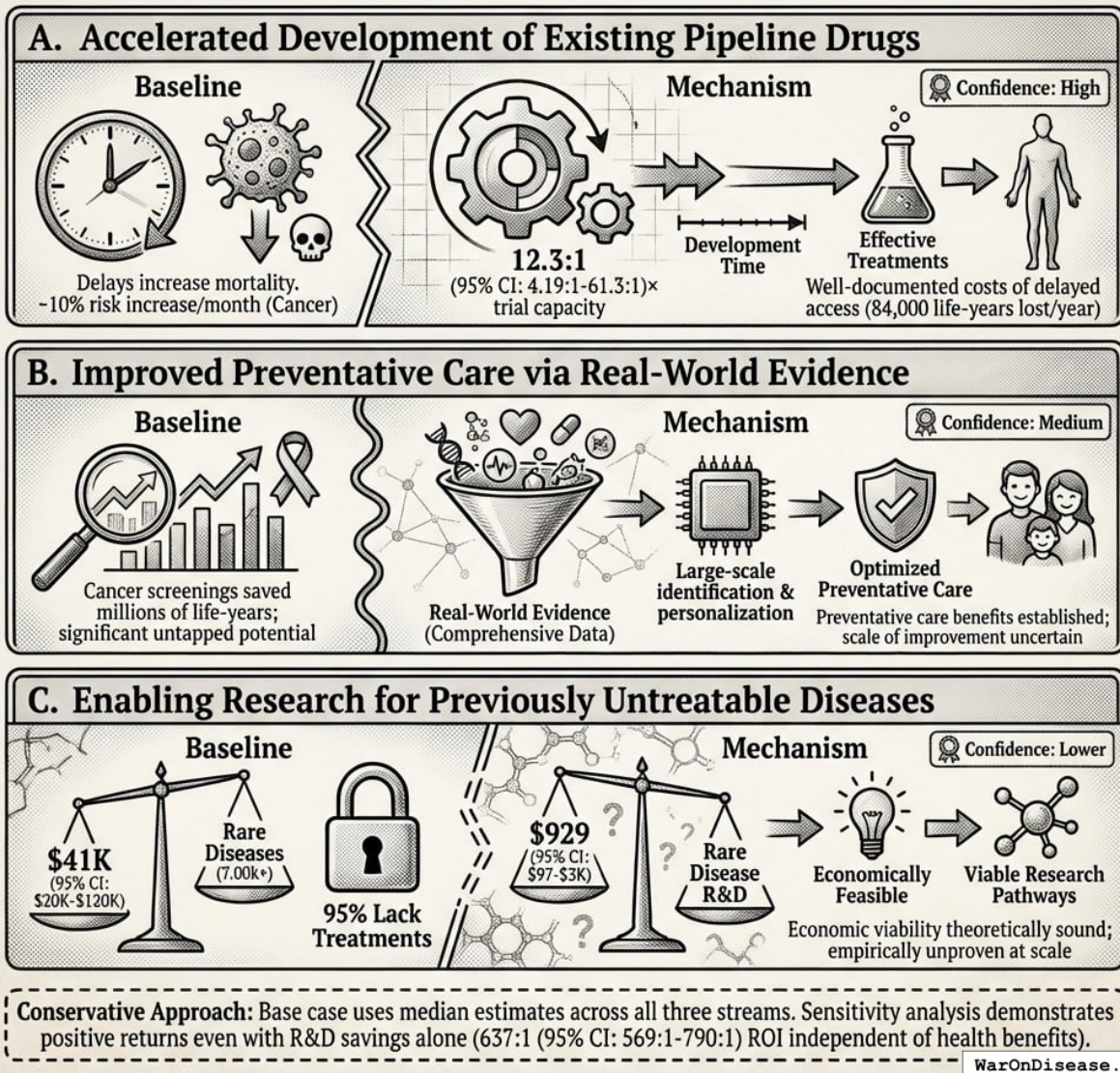


Figure 39: A breakdown of the three primary value streams (accelerated drug development, preventative care optimization, and rare disease research) that contribute to the total projected DALY impact.

12.3.2 DALY Sensitivity Analysis

The following auto-generated sensitivity analyses show how cost-effectiveness varies based on uncertainty in input parameters. These use Monte Carlo simulation with uncertainty propagation from parameter distributions in `dih_models/parameters.py`.

Key DALY Outcomes:

Sensitivity Indices for Total DALYs from Elimination of Efficacy Lag Plus Earlier

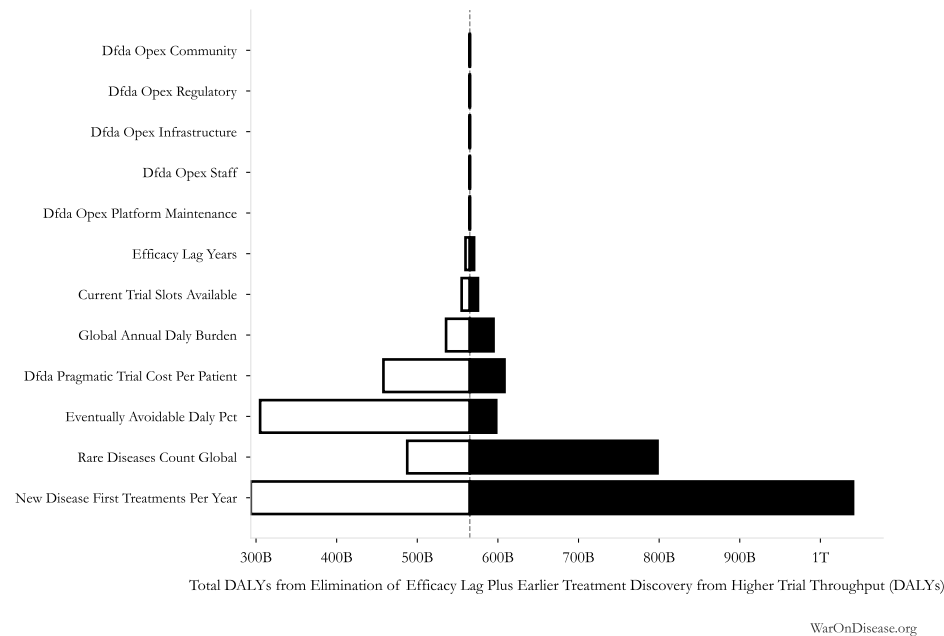
Treatment Discovery from Higher Trial Throughput

Regression-based sensitivity showing which inputs explain the most variance in the output.

Input Parameter	Sensitivity Coefficient	Interpretation
dFDA Trial Capacity Plus Efficacy Lag Years	0.9001	Strong driver
Eventually Avoidable DALY %	0.4864	Moderate driver
Global Annual DALY Burden	0.0433	Minimal effect

Interpretation: Standardized coefficients show the change in output (in SD units) per 1 SD change in input. Values near ±1 indicate strong influence; values exceeding ±1 may occur with correlated inputs.

Sensitivity Analysis: Total DALYs from Elimination of Efficacy Lag Plus Earlier Treatment Discovery from Higher Trial Throughput



12.4 Comparative Cost-Effectiveness - A Decentralized Framework vs Other Interventions

To provide context for the impact of a decentralized framework’s infrastructure, the chart below visualizes its cost-effectiveness against other well-understood public health programs. The metric used is **Quality-Adjusted Life Years (QALYs) Gained per \$1 Million of Spending**. A higher number signifies greater cost-effectiveness.

For standard interventions, this value is calculated as \$1,000,000 / ICER, where the ICER (Incremental Cost-Effectiveness Ratio) is the cost to gain one QALY. For **dominant** interventions that are both more effective and less expensive, the ICER is negative, and this metric isn’t strictly applicable. For these cases, an illustrative range is used to represent their high value.

All data used in the chart is derived from the table and sources below.

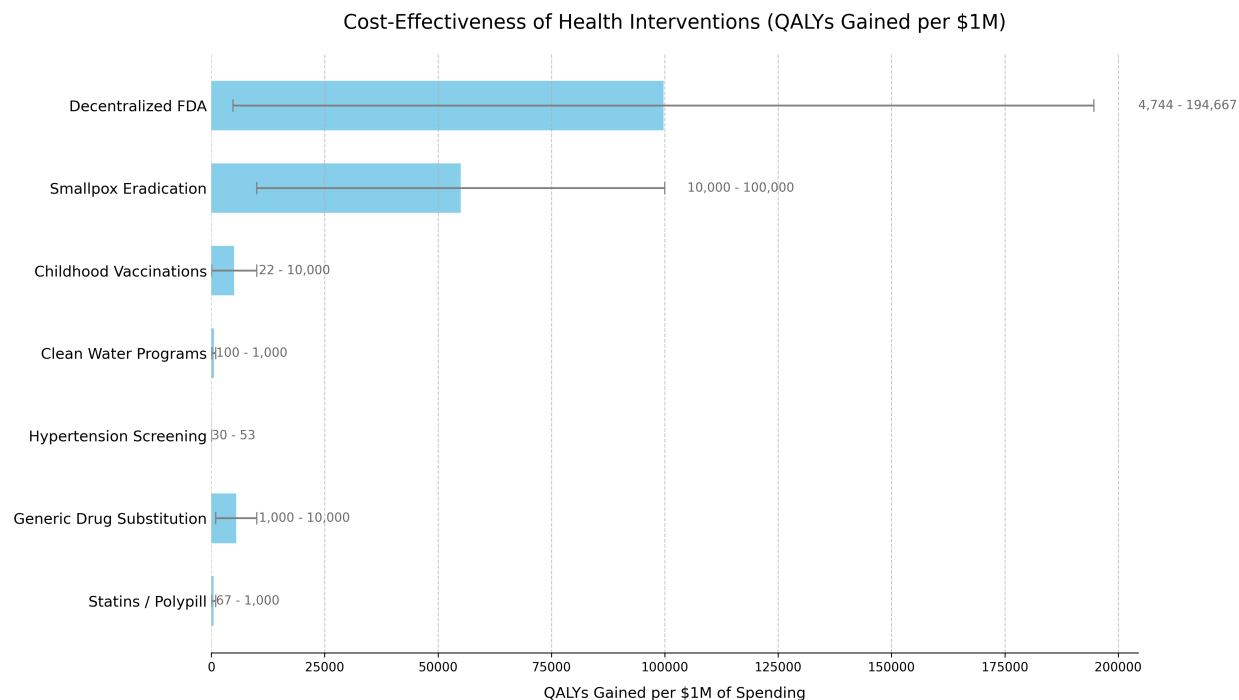


Figure 40: Cost-Effectiveness of Health Interventions (QALYs Gained per \$1M)

The following table provides the data and sources that support the chart. The list is ordered to match the chart's presentation.

Intervention	QALYs Gained per \$1M Spending ¹	Typical ICER Range (Cost per QALY Gained)	Classification	Source / Evidence
Decentralized Framework for Drug Assessment	Dominant	Cost-Saving + Health Gain	Dominant	This analysis's Sensitivity Analysis . Based on \$18.9M (95% CI: \$11M-\$26.5M)-\$40M (95% CI: \$27.5M-\$55.4M) annual costs generating 565B DALYs (95% CI: 361B DALYs-877B DALYs) from ~212 years (95% CI: 135 years-355 years)-year timeline shift.
Smallpox Eradication	100,000+ ³	Dominant (Cost-Saving)	Dominant	The \$300M program (1967-1980) prevents 5M annual deaths. Benefit-cost ratio exceeds 100:1. Standard ICER calculation is impractical due to its uncommon scale. (WHO, 2010 ¹⁴²)

Inter- vention	QALYs Gained per \$1M Spending ¹	Typical ICER Range (Cost per QALY Gained)	Classi- fica- tion	Source / Evidence
Child- hood Vacci- na- tions	22+ ³	Often Domi- nant to ~ \$100,000	Dom- inant / Highly Cost- Effective	CDC estimates routine childhood vaccinations prevent 32M hospitalizations and 1.1M deaths among 1994-2023 US birth cohorts, with \$2.9T in societal cost savings. (CDC, 2023 ¹⁴³)
Clean Water Pro- grams	100	~ \$1,000 - \$10,000	Highly Cost- Effective	WHO estimates household water treatment costs \$100-\$500/DALY averted. Community water supply improvements cost \$200/DALY. (WHO, 2004 ¹⁴⁴)
Hy- per- ten- sion Screen- ing	30 - 50	~ \$20,000 - \$33,000	Highly Cost- Effective	Recent US studies show pharmacist-led hypertension management has ICERs in the \$20,000-\$33,000 range per QALY gained, falling within standard willingness-to-pay thresholds. (JAMA Netw Open, 2023 ¹⁴⁵)
Generic Drug Sub- stitu- tion	+ ³	Domi- nant (Cost- Saving)	Dom- inant	By definition cost-saving when therapeutic equivalence is maintained, with typical savings of 30-80% versus brand-name drugs. (WHO, 2015 ¹⁴⁶)
Statins / Polyp- ill	67+ ³	Cost- Saving to ~ \$15,000	Dom- inant / Highly Cost- Effective	Cost-saving in high-risk populations. ICERs range from dominant to \$15k/QALY in lower-risk groups. (eClinicalMedicine, 2022 ¹⁴⁷)
Prag- matic Trials (RE- COV- ERY model)	~250,000	\$4.00 (95% CI: \$1.71- \$10)/QALY	Highly Cost- Effective	UK RECOVERY trial: \$20M (95% CI: \$15M-\$25M) spent, saving 1.00M lives (95% CI: 500k lives-2.00M lives) globally via dexamethasone discovery. 12.5kx (95% CI: 2.3kx-51.5kx) more efficient than standard research. (Note: RECOVERY's \$500 (95% CI: \$400-\$2.50K)/patient benefited from NHS infrastructure; ADAPTABLE achieved \$929 (95% CI: \$929-\$1.40K)/patient in US settings.)
NIH Stan- dard Re- search Port- folio	~20	\$50K (95% CI: \$20K- \$100K)/QALY	Ineffi- cient Base- line	Standard NIH-funded research. Represents current status quo efficiency. ⁷⁶

12.5 Methodology Notes

¹ QALYs per \$1M Calculation:

- For a decentralized framework: (Annual QALYs Gained) / (Annual Cost in Millions)
- Ranges reflect conservative to optimistic scenarios accounting for parameter uncertainties

² Cost-Dominant Interpretation:

- All scenarios for the framework show extremely low cost per DALY while generating net economic benefits that exceed costs
- The framework is “dominant” - more effective and less costly than the status quo

³ Dominant Interventions:

- For cost-saving (dominant) interventions, standard QALY/\$1M calculations are not applicable
- Values shown are illustrative to demonstrate relative cost-effectiveness
- Upper bounds represent the exceptional value of these interventions

12.6 Data Limitations

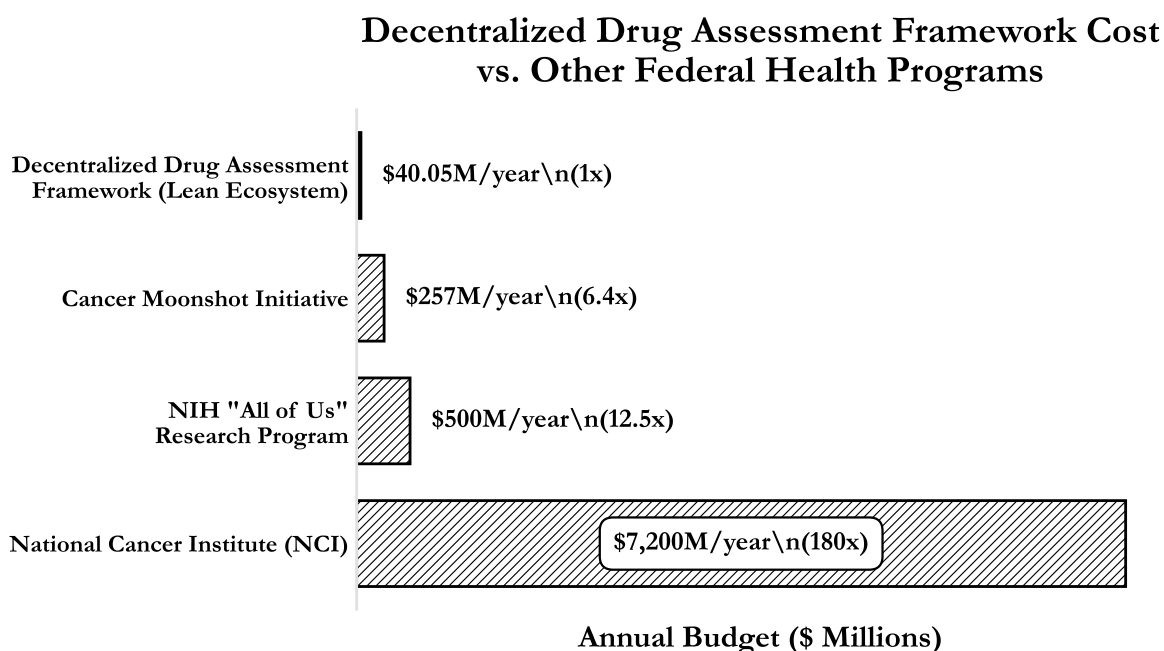
- Historical interventions (e.g., smallpox) use retrospective analyses
- Direct comparisons between interventions should consider contextual differences
- All costs are in 2023 USD, adjusted using appropriate health inflation indices
- QALY calculations use standard health state utility weights where available

13 Comparison to Other Major Public Investments

To provide context for the estimated costs of a decentralized framework, it is useful to compare them to other significant U.S. government investments in health and technology. The projected ‘Lean Ecosystem’ cost for a decentralized framework for drug assessment of approximately **\$40M (95% CI: \$27.5M-\$55.4M) per year** (covering Core framework operations plus medium-scope broader initiatives) is modest in comparison to other major federal projects.

Initiative / Project	Approximate Cost / Budget (Annualized)	Comparison to Framework’s Annual Cost	Source / Note
Decentralized Framework (Lean Ecosystem)	~\$40M (95% CI: \$27.5M-\$55.4M) / year	1x (Baseline)	This analysis
Cancer Moonshot Initiative	~\$257 Million / year (\$1.8B over 7 years)	~6.4x	21st Century Cures Act ¹⁴⁸
NIH “All of Us” Research Program	~\$500M / year (FY23 Approx. Budget)	~12.5x	NIH Budget

Initiative / Project	Approximate Cost / Budget (Annualized)	Comparison to Framework's Annual Cost	Source / Note
Health-Care.gov (Initial Build)	~\$1.7 - \$2.1 Billion ¹⁴⁹ (Total Upfront Cost)	~42x - 52x (of one year's cost)	GAO Reports / Public Reporting ¹⁴⁹
National Cancer Institute (NCI)	~\$7.2 Billion / year ¹⁵⁰ (FY25 Budget)	~180x	NCI Budget Data ¹⁵⁰



WarOnDisease.org

Key Takeaway: The estimated annual cost of this initiative is an order of magnitude smaller than the budgets for other major national health priorities like the “All of Us” program or the Cancer Moonshot. It represents approximately **0.55%** of the NCI’s annual budget (calculated from the framework’s [annual cost](#) and NCI budget¹⁵⁰). This comparison underscores that such an infrastructure is not only a high-leverage investment (due to its massive ROI) but also a remarkably cost-effective one relative to the scale of federal health and technology spending.

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Mean exclusion rate: 86.1% across 158 antidepressant efficacy trials (range: 44.4% to 99.8%) More than 82% of real-world depression patients would be ineligible for antidepressant registration trials Exclusion rates increased over time: 91.4% (2010-2014) vs. 83.8% (1995-2009) Most common exclusions: comorbid psychiatric disorders, age restrictions, insufficient depression severity, medical conditions Emergency psychiatry patients: only 3.3% eligible (96.7% excluded) when applying 9 common exclusion criteria Only a minority of depressed patients seen in clinical practice are likely to be eligible for most AETs Note: Generalizability of antidepressant trials has decreased over time, with increasingly stringent exclusion criteria eliminating patients who would actually use the drugs in clinical practice Additional sources: <https://pubmed.ncbi.nlm.nih.gov/26276679/> | <https://pubmed.ncbi.nlm.nih.gov/26164052/> | <https://www.wolterskluwer.com/en/news/antidepressant-trials-exclude-most-real-world-patients-with-depression>
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Heart failure alone: \$108 billion/year (2012 global analysis, 197 countries) US CVD: \$555B. (2016) → projected \$1.8T by 2050 LMICs total CVD loss: \$3.7T cumulative (2011-2015, 5-year period) CVD is costliest disease category in most developed nations Note: No single \$2.1T global figure found; estimates vary widely by scope and year Additional sources: [https://www.internationaljournalofcardiology.com/article/S0167-5273\(13](https://www.internationaljournalofcardiology.com/article/S0167-5273(13) | <https://www.aha-journals.org/doi/10.1161/CIR.0000000000001258>
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